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(54) SUSTAINABLE AND MODULAR PACKAGING SYSTEMS FOR CONTAINERS FOR PHARMACEUTICAL USE

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(73) Assignee: **AMGEN INC.**, Thousand Oaks, CA (US)

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Related U.S. Application Data

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Publication Classification

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B65D 5/50 (2006.01)

B65D 5/10 (2006.01)

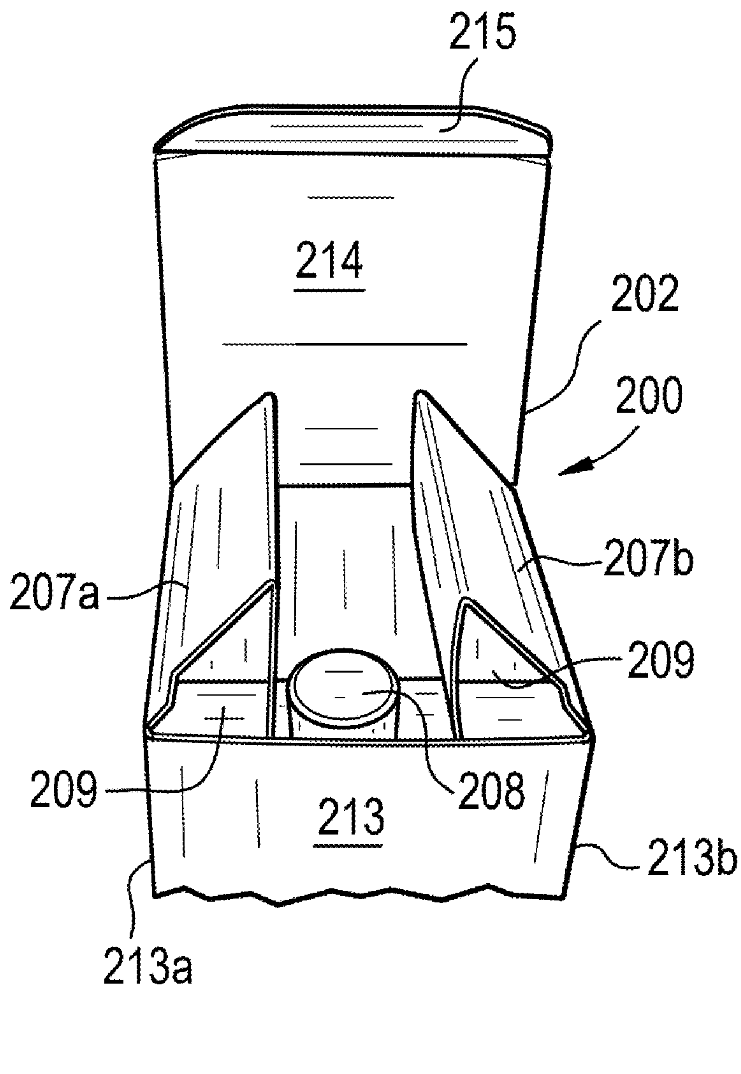
B65D 5/54 (2006.01)

(52) **U.S. Cl.**

CPC ***B65D 5/5038*** (2013.01); ***B65D 5/10*** (2013.01); ***B65D 5/5415*** (2013.01)

(57) **ABSTRACT**

Systems for packaging container(s) is disclosed. The system comprises a carton having a storage configuration where the carton is substantially flat and an assembled configuration. The carton comprises a bottom box, a lid coupled to the bottom box, a first flap extending from the first sidewall of the bottom box, and a second flap extending from the second sidewall of the bottom box. The first and second flaps define a set of cavities. The system also comprises an insert configured to be disposed within the bottom box of the carton. The insert has a storage configuration where the insert is substantially flat and an assembled configuration. The insert defines at least one cavity dimensioned to house at least one container.



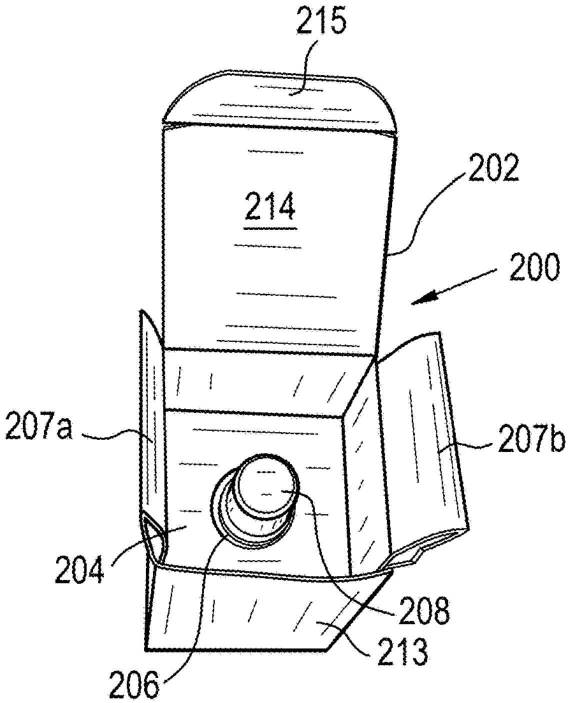


FIG. 1A

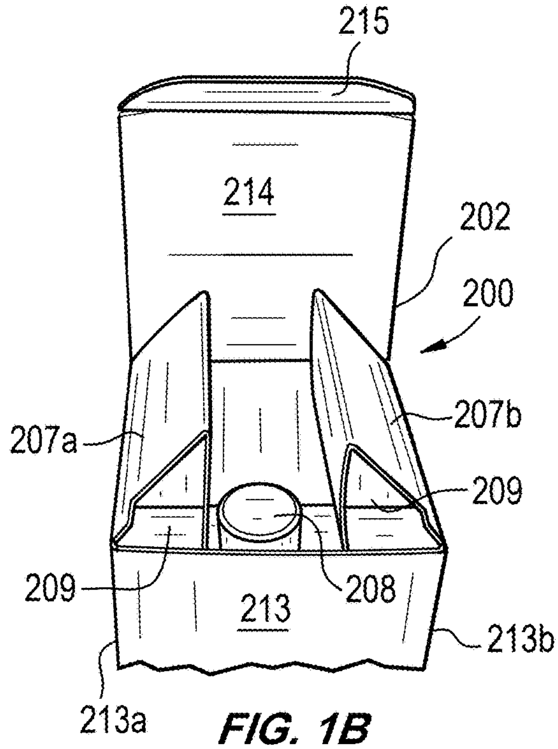


FIG. 1B

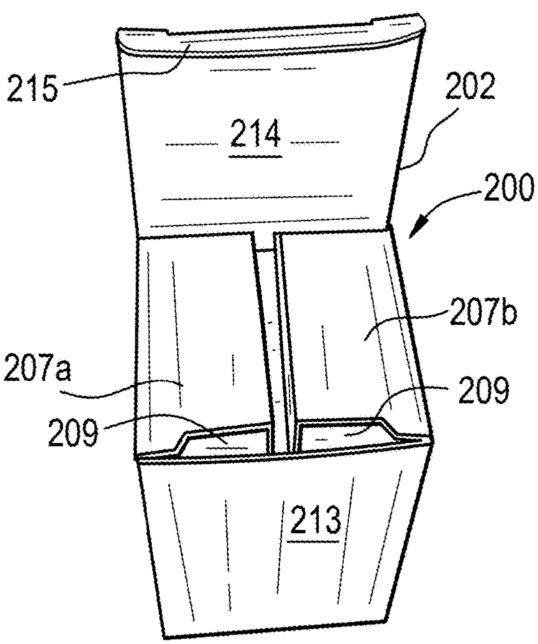


FIG. 1C

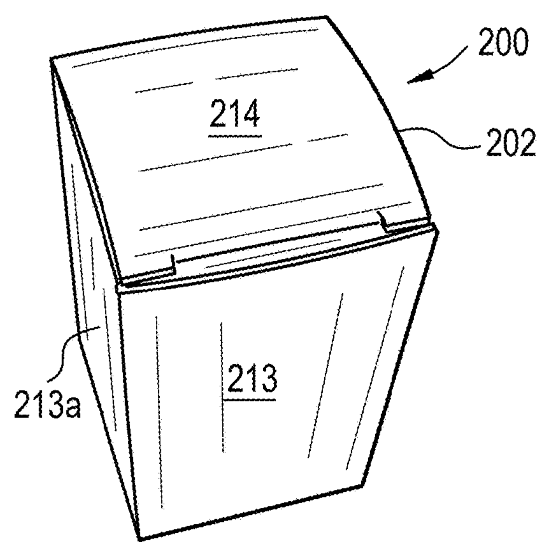


FIG. 1D

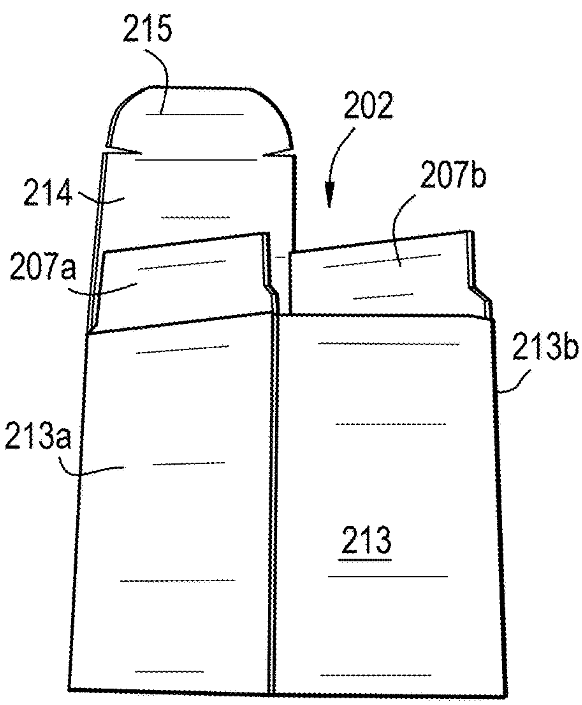


FIG. 2A

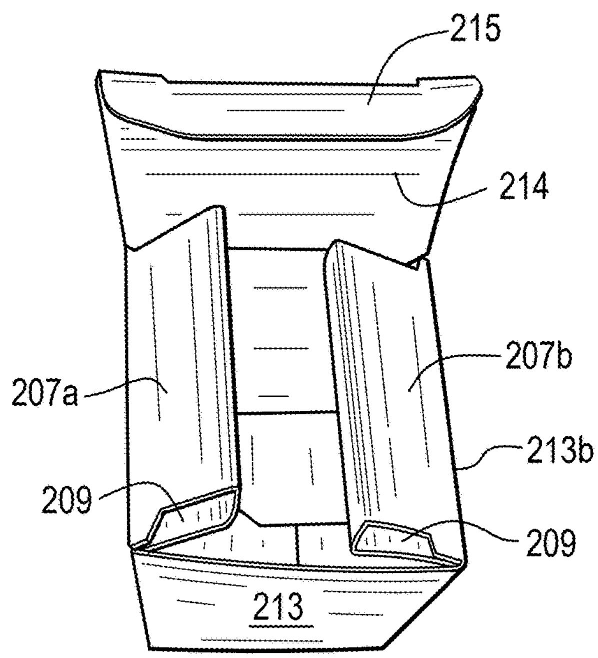


FIG. 2B

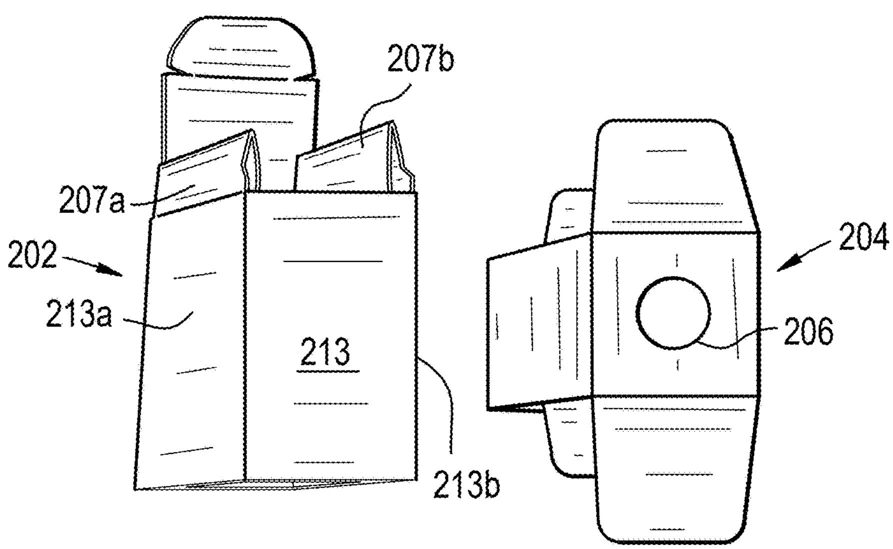


FIG. 2C

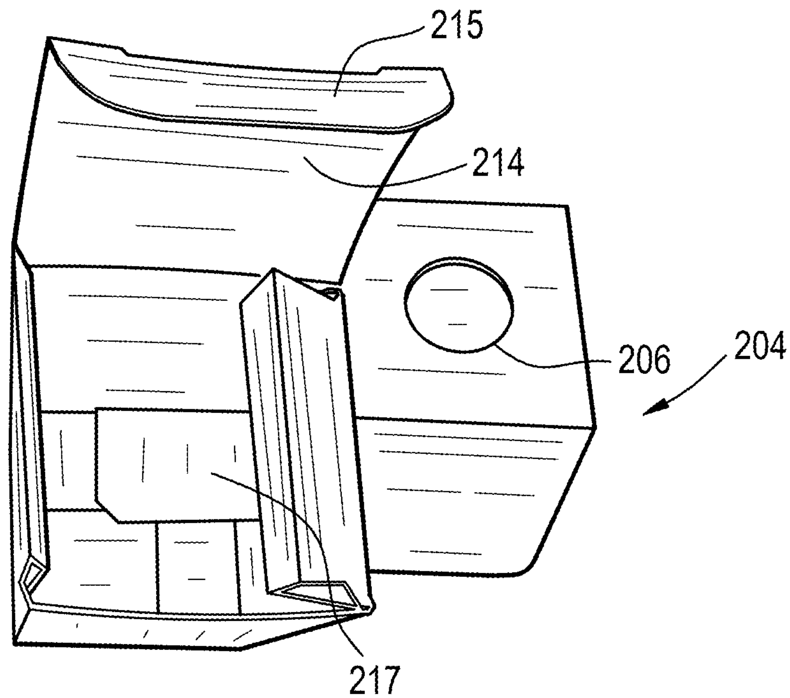


FIG. 2D

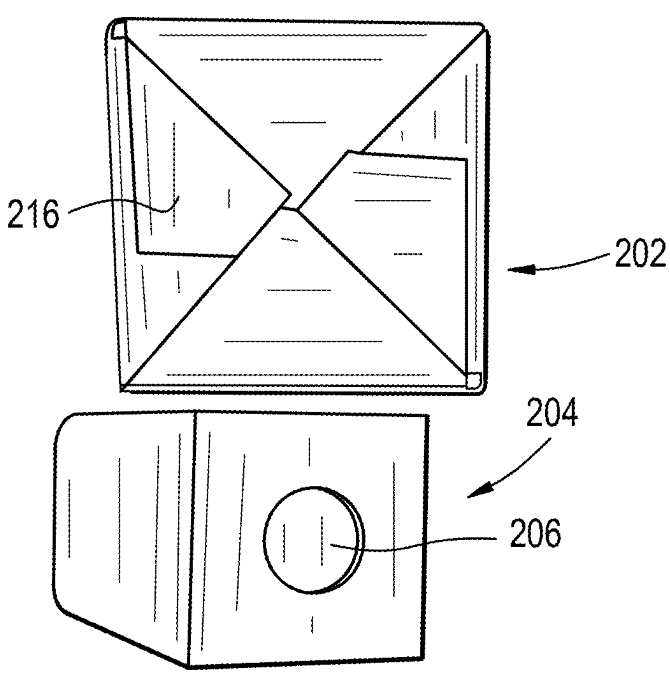


FIG. 2E

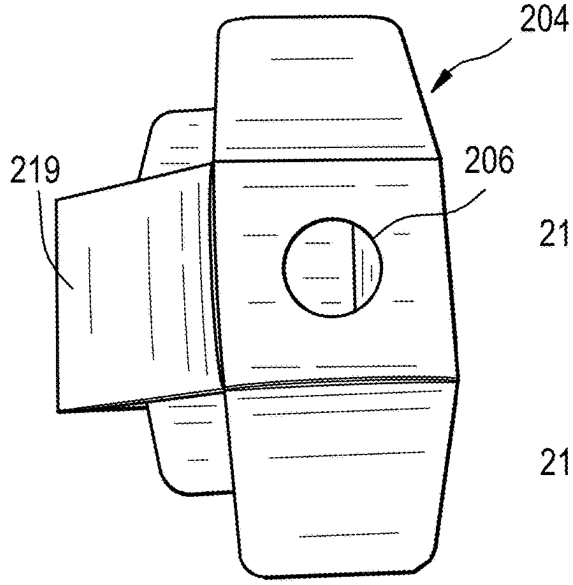


FIG.3A

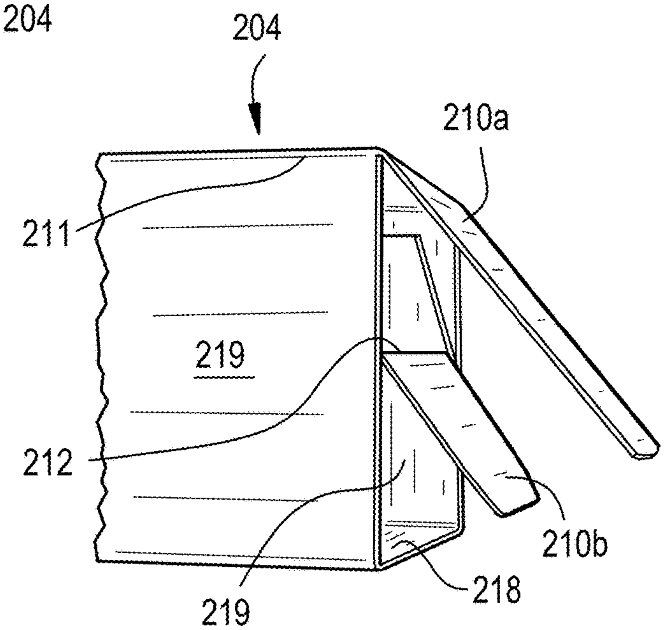


FIG. 3B

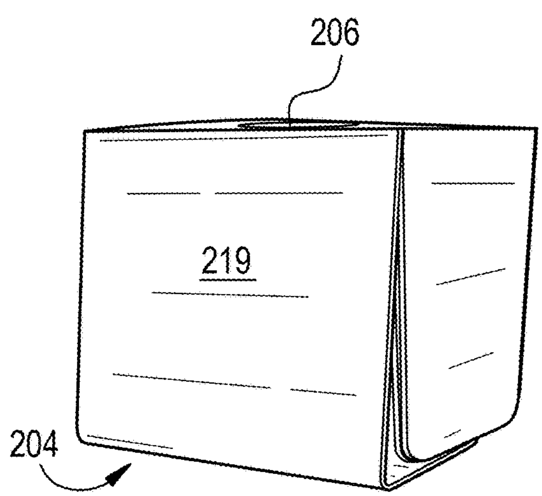


FIG. 3C

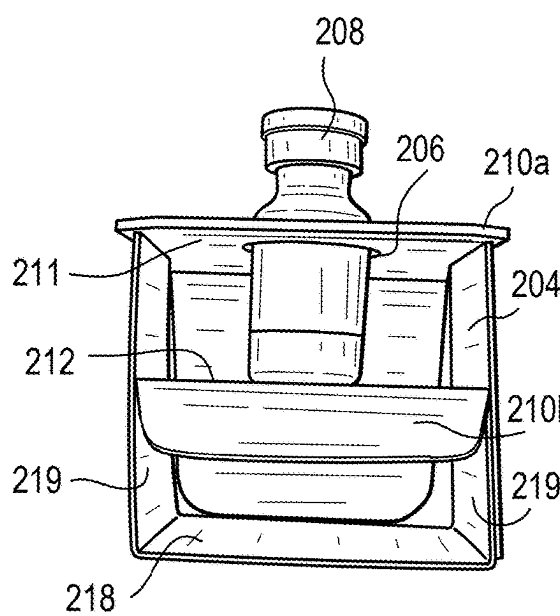


FIG. 3D

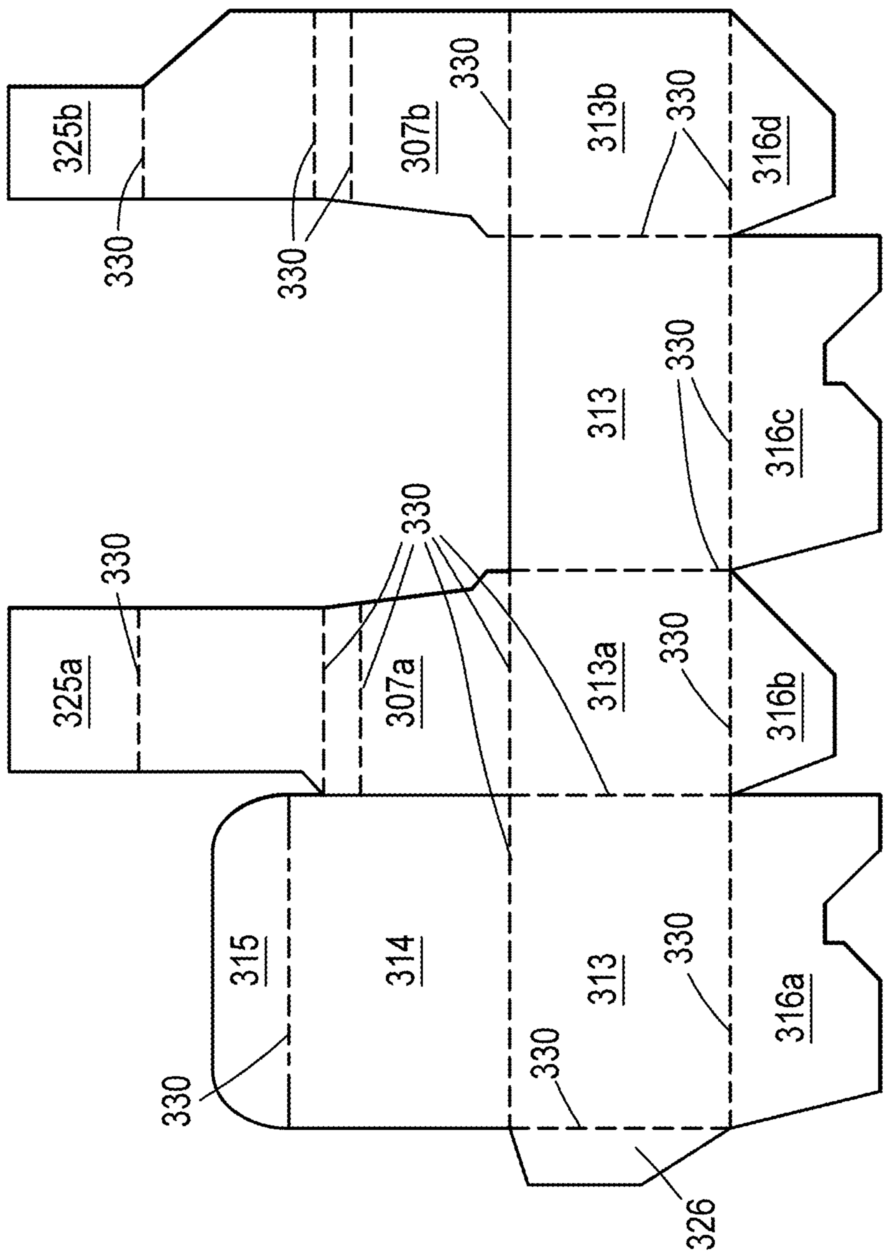

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FIG. 4A

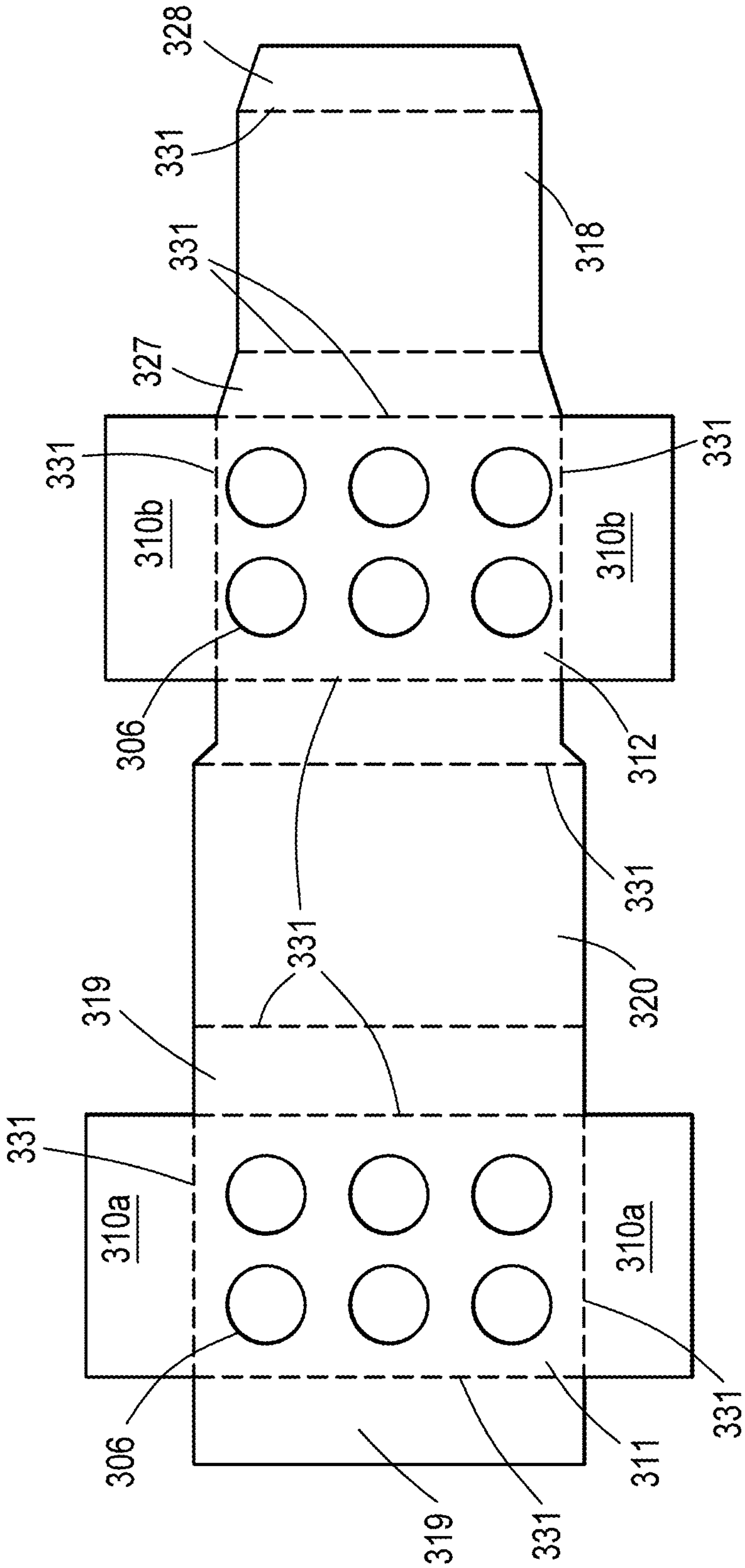
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FIG. 4B

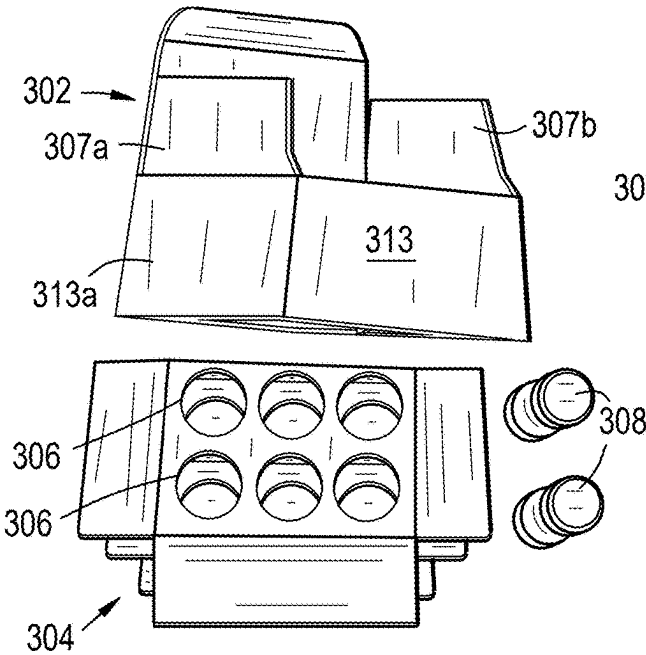

FIG. 5A

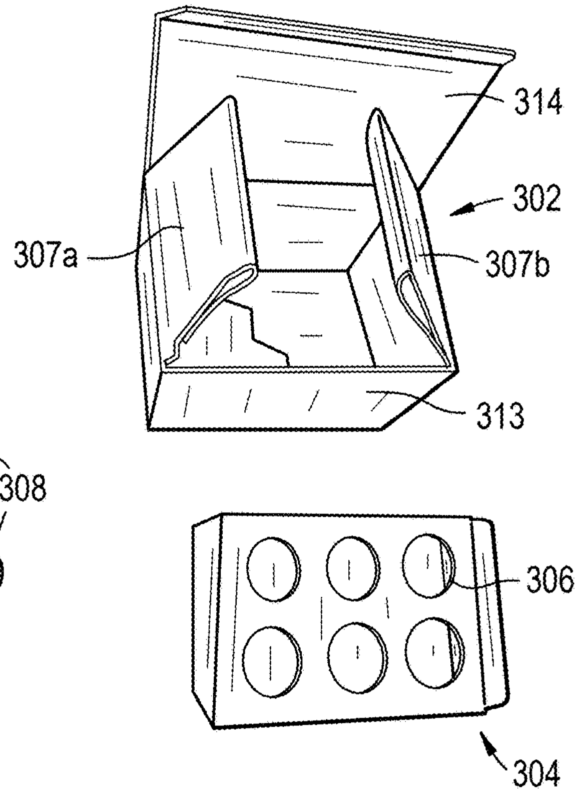

FIG. 5B

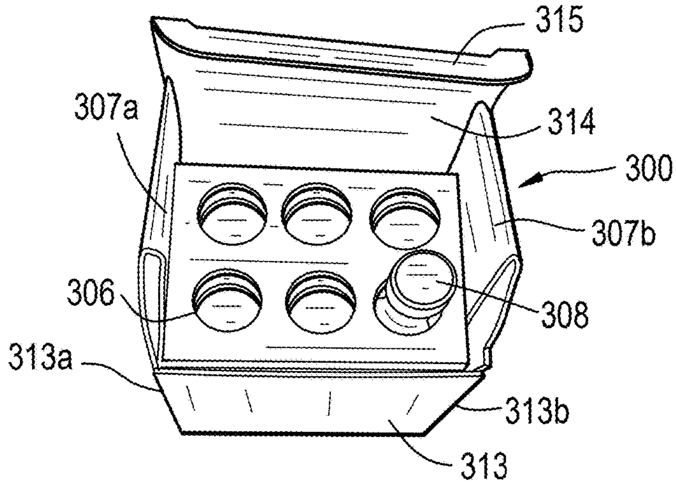

FIG. 5C

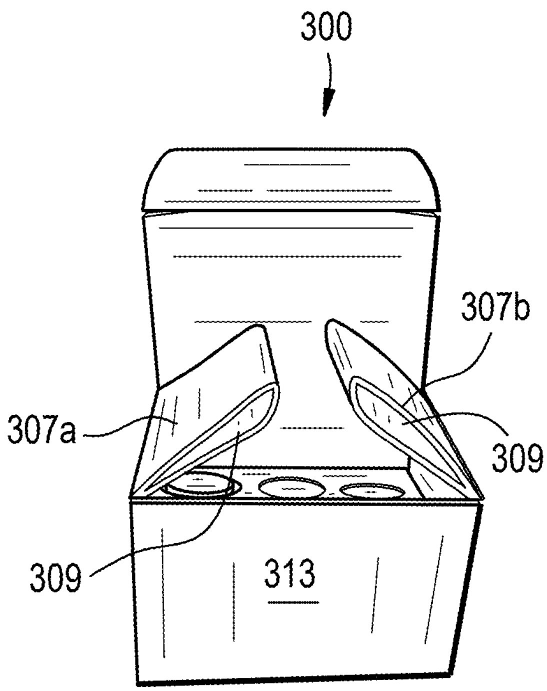

FIG. 5D

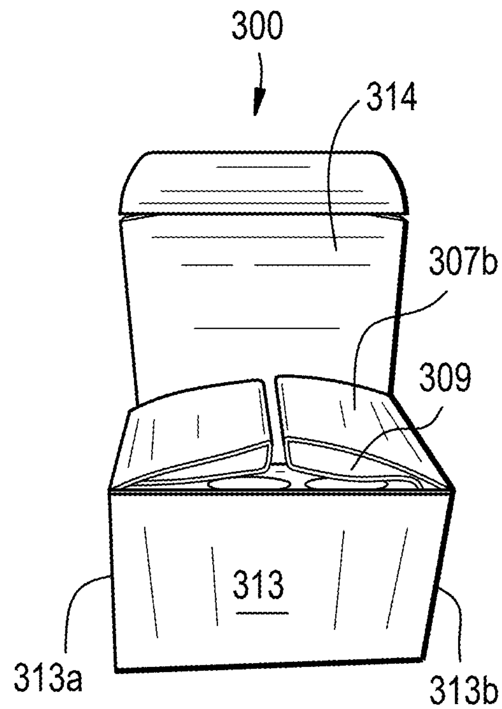

FIG. 5E

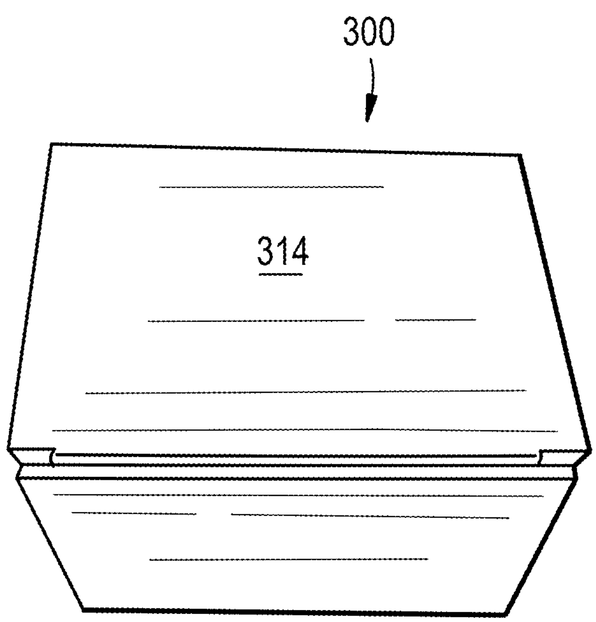

FIG. 5F

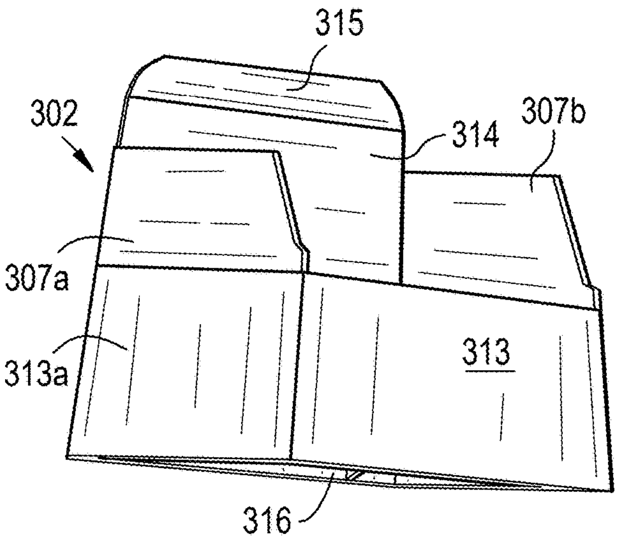

FIG. 6A

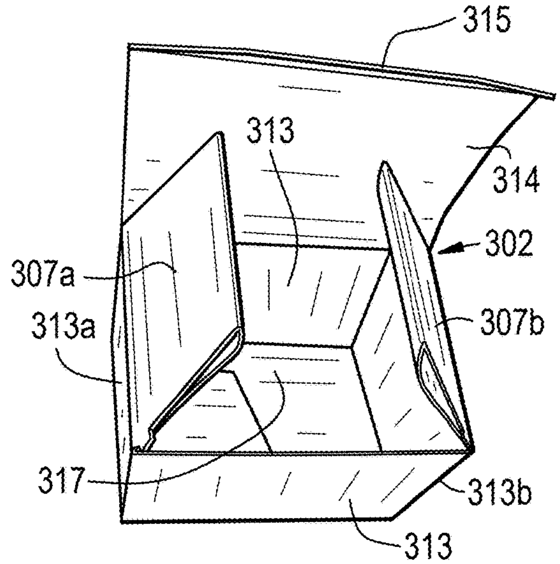

FIG. 6B

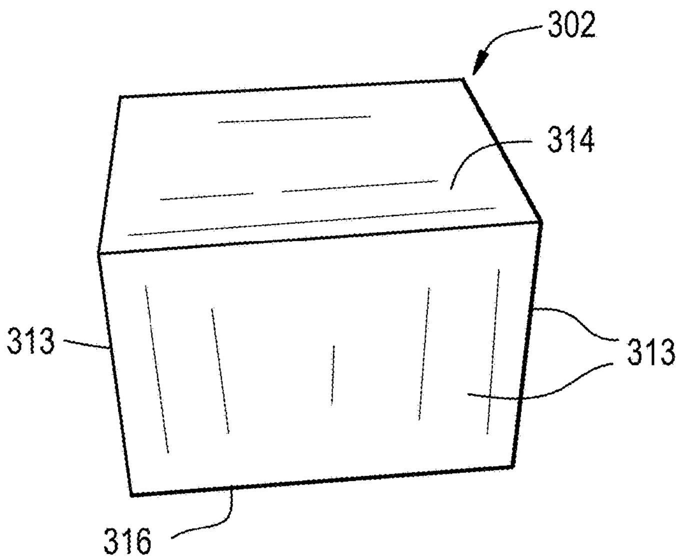

FIG. 6C

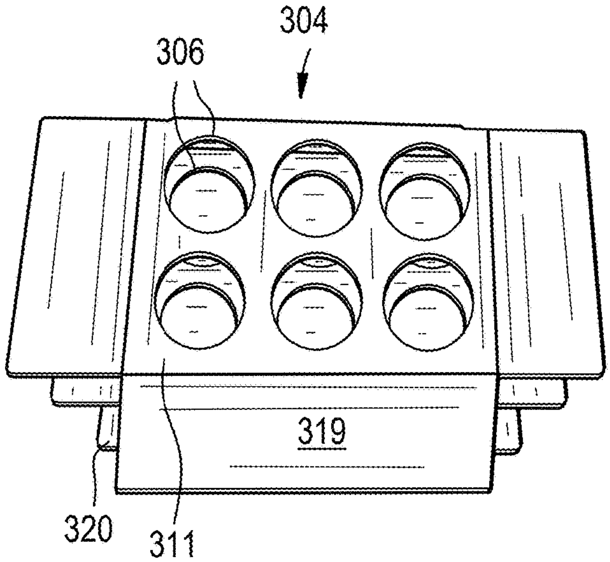

FIG. 7A

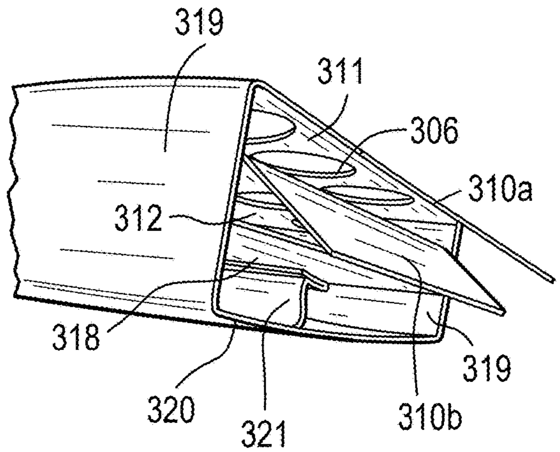

FIG. 7B

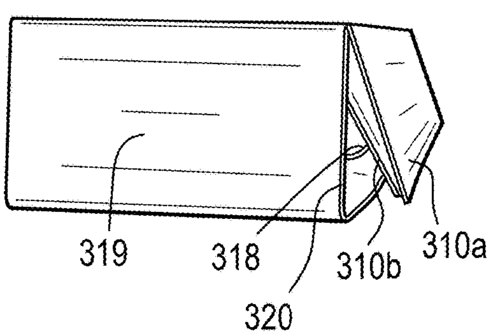

FIG. 7C

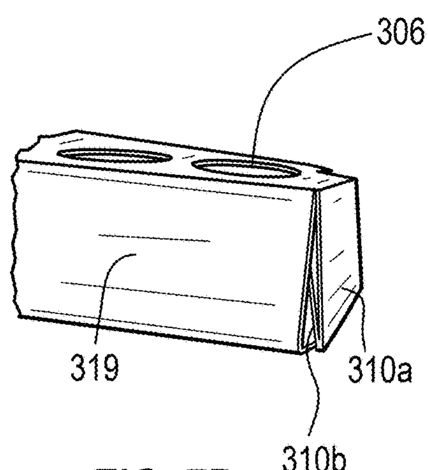

FIG. 7D

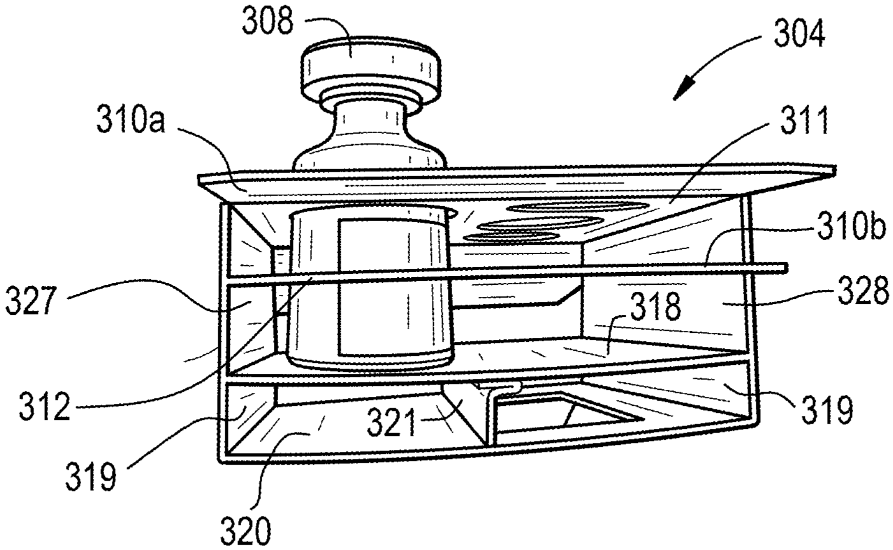

FIG. 8

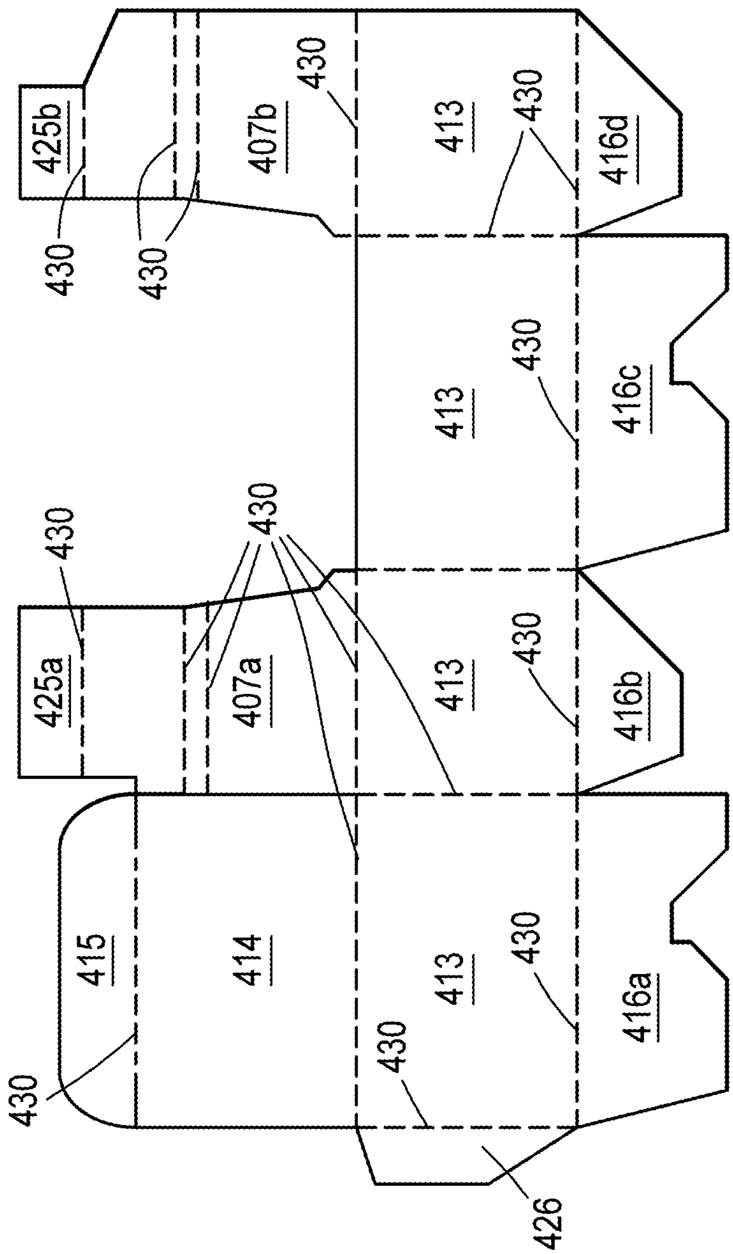
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FIG. 9A

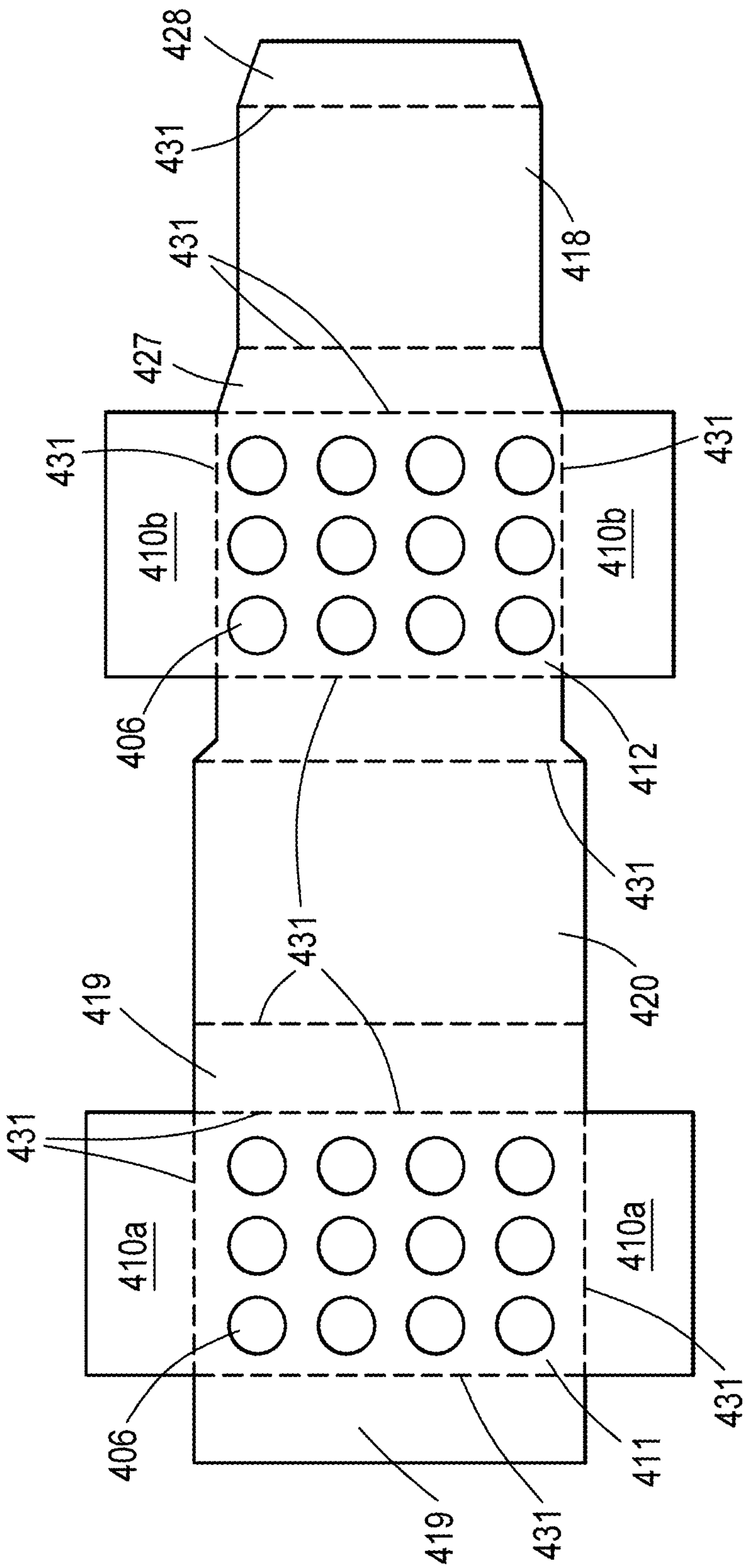

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FIG. 9B

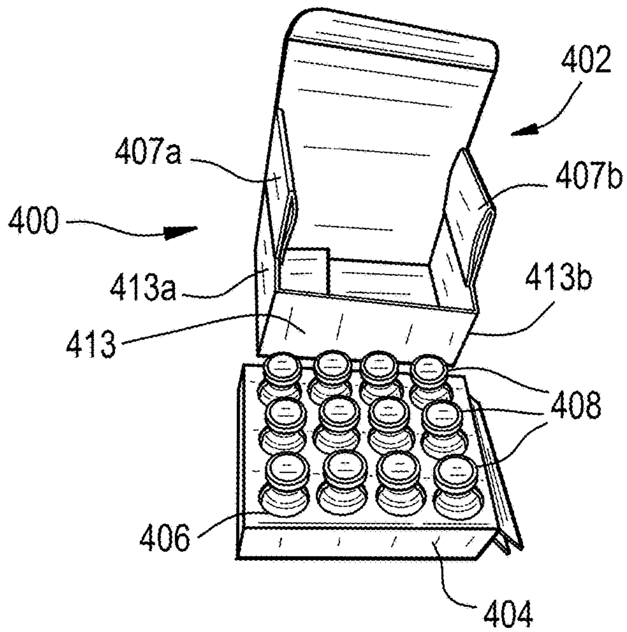

FIG. 10A

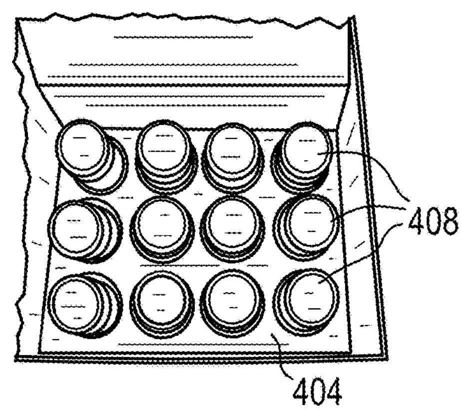

FIG. 10B

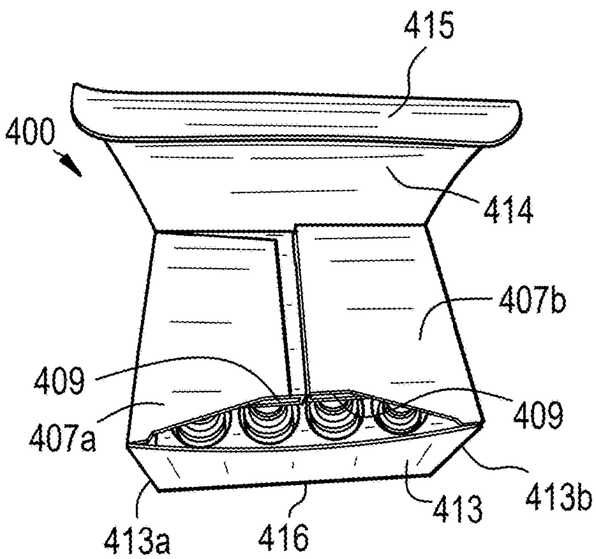

FIG. 10C

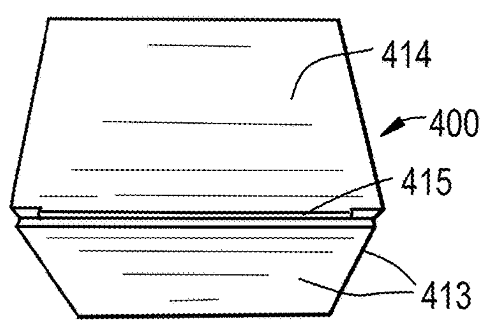

FIG. 10D

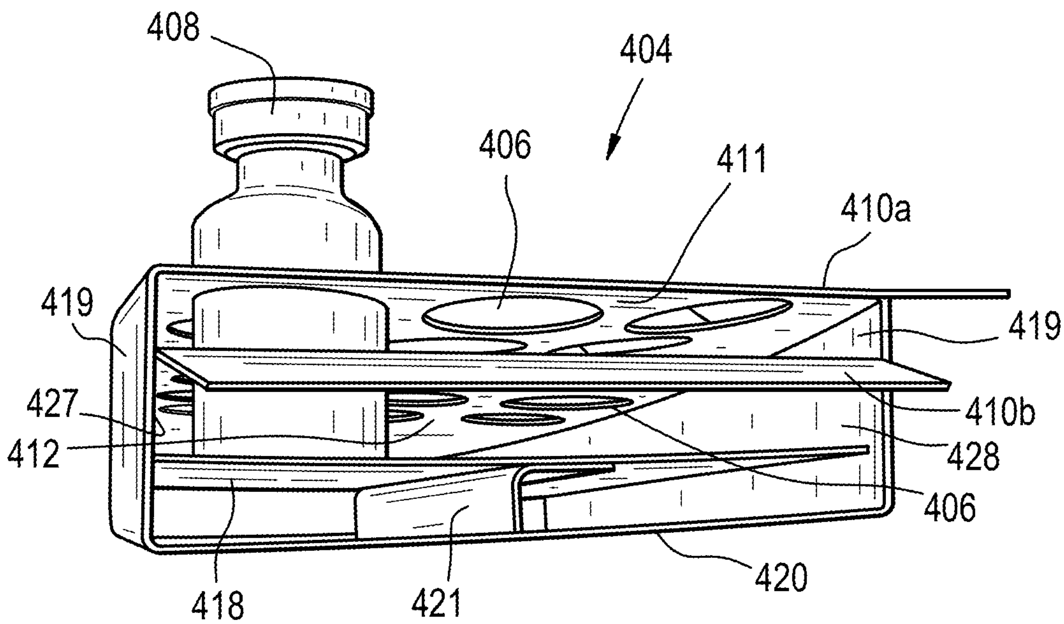

FIG. 11

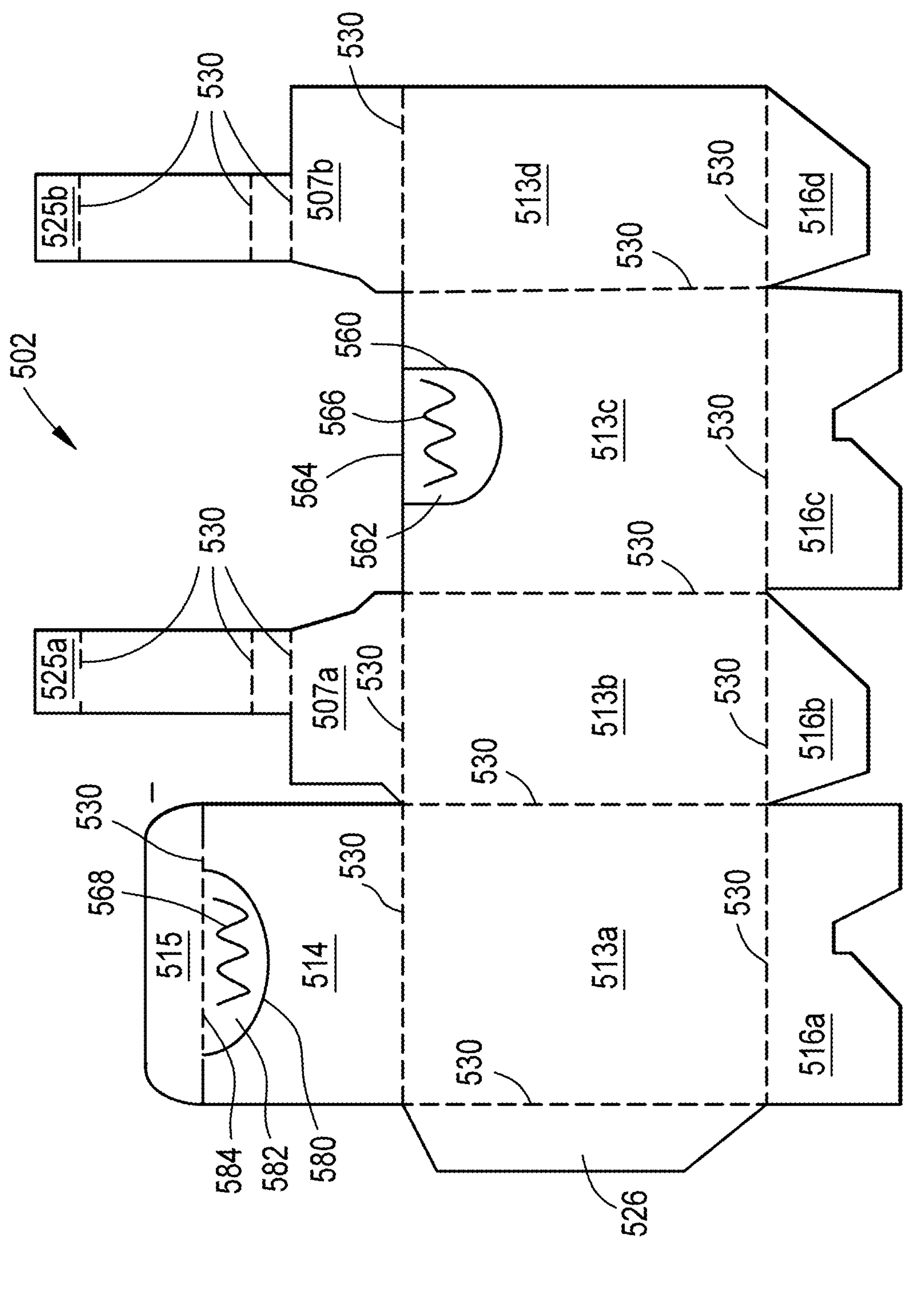
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FIG. 12A

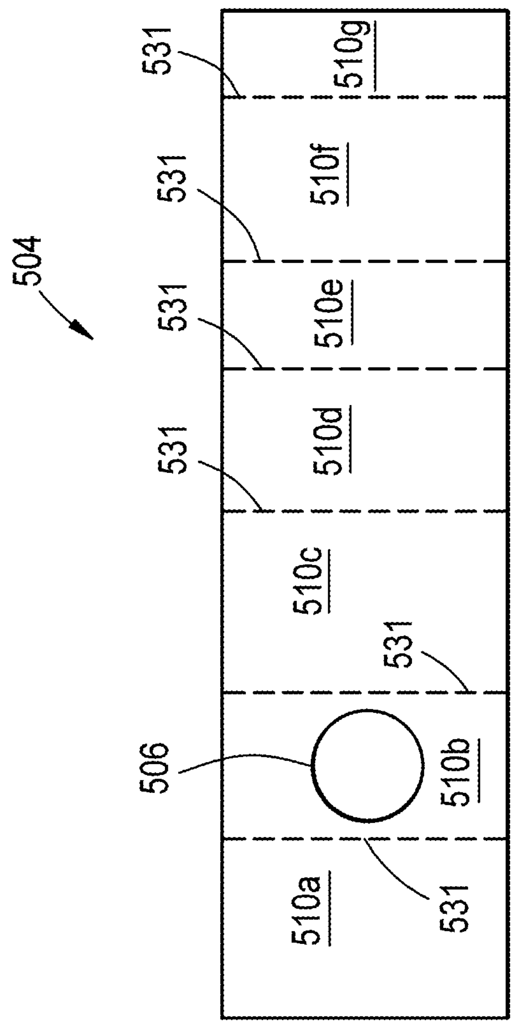

FIG. 12B

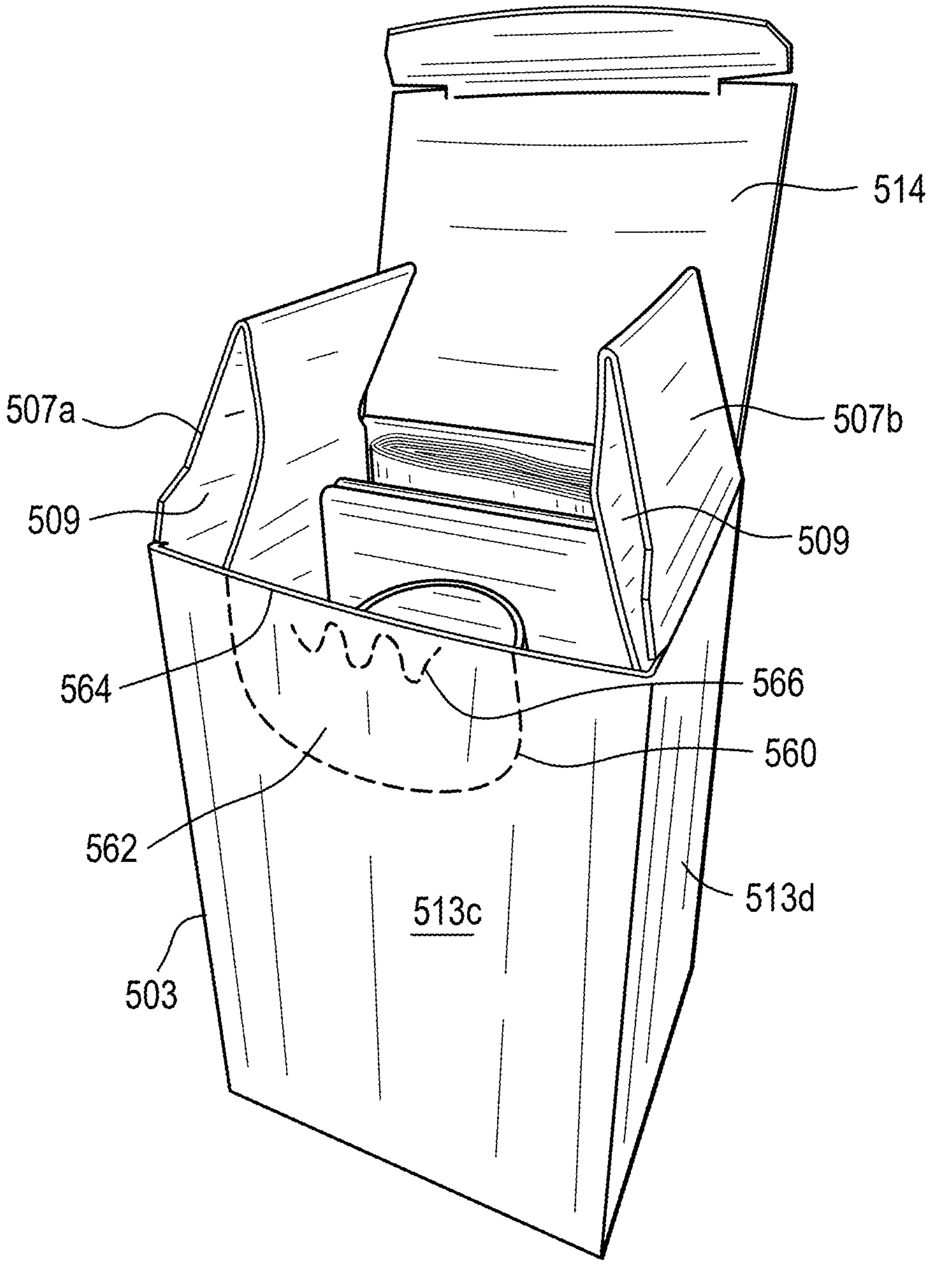

FIG. 13A

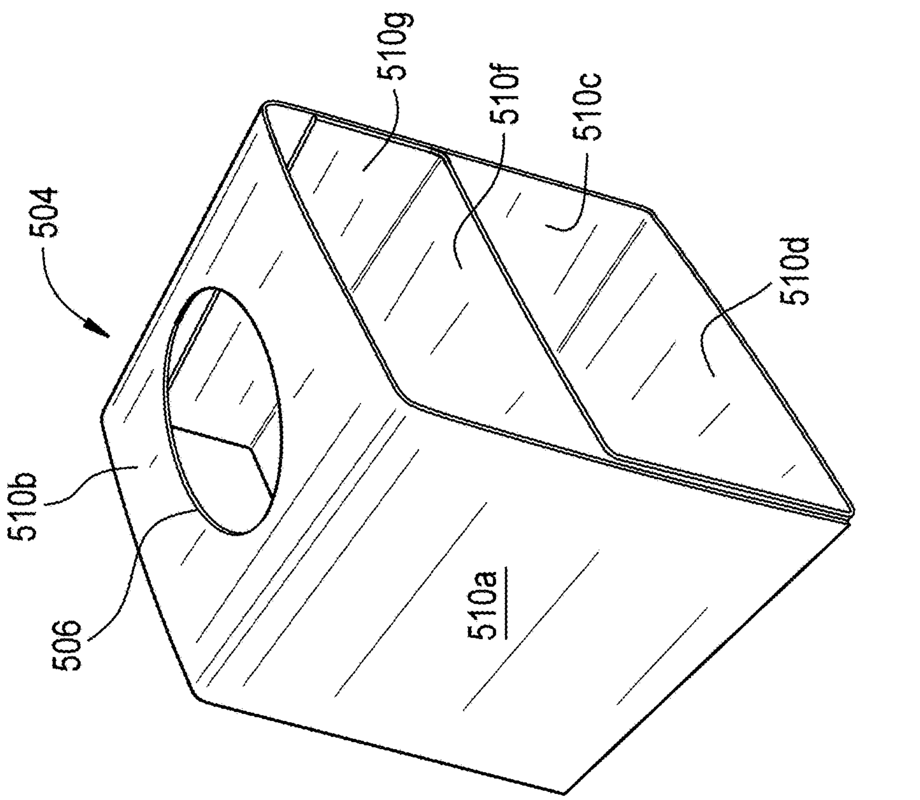

FIG. 13C

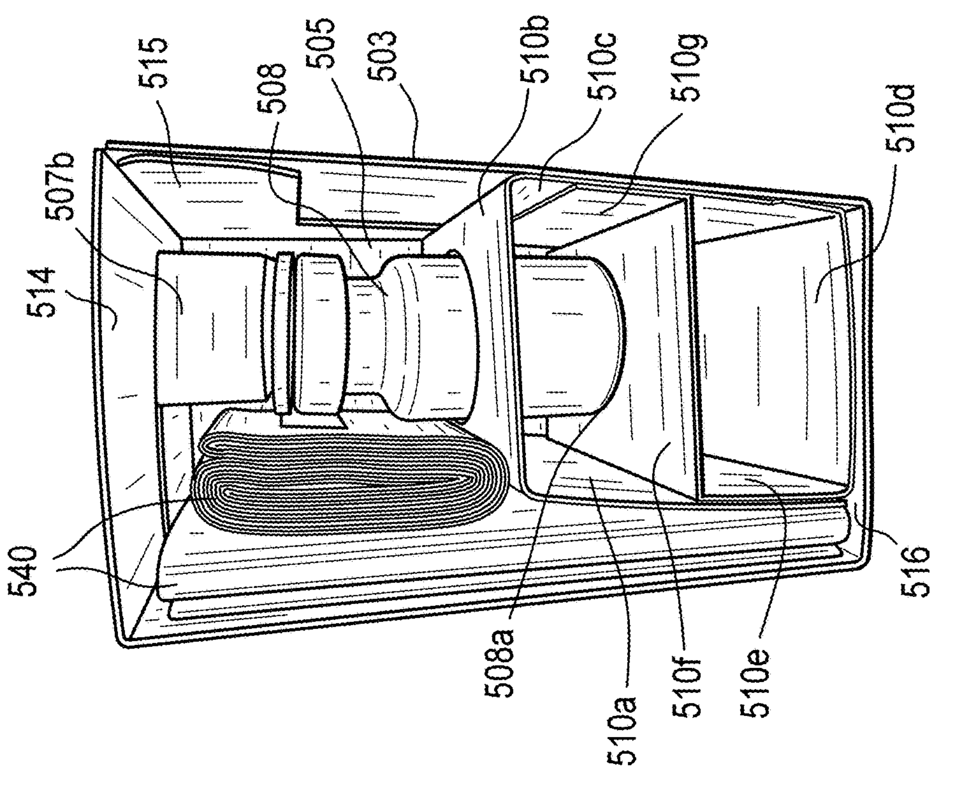

FIG. 13B

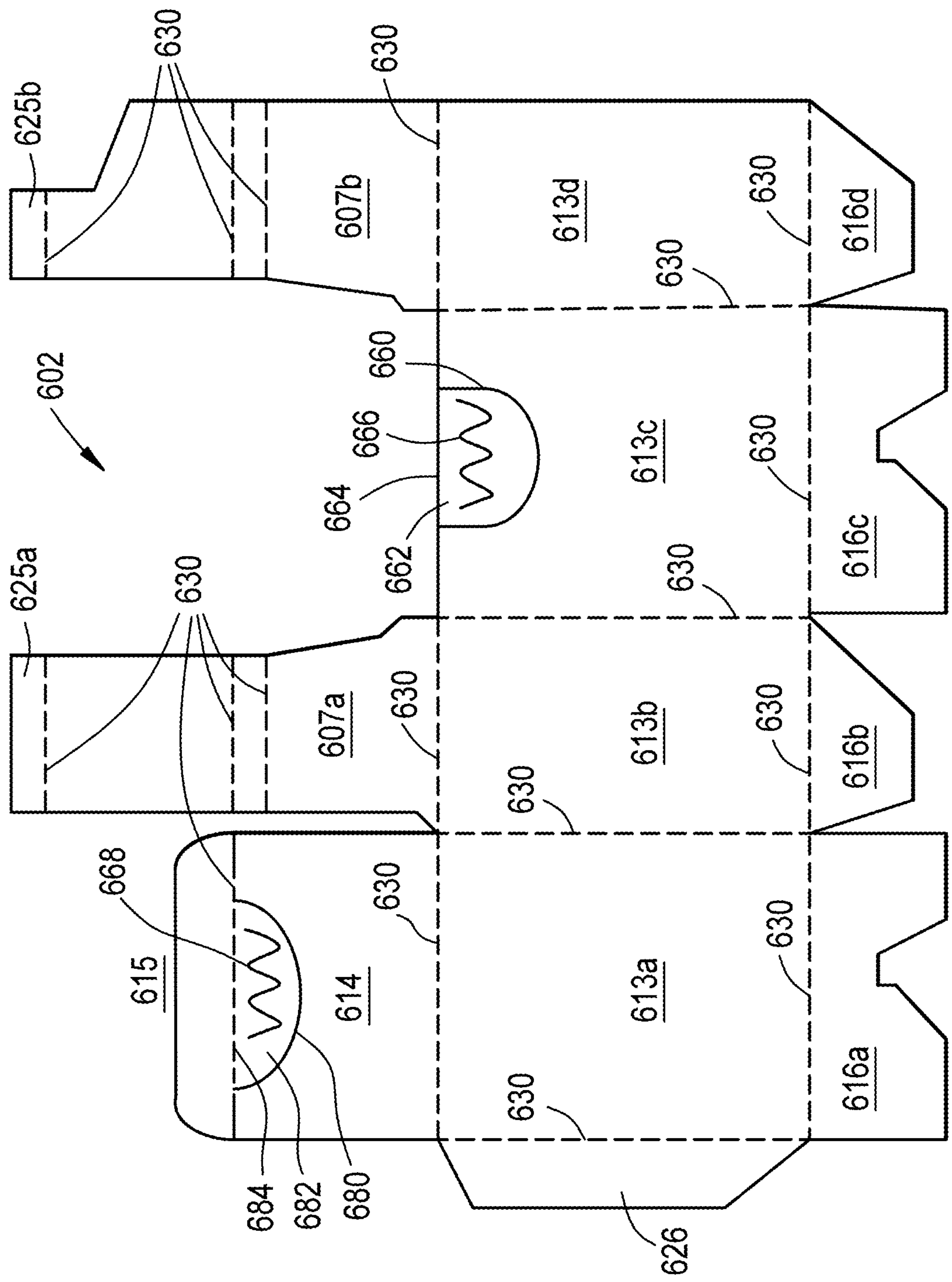
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FIG. 14A

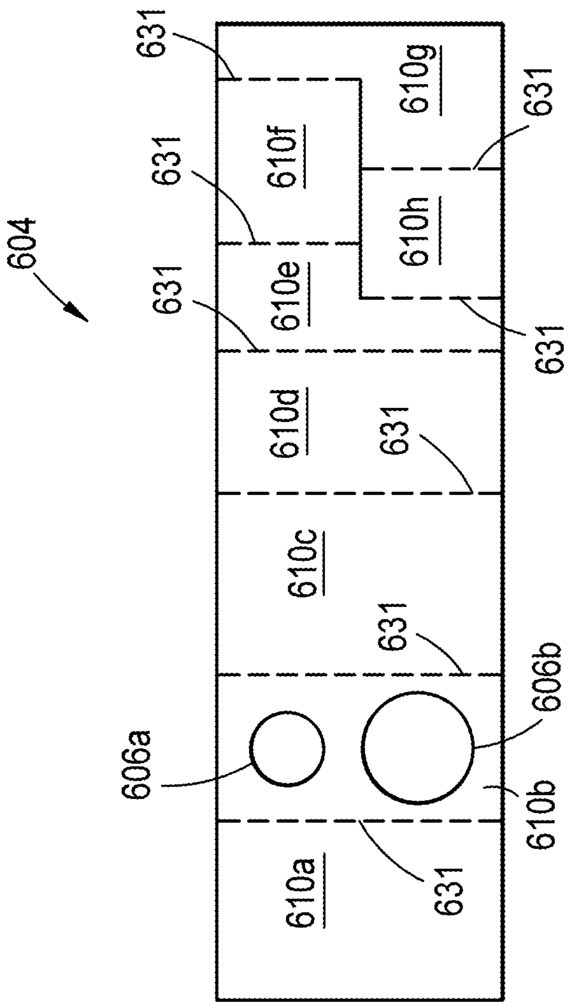
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FIG. 14B

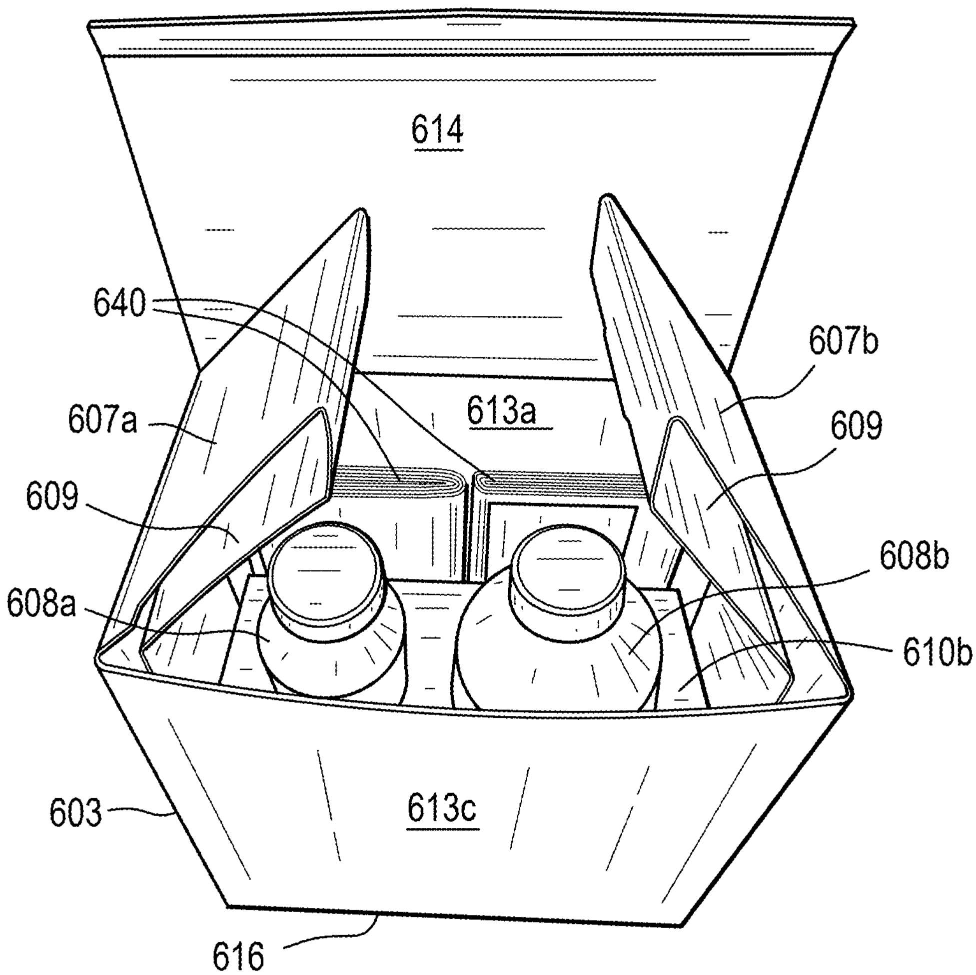

FIG. 15A

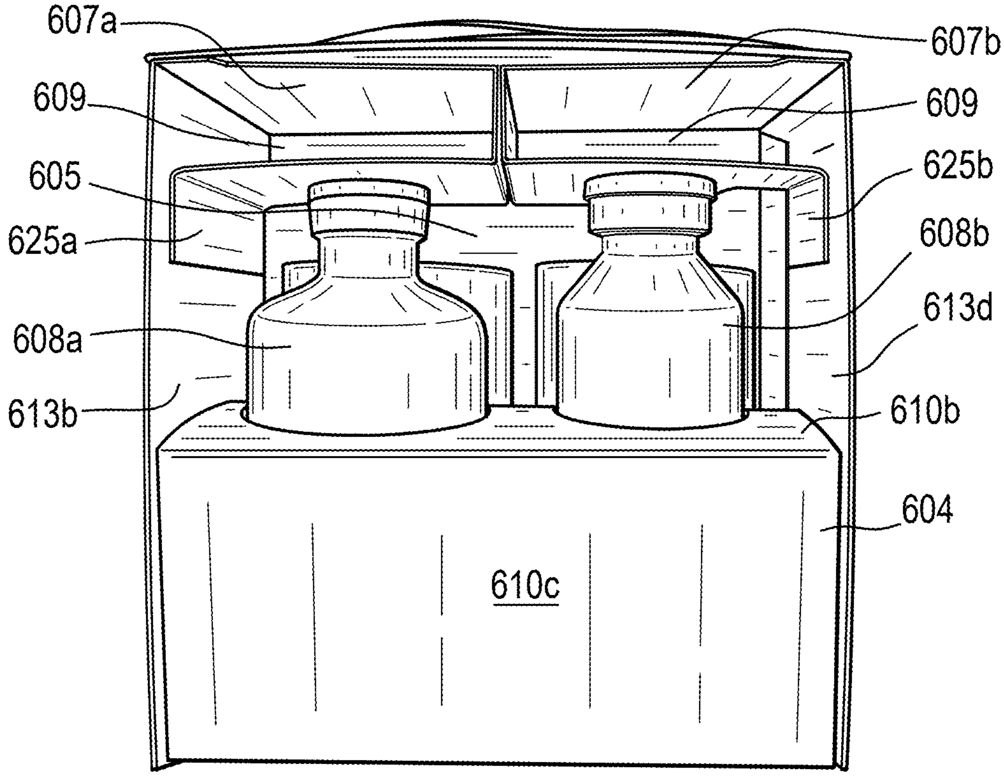
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FIG. 15B

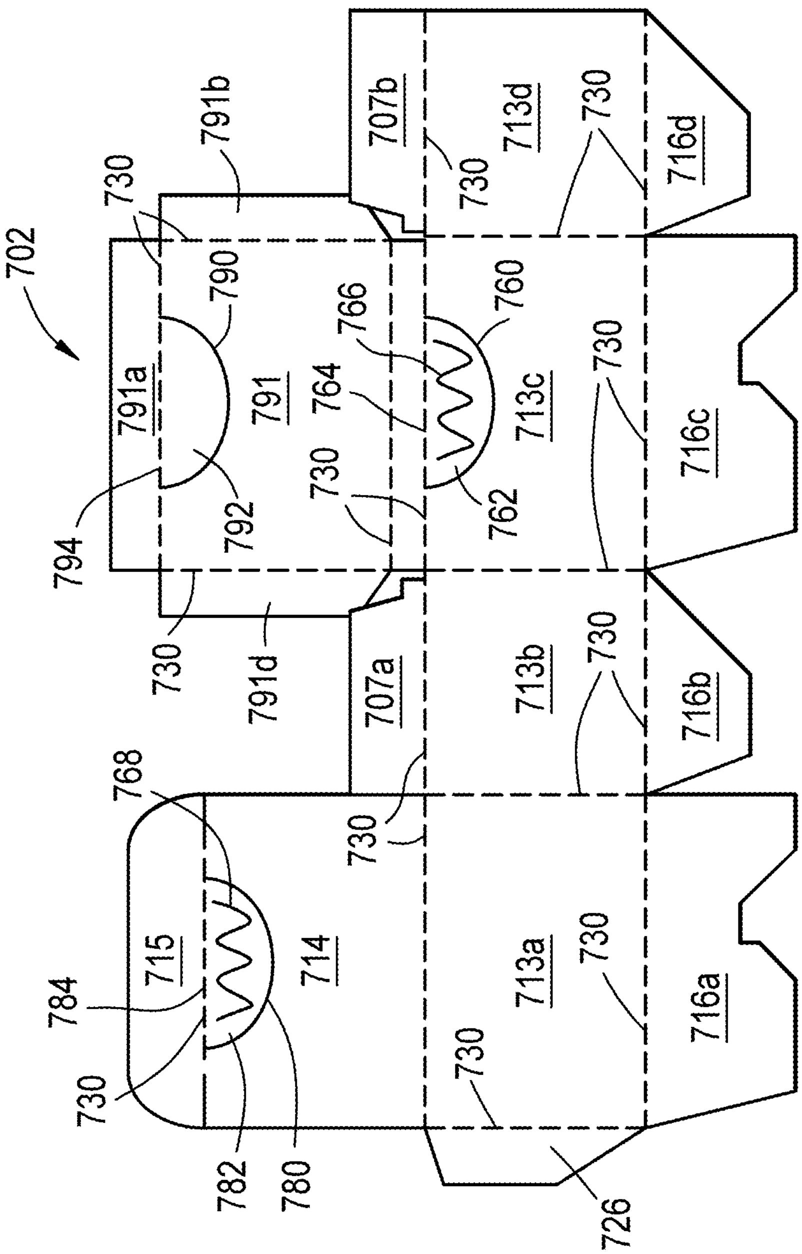
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FIG. 16A

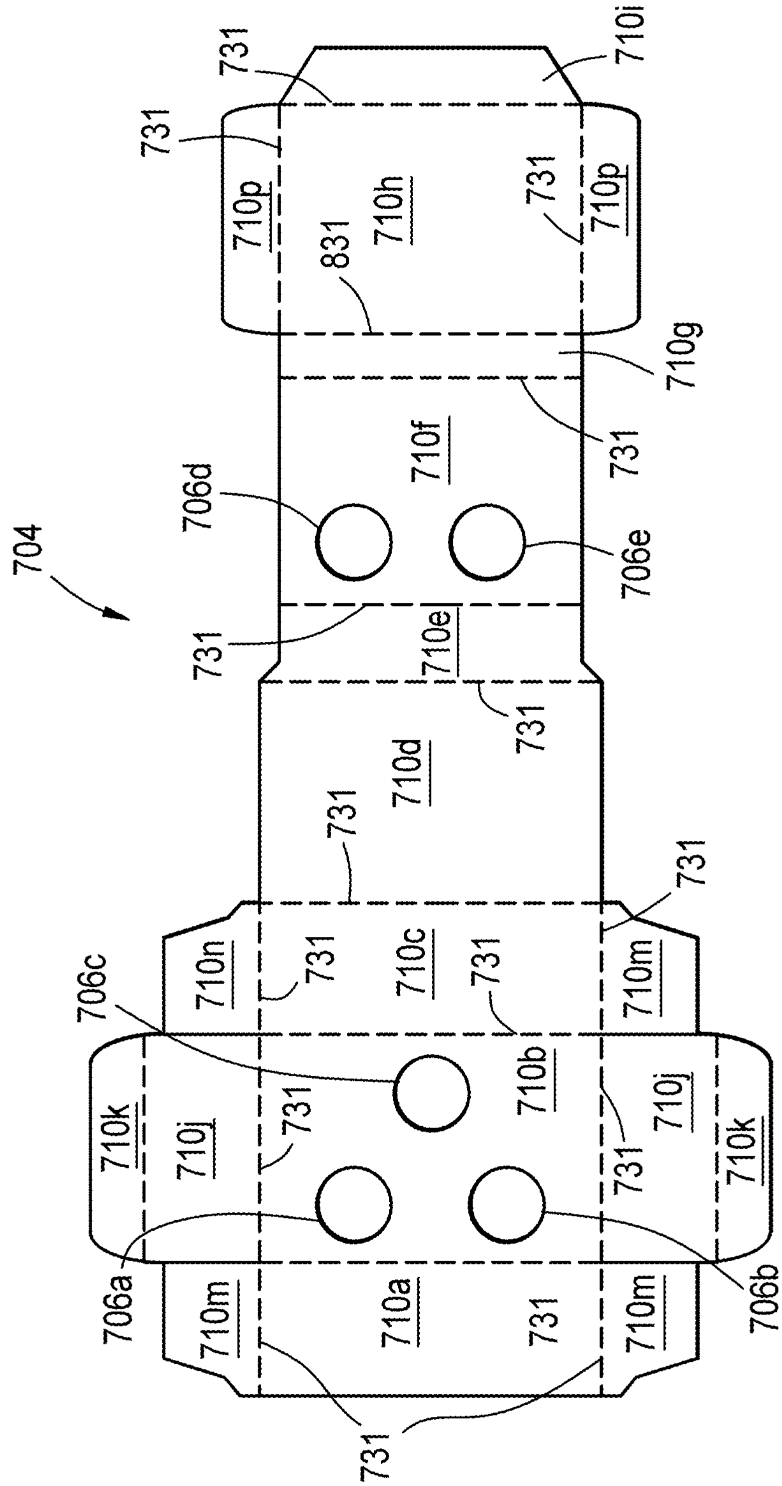
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FIG. 16B

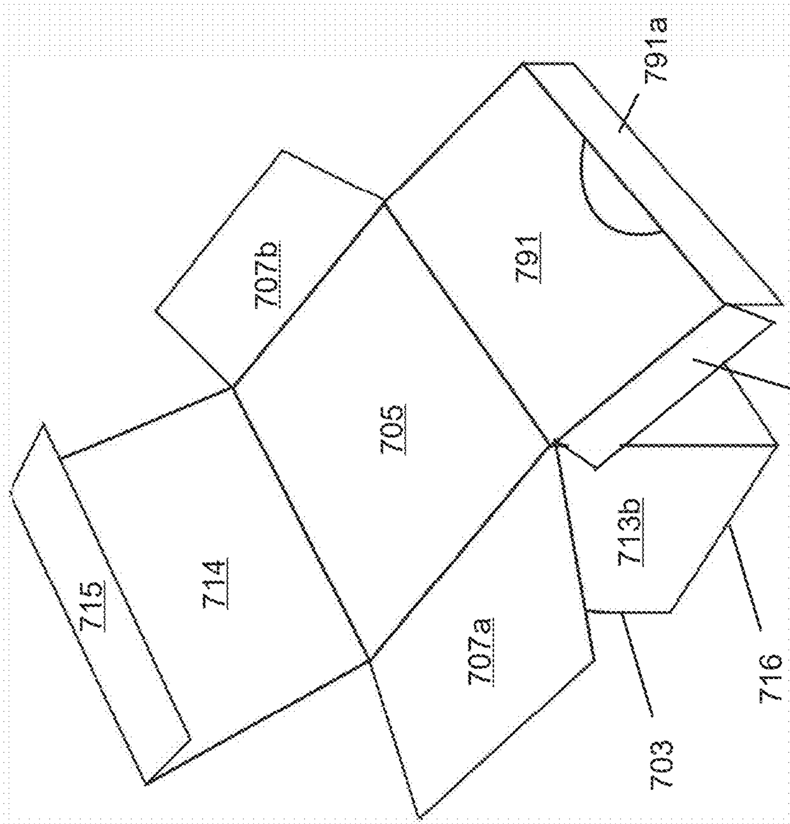


FIG. 17A

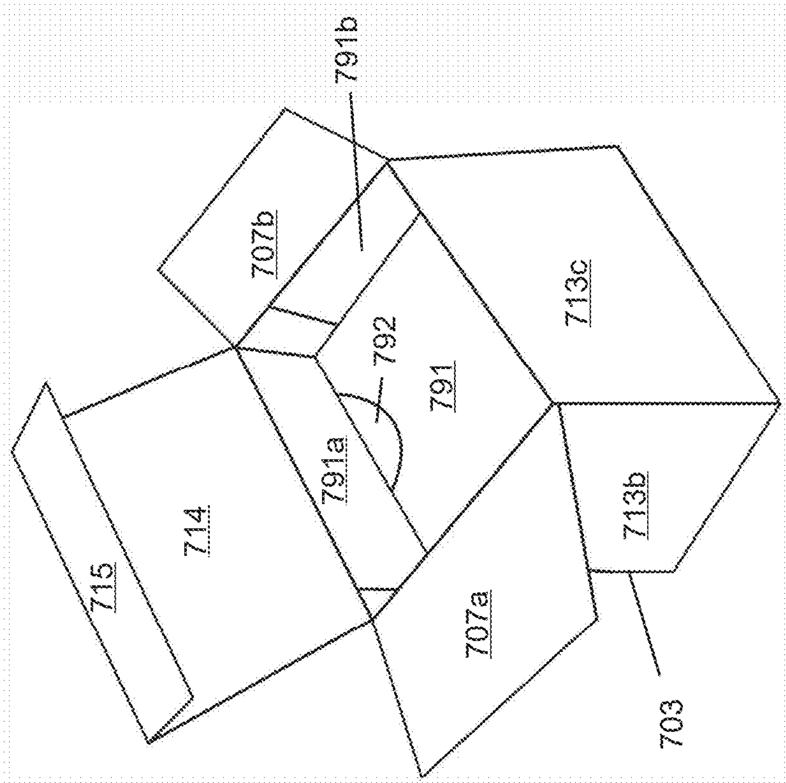


FIG. 17B

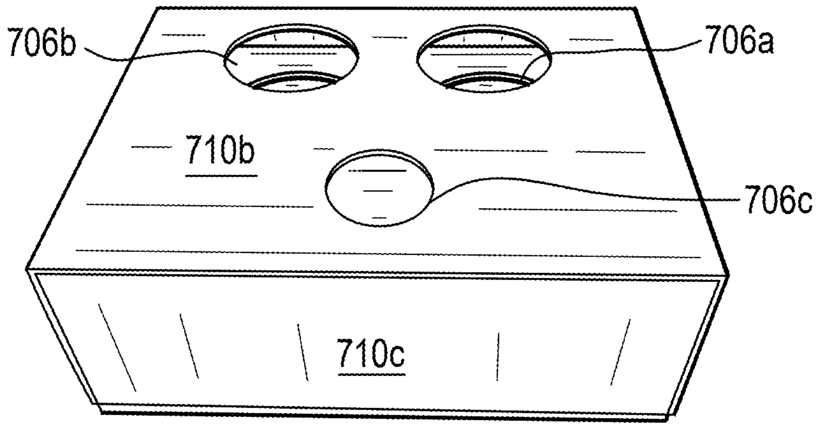


FIG. 18A

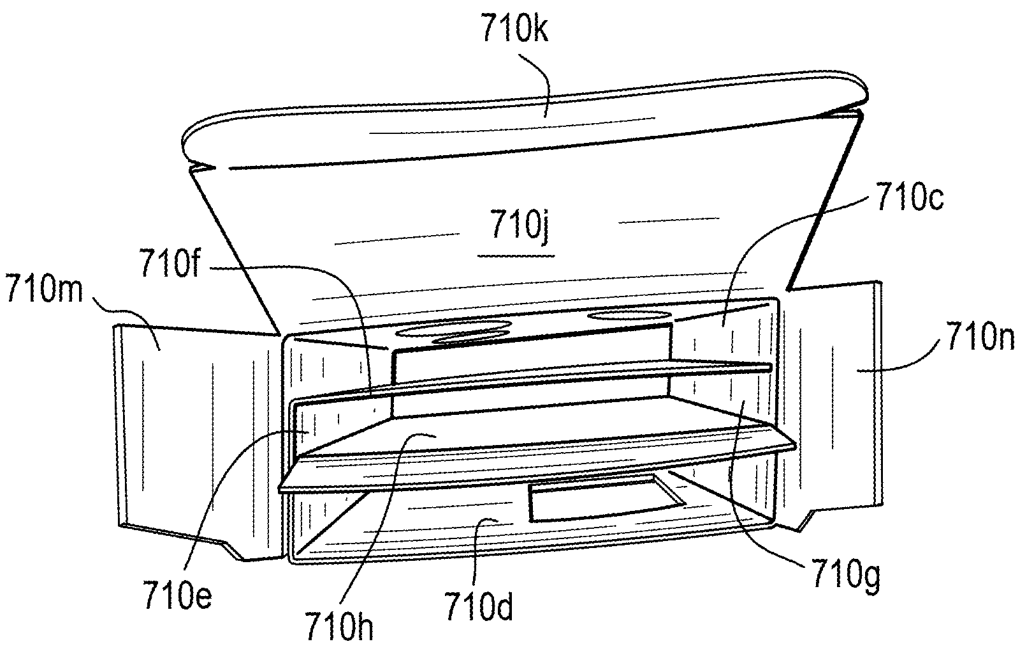


FIG. 18B

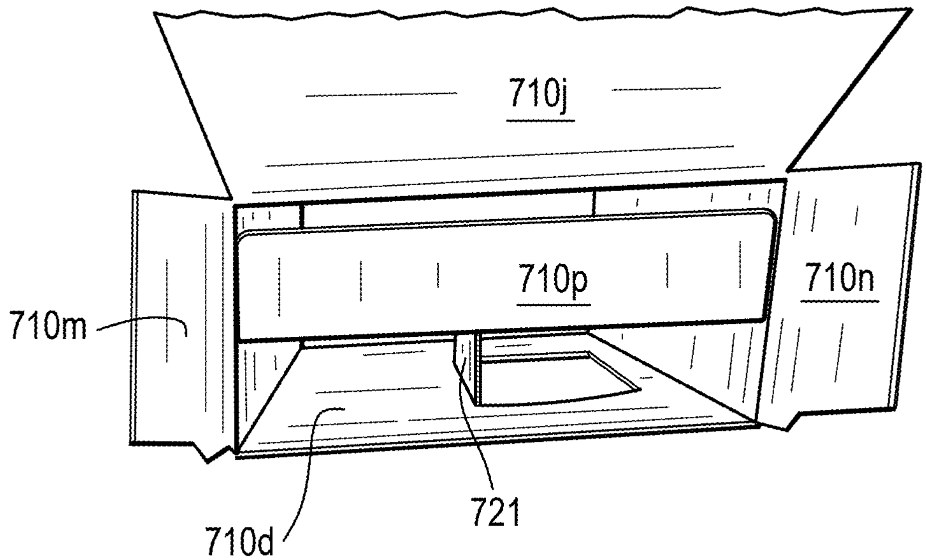


FIG. 18C

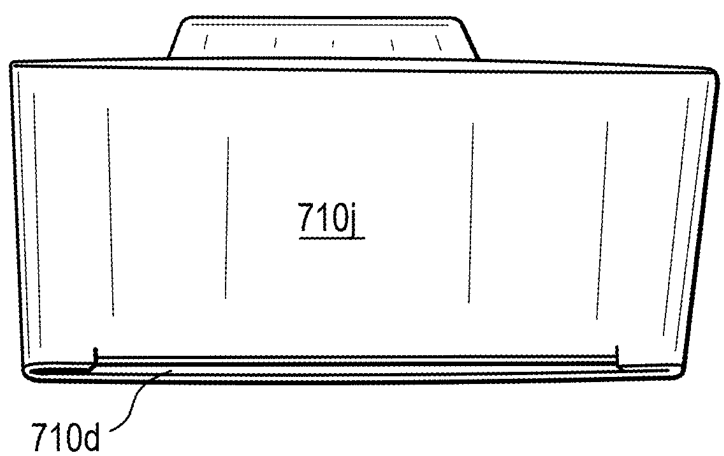
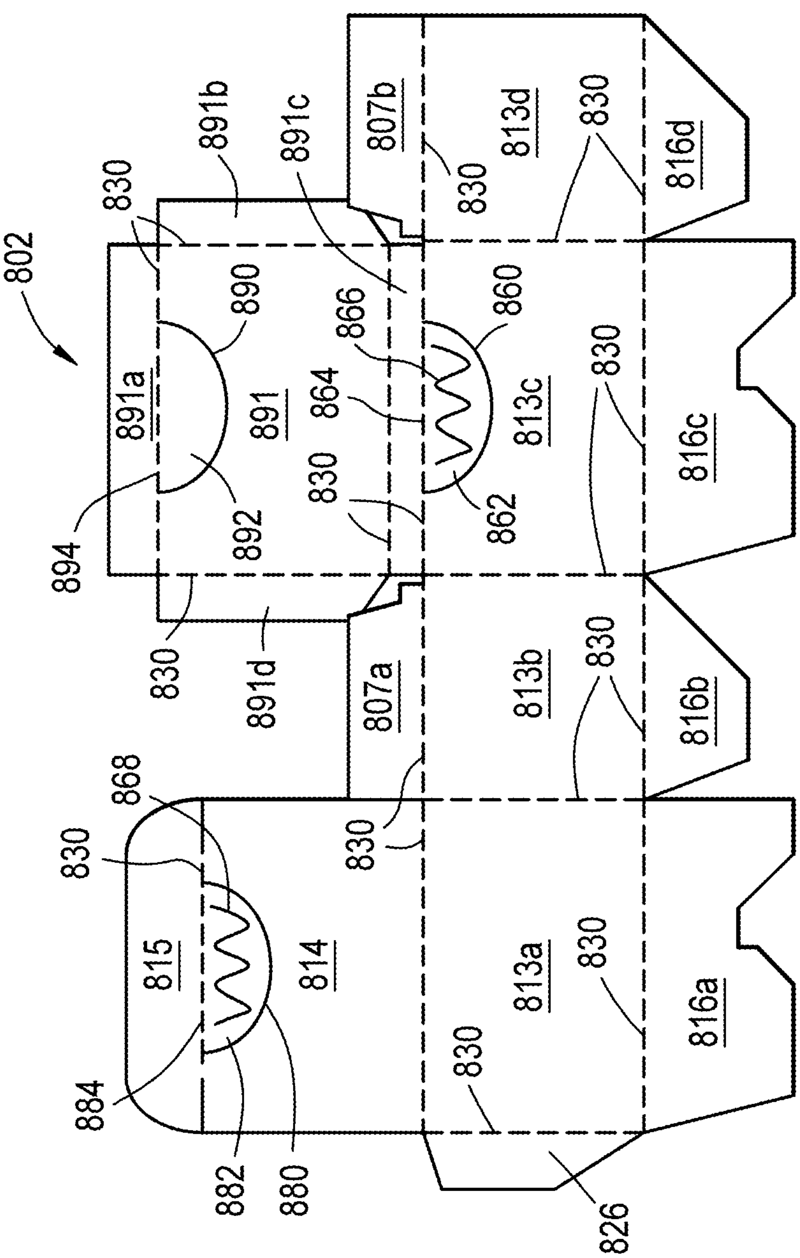


FIG. 18D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826

FIG. 19A

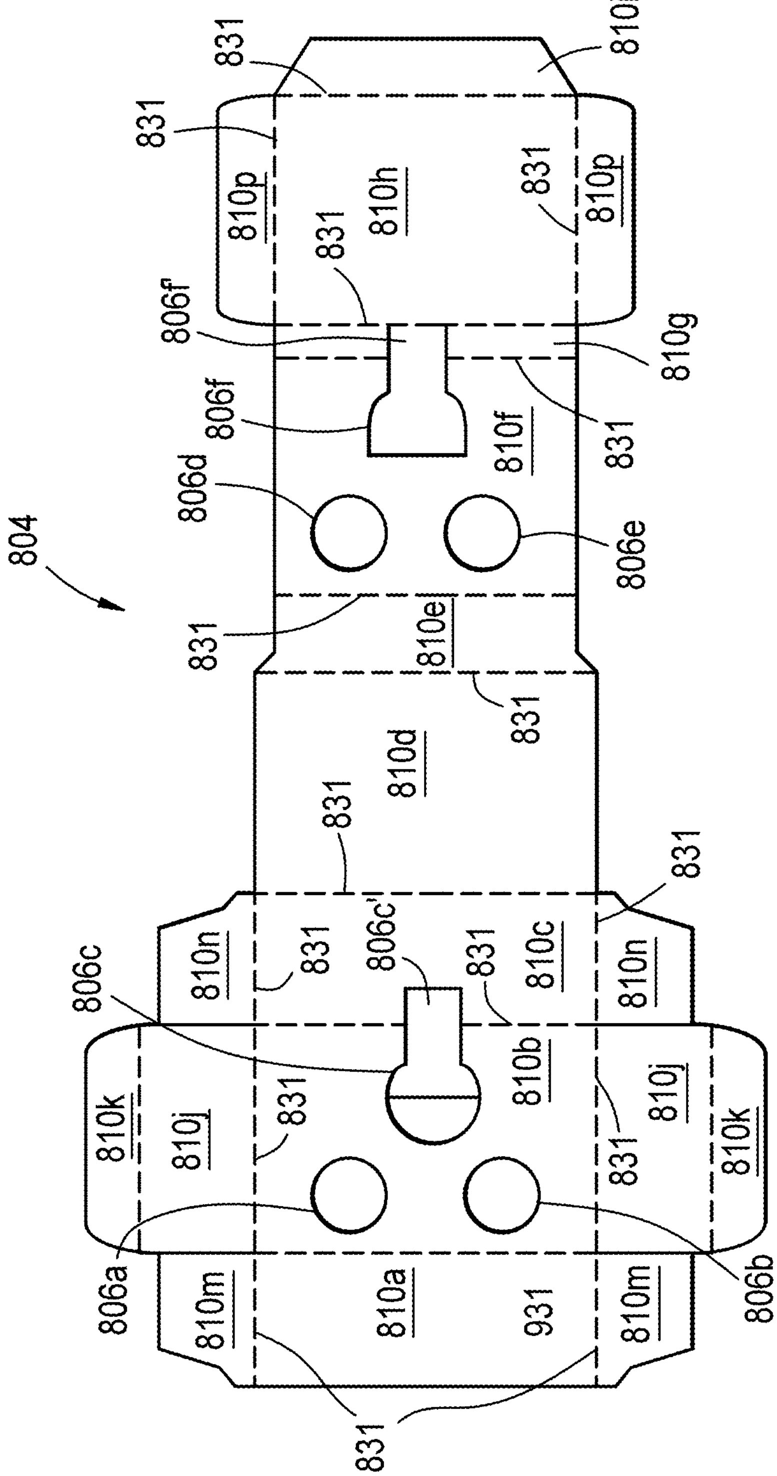
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FIG. 19B

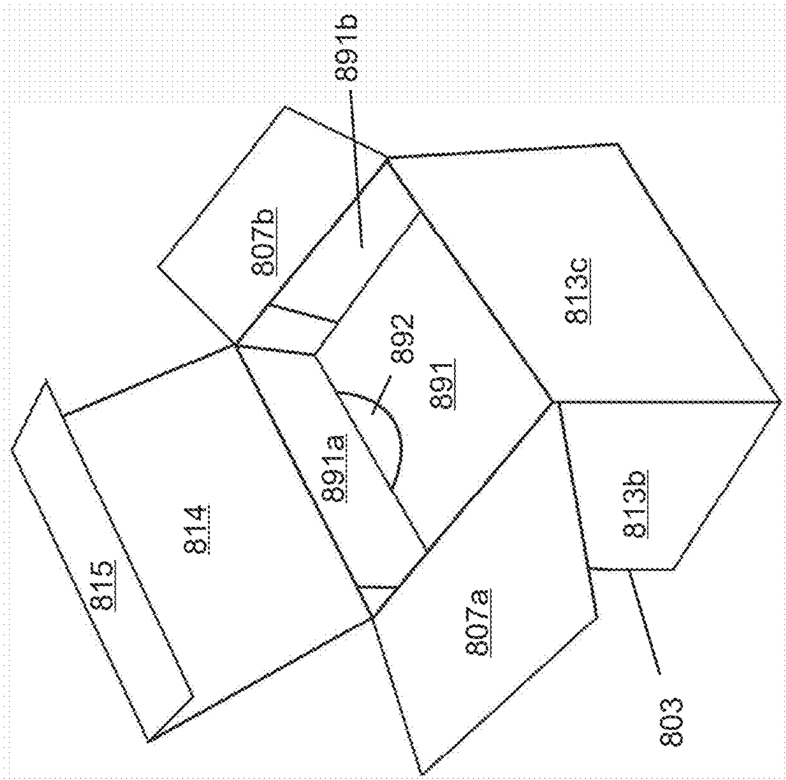

FIG. 20B

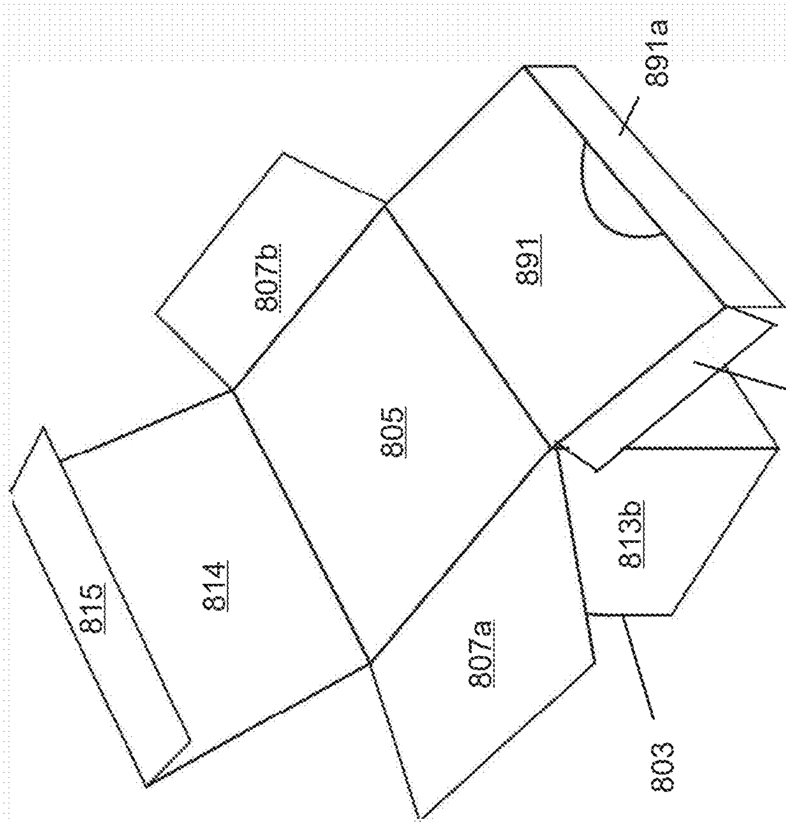

FIG. 20A

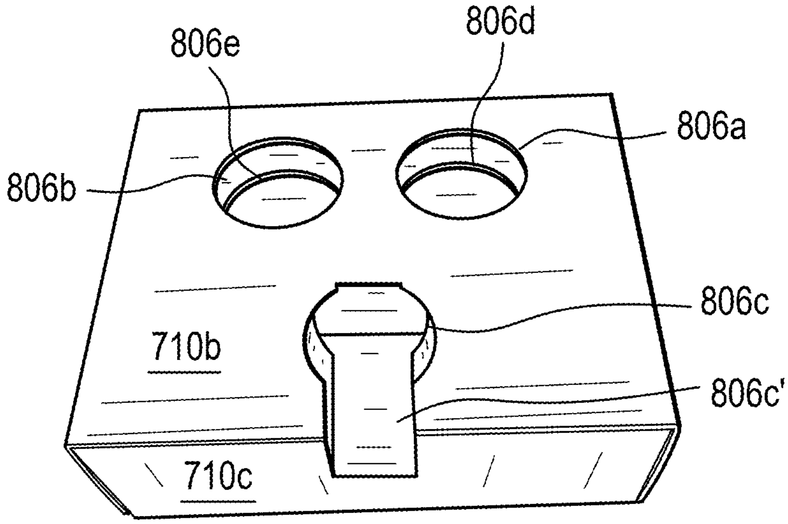

FIG. 21A

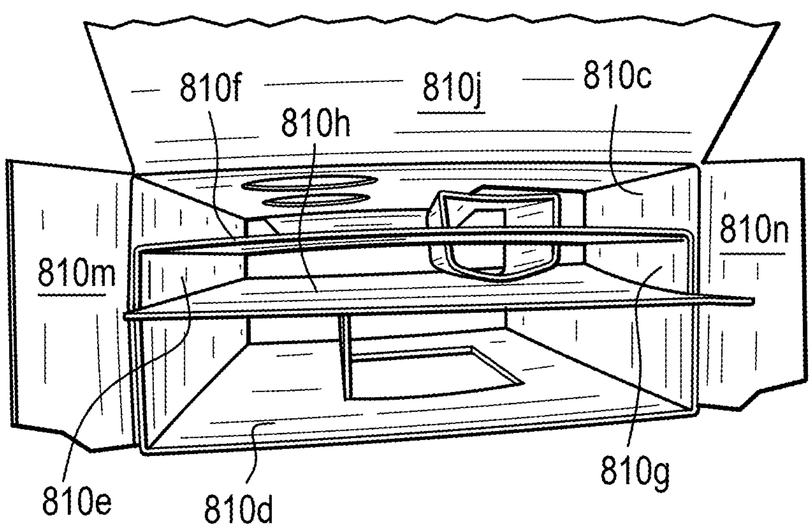

FIG. 21B

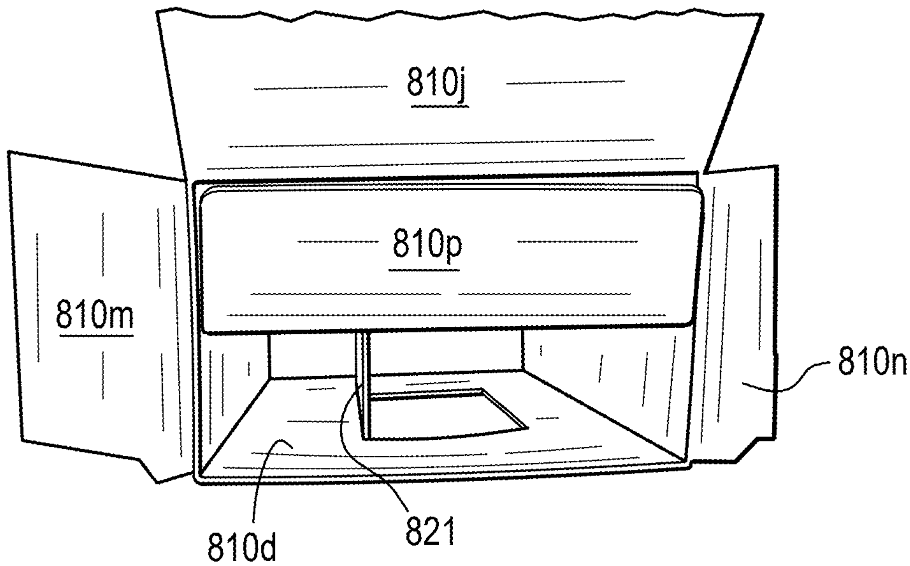

FIG. 21C

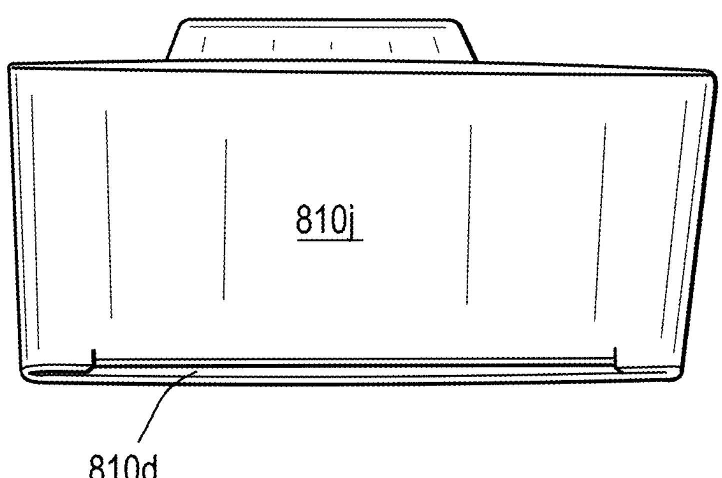

FIG. 21D

SUSTAINABLE AND MODULAR PACKAGING SYSTEMS FOR CONTAINERS FOR PHARMACEUTICAL USE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Priority is claimed to U.S. Provisional Patent Application No. 63/559,377, filed Feb. 29, 2024, the entire contents of which are hereby incorporated by reference herein.

FIELD OF DISCLOSURE

[0002] The present disclosure generally relates to packaging systems for containers for clinical, medical, or pharmaceutical use, and, more particularly, sustainable and modular systems for packaging containers, such as glass vials, for clinical, medical, or pharmaceutical use.

BACKGROUND

[0003] Containers, such as glass vials, are commonly used in the medical and pharmaceutical field to hold various quantities of medicament. These containers are frequently transported from one site to another, particularly when they are manufactured in one site, filled in another site, and used in yet another site. When transporting the containers from one site to another, they are generally grouped together in packaging systems, such as trays and/or boxes, either manually or by automated means. By way of example, sterilized glass vials containing medicament therein are placed in packaging boxes, which are then sealed. After being transported to their destination, the packaging boxes are opened so that each container can be removed and used thereafter.

[0004] Trays and/or boxes for packaging and transporting containers of medicament, such as glass vials, exist in the industry. However, conventional packaging systems do not provide sufficient protection from lateral impact between each of the containers to prevent damage or breakage of the containers that are commonly made of fragile material, such as glass. Moreover, conventional packaging systems do not protect the containers from top load impact, particularly when the packages are stacked on top of each other, or bottom impact, particularly when the packages are placed on hard surface. Thus, containers transported in these conventional packaging systems are prone to damage or breakage.

[0005] In addition, International Organization of Standardization (ISO) determines standard dimensions and capacities for injectable containers, such as glass vials. By way of example, ISO 8362-1 indicates the standard capacities of injectable vials made of glass. According to ISO 8362-1, 2R vials have a brimful capacity of 4 cc (4 ml), 6R vials have a brimful capacity of 10 cc (10 ml), and 10R vials have a brimful capacity of 13.5 cc (13.5 ml). ISO 8362-1 also indicates the standard dimensions of glass vials. For example, ISO 8362-1 indicates that 2R glass vials have a body with an outer diameter of 16 mm±0.15 mm and a height of 35 mm±0.5 mm, 6R vials have a body with an outer diameter of 22 mm±0.2 mm and a height of 40 mm±0.5 mm, and 10R vials have a body with an outer diameter of 24 mm±0.2 mm and a height of 45 mm±0.5 mm.

[0006] Because conventional packaging systems are fixed in dimensions and sizes and because ISO 8362-1 establishes the standard dimensions of containers, such as glass vials, conventional packaging systems may only allow one size (e.g., height, outer diameter) of containers to be grouped into respective packages. For example, conventional boxes for packaging glass vials of medicament do not allow ISO 2R and 4cc vials to be grouped together with ISO 6R, ISO 10R, or 10 cc vials in the same box because of the differences in standard dimensions of the vials. Accordingly, conventional packaging systems do not provide the flexibility in use in that it is not suitable for multiple containers of different sizes to be packaged together into one packaging system. This limitation can have a direct consequence on the number of containers that can be packaged into each packaging system. This limitation can also have a direct consequence on the number of packaging systems having to be used to transport a given number of containers and, thus, the cost of transporting the containers from one site to another.

[0007] Furthermore, conventional packaging systems that attempt to provide containers therein with protection from damage generally include foam inserts or plastic trays. The foam inserts and plastic trays prevent containers from contacting each other and, thus, provide protection from lateral impact. However, materials such as foam or plastic are not sustainable and harmful to the environment.

[0008] The present disclosure sets forth systems embodying advantageous alternatives to existing systems for packaging multiple containers for clinical, medical, or pharmaceutical use, and that may address one or more of the challenges or needs mentioned herein, as well as provide other benefits and advantages.

SUMMARY

[0009] One aspect of the present disclosure provides a system for packaging one or more containers. The system may comprise a carton having a storage configuration where the carton is substantially flat, and an assembled configuration. The carton may comprise a bottom box having a first bottom wall, a first sidewall, a second sidewall, a first back wall, and a first front wall when the carton is in the assembled configuration. The carton may also comprise a lid coupled to the bottom box and movable between an open position and a closed position, a first flap extending from the first sidewall of the bottom box, and a second flap extending from the second sidewall of the bottom box. The second sidewall may be opposite the first sidewall and facing the first sidewall, and the first flap and the second flap may define a set of cavities. The system may also comprise an insert configured to be disposed within the bottom box of the carton. The insert may have a storage configuration where the insert is substantially flat and an assembled configuration. The insert may define at least one cavity dimensioned to house at least one container. In the assembled configuration, the insert may comprise a top wall defining the at least one cavity, a second bottom wall, a second back wall, a second front wall, and a middle wall positioned between the top wall and the second bottom wall.

[0010] In some embodiments, the insert may define a first cavity dimensioned to house a first container and a second cavity dimensioned to house a second container. In some embodiments, a diameter of the first cavity may be smaller than a diameter of the second cavity, and a diameter of the first container may be smaller than a diameter of the second container. In some embodiments, a bottom surface of the at least one container may be configured to abut the middle wall of the insert when the at least one container is housed within the at least one cavity.

[0011] In some embodiments, the carton may further comprise a release tab comprising a portion of at least one of the lid and the bottom box and being defined by a perforated line in at least one of the lid and the bottom box. When the lid is in the closed position, the release tab may be secured to at least one of the lid and the bottom box via the perforation. When the lid is in the open position, the release tab may be separated along the perforated line from at least one of the lid and the bottom box. In some embodiments, the release tab may be a push tab that detaches the release tab from at least one of the lid and the bottom box. In some embodiments, the system may further comprise an adhesive positioned on at least a part of the release tab, and the release tab may include a kiss cut to improve securement between the release tab and the adhesive.

[0012] In some embodiments, when the insert is disposed within the bottom box of the carton, the second bottom wall may be configured to abut the first bottom wall, the second back wall may be configured to abut the first back wall, and the second front wall may be configured to abut the first front wall. In some embodiments, the at least one container may include a drug vial. The drug vial may be one of: an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In other embodiments, the first container may be one of: an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial, and the second container may be a different one of: an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In some embodiments, the drug vial may be filled or pre-filled with a drug. The drug may comprise tarlatamab. In some embodiments, the carton and the insert may be made of paperboard or a combination of synthetic and natural materials.

[0013] Another aspect of the present disclosure provides a system for packaging a plurality of containers. The system may comprise a carton having a storage configuration where the carton is substantially flat, and an assembled configuration. The carton may comprise a bottom box having a first bottom wall, a first sidewall, a second sidewall, a first back wall, and a first front wall when the carton is in the assembled configuration. The carton may also comprise a lid coupled to the bottom box and movable between an open position and a closed position, a tray coupled to the bottom box and movable between a first position and a second position, a first flap extending from the first sidewall of the bottom box, and a second flap extending from the second sidewall of the bottom box. The second sidewall may be opposite the first sidewall and face the first sidewall. The system may also comprise an insert configured to be disposed within the bottom box of the carton. The insert may have a storage configuration where the insert is substantially flat and an assembled configuration. The insert may define a plurality of cavities, and each cavity may be dimensioned to house a respective one of a plurality of containers. In the assembled configuration, the insert may comprise a top wall, a second bottom wall, a third sidewall, a fourth sidewall, a second back wall, a second front wall, and a first middle wall positioned between the top wall and the second bottom wall. The top wall and the first middle wall may define the plurality of cavities.

[0014] In some embodiments, the tray may be configured to define a second bottom box disposed within the bottom box when the tray is in the second position. In some embodiments, the insert may further comprise a first pair of opposing flaps coupled to the second front wall and a second pair of opposing flaps coupled to the second back wall. The insert may further comprise a third pair of opposing flaps coupled to the top wall, and the third pair of opposing flaps may be movable between an open position and a closed position. In the closed position, the third pair of opposing flaps may define the third sidewall and the fourth sidewall of the insert.

[0015] In some embodiments, when the insert is disposed within the bottom box of the carton, the second bottom wall may be configured to abut the first bottom wall, the third sidewall may be configured to abut the first sidewall, the fourth sidewall may be configured to abut the second sidewall, the second back wall may be configured to abut the first back wall, and the second front wall may be configured to abut the first front wall. In some embodiments, the insert may further comprise a second middle wall positioned between the first middle wall and the second bottom wall. At least one of the first middle wall or the second middle wall may be configured to abut bottom surfaces of the plurality of containers when the plurality of containers are housed within the plurality of cavities.

[0016] In some embodiments, the insert may further comprise a vertical rib support extending between the second middle wall and the second bottom wall, and the vertical rib support may be configured to maintain a predefined distance between the second middle wall and the second bottom wall. In some embodiments, a number of cavities defined by the top wall may be different from a number of cavities defined by the first middle wall. In other embodiments, a number of cavities defined by the top wall may be equal to a number of cavities defined by the first middle wall. In some embodiments, each of the plurality of cavities may have equal dimensions. In other embodiments, at least one of the plurality of cavities may have different dimensions than a remainder of the plurality of cavities.

[0017] In some embodiments, the carton may further comprise a release tab comprising a portion of at least one of the lid and the bottom box and may be defined by a perforated line in at least one of the lid and the bottom box. When the lid is in the closed position, the release tab may be secured to at least one of the lid and the bottom box via the perforation. When the lid is in the open position, the release tab may be separated along the perforated line from at least one of the lid and the bottom box. In some embodiments, the release tab may be a push tab that detaches the release tab from at least one of the lid and the bottom box. In some embodiments, the system may further comprise an adhesive positioned on at least a part of the release tab, and the release tab may include a kiss cut to improve securement between the release tab and the adhesive.

[0018] In some embodiments, at least one of the plurality of containers may include a drug vial. The drug vial may be one of: an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In some embodiments, the plurality of containers may include a first drug vial and a second drug vial. The first drug vial may be one of: an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial, and the second drug vial may be a different one of: an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In some embodiments, the drug vial may be filled or pre-filled with a drug, and the drug may comprise tarlatamab. In some embodiments, the carton and the insert may be made of paperboard or a combination of synthetic and natural materials.

[0019] Another aspect of the present disclosure provides a system for packaging a plurality of containers. The system may comprise a carton having a storage configuration where the carton is substantially flat, and an assembled configuration. The carton may comprise a plurality of sidewalls, a top wall, a bottom wall, a first flap extending from a first sidewall of the plurality of sidewalls, and a second flap extending from a second sidewall of the plurality of sidewalls, the second sidewall being opposite the first sidewall and facing the first sidewall. The first flap and the second flap define a first set of cavities. The system also comprises an insert configured to be disposed within the carton, and the insert has a storage configuration where the insert is substantially flat and an assembled configuration. In the assembled configuration, the insert comprises a first layer, a second layer positioned at a first distance below the first layer, and a third layer positioned at a second distance below the second layer. The first layer comprises at least one hole configured to house a container and a first set of flaps extending from opposite sides of the first layer, and the second layer is configured to support a bottom surface of the container and comprises a second set of flaps extending from opposite sides of the second layer. The first set of flaps of the first layer and the second set of flaps of the second layer are configured to fold down.

[0020] In some embodiments, in the assembled configuration, the carton may comprise four side walls, the top wall, and the bottom wall, and at least the four side walls and the bottom wall may define a second cavity inside the carton. The second cavity inside the carton may be configured to accommodate the insert. In the assembled configuration, the insert may comprise two opposite side walls connecting the first layer, the second layer, and the third layer. In some embodiments, when the insert is in the assembled configuration, the first set of flaps of the first layer and the second set of flaps of the second layer may be configured to fold down to define two additional side walls of the insert. In some embodiments, when the insert is in the assembled configuration and disposed within the carton, an inner surface of the first set of flaps of the first layer may be configured to abut an outer surface of the second set of flaps of the second layer, and an outer surface of the first set of flaps of the first layer may be configured to abut an inner surface of two side walls of the four side walls of the carton.

[0021] In some embodiments, the first set of cavities may comprise substantially rectangular air cavities, and the first set of cavities may be configured to provide a distance between the top wall of the carton and a top surface of the container to protect the container from top load. In the assembled configuration, the insert may comprise two opposite side walls connecting the first layer, the second layer, and the third layer, and, in the assembled configuration, the first set of flaps of the first layer and the second set of flaps of the second layer may be configured to fold down to define two additional side walls of the insert. In some embodiments, the first layer, the second layer, the third layer, the two opposite side walls connecting the first layer, the second layer, and the third layer, and the two additional side walls of the insert may be formed integrally.

[0022] In some embodiments, the second layer of the insert may be raised the second distance from the third layer of the insert when the insert is disposed within the carton such that the bottom surface of the container is raised the second distance from the bottom wall of the carton. In some embodiments, the plurality of sidewalls, the top wall, the bottom wall, the first flap, and the second flap of the carton may be formed integrally. In some embodiments, the container may be a vial, and the vial may be one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial.

[0023] Another aspect of the present disclosure provides a system for packaging a plurality of containers. The system may comprise a carton having a storage configuration where the carton is substantially flat, and an assembled configuration. The carton may comprise a plurality of sidewalls, a top wall, a bottom wall, a first flap extending from a first sidewall of the plurality of sidewalls, and a second flap extending from a second sidewall of the plurality of sidewalls, the second sidewall being opposite the first sidewall and facing the first sidewall. The first flap and the second flap define a first set of cavities. The system also comprises an insert configured to be disposed within the carton, and the insert has a storage configuration where the insert is substantially flat and an assembled configuration. In the assembled configuration, the insert comprises a first layer, a second layer positioned at a first distance below the first layer, a third layer positioned at a second distance below the second layer, and a fourth layer positioned at a third distance below the third layer. The first layer comprises a first set of holes configured to accommodate a plurality of containers, and each of the first set of holes is configured to accommodate a first portion of a respective container of the plurality of containers. The first layer also comprises a first set of flaps extending from opposite sides of the first layer. The second layer comprises a second set of holes configured to accommodate the plurality of containers, and each of the second set of holes is configured to accommodate a second portion of the respective container of the plurality of containers. The second layer also comprises a second set of flaps extending from opposite sides of the second layer. The third layer is configured to support a bottom surface of the plurality of containers, and the first set of flaps of the first layer and the second set of flaps of the second layer are configured to fold down.

[0024] In some embodiments, the insert may further comprise a vertical rib support extending between the third layer and the fourth layer. The vertical rib support may be configured to maintain the third distance between the third layer and the fourth layer. In other embodiments, the first set of holes in the first layer and the second set of holes in the second layer may be vertically aligned. In the assembled configuration, the carton may comprise four sidewalls, the top wall, and the bottom wall, and at least the four side walls and the bottom wall may define a second cavity inside the carton. The second cavity inside the carton may be configured to accommodate the insert.

[0025] In some embodiments, in the assembled configuration, the insert may comprise two opposite sidewalls connecting the first layer, the second layer, the third layer, and the fourth layer. When the insert is in the assembled configuration, the first set of flaps of the first layer and the second set of flaps of the second layer may be configured to fold down to define two additional sidewalls of the insert. In some embodiments, when the insert is in the assembled configuration and disposed within the carton, an inner surface of the first set of flaps of the first layer may be configured to abut an outer surface of the second set of flaps of the second layer, and an outer surface of the first set of flaps of the first layer may be configured to abut an inner surface of two side walls of the four side walls of the carton.

[0026] In some embodiments, the first set of cavities may comprise substantially rectangular air cavities, and the first set of cavities may be configured to provide a distance between the top wall of the carton and a top surface of the plurality of containers to protect the plurality of containers from top load. In the assembled configuration, the insert may comprise two opposite sidewalls connecting the first layer, the second layer, the third layer, and the fourth layer, and, in the assembled configuration, the first set of flaps of the first layer and the second set of flaps of the second layer may be configured to fold down to define two additional sidewalls of the insert.

[0027] In some embodiments, the first layer, the second layer, third layer, the fourth layer, the two opposite sidewalls connecting the first layer, the second layer, the third layer, and the fourth layer, and the two additional sidewalls of the insert may be formed integrally. In other embodiments, the third layer of the insert may be raised the third distance from the fourth layer of the insert when the insert is disposed within the carton such that the bottom surface of the plurality of containers is raised the third distance from the bottom wall of the carton. In some embodiments, the plurality of sidewalls, the top wall, the bottom wall, the first flap, and the second flap of the carton may be formed integrally.

[0028] Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of the embodiments exemplifying the best mode of carrying out the disclosure as presently perceived.

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] It is believed that the disclosure will be more fully understood from the following description taken in conjunction with the accompanying drawings. Some of the drawings may have been simplified by the omission of selected elements for the purpose of more clearly showing other elements. Such omissions of elements in some drawings are not necessarily indicative of the presence or absence of particular elements in any of the exemplary embodiments, except as may be explicitly delineated in the corresponding written description. Also, none of the drawings is necessarily to scale.

[0030] FIG. **1**A illustrates an exemplary system for packaging a container, in accordance with various embodiments of the present disclosure.

[0031] FIG. **1**B illustrates another view of the exemplary system of FIG. **1**A, in accordance with various embodiments of the present disclosure.

[0032] FIG. **1**C illustrates another view of the exemplary system of FIG. **1**A, in accordance with various embodiments of the present disclosure.

[0033] FIG. **1**D illustrates another view of the exemplary system of FIG. **1**A, in accordance with various embodiments of the present disclosure.

[0034] FIG. **2**A illustrates an exemplary carton of the exemplary system of FIG. **1**A, in accordance with various embodiments of the present disclosure.

[0035] FIG. **2**B illustrates a top view of the exemplary carton of FIG. **2**A, in accordance with various embodiments of the present disclosure.

[0036] FIG. **2**C illustrates the exemplary carton of FIG. **2**A with an exemplary insert, in accordance with various embodiments of the present disclosure.

[0037] FIG. **2**D illustrates a top view of the exemplary carton of FIG. **2**A and the exemplary insert of FIG. **2**C, in accordance with various embodiments of the present disclosure.

[0038] FIG. **2**E illustrates a bottom view of the exemplary carton of FIG. **2**A and the exemplary insert of FIG. **2**C, in accordance with various embodiments of the present disclosure.

[0039] FIG. **3**A illustrates another view of the exemplary insert of FIG. **2**C, in accordance with various embodiments of the present disclosure.

[0040] FIG. **3**B illustrates a side view of the exemplary insert of FIG. **3**A, in accordance with various embodiments of the present disclosure.

[0041] FIG. **3**C illustrates another side view of the exemplary insert of FIG. **3**A, in accordance with various embodiments of the present disclosure.

[0042] FIG. **3**D illustrates another side view of the exemplary insert of FIG. **3**A, in accordance with various embodiments of the present disclosure.

[0043] FIG. **4**A illustrates a plan view of a die cut of an exemplary carton of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0044] FIG. **4**B illustrates a plan view of a die cut of an exemplary insert of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0045] FIG. **5**A illustrates the exemplary carton of FIG. **4**A and the exemplary insert of FIG. **4**B in their storage configurations, in accordance with various embodiments of the present disclosure.

[0046] FIG. **5**B illustrates the exemplary carton and the exemplary insert of FIG. **5**A in their assembled configurations, in accordance with various embodiments of the present disclosure.

[0047] FIG. **5**C illustrates an exemplary system for packaging a plurality of containers in its assembled configuration with the exemplary carton and the exemplary insert of FIG. **5**B, in accordance with various embodiments of the present disclosure.

[0048] FIG. **5**D illustrates another view of the exemplary system of FIG. **5**C, in accordance with various embodiments of the present disclosure.

[0049] FIG. **5**E illustrates another view of the exemplary system of FIG. **5**C, in accordance with various embodiments of the present disclosure.

[0050] FIG. **5**F illustrates another view of the exemplary system of FIG. **5**C, in accordance with various embodiments of the present disclosure.

[0051] FIG. **6**A illustrates the exemplary carton of FIG. **5**A in its storage configuration, in accordance with various embodiments of the present disclosure.

[0052] FIG. **6**B illustrates the exemplary carton of FIG. **5**A in its assembled configuration, in accordance with various embodiments of the present disclosure.

[0053] FIG. **6**C illustrates another view of the exemplary carton of FIG. **5**A in its assembled configuration, in accordance with various embodiments of the present disclosure.

[0054] FIG. 7A illustrates the exemplary insert of FIG. 5A in its storage configuration, in accordance with various embodiments of the present disclosure.

[0055] FIG. 7B illustrates a side view of the exemplary insert of FIG. 5A in its assembled configuration, in accordance with various embodiments of the present disclosure.

[0056] FIG. 7C illustrates another side view of the exemplary insert of FIG. 5A in its assembled configuration, in accordance with various embodiments of the present disclosure.

[0057] FIG. 7D illustrates another side view of the exemplary insert of FIG. 5A in its assembled configuration, in accordance with various embodiments of the present disclosure.

[0058] FIG. 8 illustrates a side plan view of the exemplary insert of FIG. 5A in its assembled configuration, in accordance with various embodiments of the present disclosure.

[0059] FIG. 9A illustrates a plan view of a die cut of an exemplary carton of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0060] FIG. 9B illustrates a plan view of a die cut of an exemplary insert of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0061] FIG. 10A illustrates the exemplary carton of FIG. 9A and the exemplary insert of FIG. 9B in their assembled configurations, in accordance with various embodiments of the present disclosure.

[0062] FIG. 10B illustrates an exemplary system for packaging a plurality of containers in its assembled configuration with the exemplary carton and the exemplary insert of FIG. 10A, in accordance with various embodiments of the present disclosure.

[0063] FIG. 10C illustrates a perspective view of the exemplary system of FIG. 10A, in accordance with various embodiments of the present disclosure.

[0064] FIG. 10D illustrates another view of the exemplary system of FIG. 10A, in accordance with various embodiments of the present disclosure.

[0065] FIG. 11 illustrates a side view of the exemplary insert of FIG. 10A, in accordance with various embodiments of the present disclosure.

[0066] FIG. 12A illustrates a plan view of a die cut of an exemplary carton of an exemplary system for packaging a container, in accordance with various embodiments of the present disclosure.

[0067] FIG. 12B illustrates a plan view of a die cut of an exemplary insert of an exemplary system for packaging a container, in accordance with various embodiments of the present disclosure.

[0068] FIG. 13A illustrates a perspective view of an exemplary system for packaging a container with the exemplary carton of FIG. 12A in an assembled configuration and the exemplary insert of FIG. 12B in an assembled configuration, in accordance with various embodiments of the present disclosure.

[0069] FIG. 13B illustrates a cross-sectional view of the exemplary system of FIG. 13A, in accordance with various embodiments of the present disclosure.

[0070] FIG. 13C illustrates a perspective view of the exemplary insert of FIG. 12B in the assembled configuration, in accordance with various embodiments of the present disclosure.

[0071] FIG. 14A illustrates a plan view of a die cut of an exemplary carton of an exemplary system for packaging at least one container, in accordance with various embodiments of the present disclosure.

[0072] FIG. 14B illustrates a plan view of a die cut of an exemplary insert of an exemplary system for packaging at least one container, in accordance with various embodiments of the present disclosure.

[0073] FIG. 15A illustrates a perspective view of an exemplary system for packaging at least one container with the exemplary carton of FIG. 14A in an assembled configuration and the exemplary insert of FIG. 14B in an assembled configuration, in accordance with various embodiments of the present disclosure.

[0074] FIG. 15B illustrates a cross-sectional view of the exemplary system of FIG. 15A, in accordance with various embodiments of the present disclosure.

[0075] FIG. 16A illustrates a plan view of a die cut of an exemplary carton of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0076] FIG. 16B illustrates a plan view of a die cut of an exemplary insert of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0077] FIG. 17A illustrates a perspective view of the exemplary carton of FIG. 16A in an assembled configuration with an exemplary tray in a first position, in accordance with various embodiments of the present disclosure.

[0078] FIG. 17B illustrates a perspective view of the exemplary carton of FIG. 17A with the exemplary tray in a second position, in accordance with various embodiments of the present disclosure.

[0079] FIG. 18A illustrates a perspective view of the exemplary insert of FIG. 16B in an assembled configuration, in accordance with various embodiments of the present disclosure.

[0080] FIG. 18B illustrates a side view of the exemplary insert of FIG. 18A, in accordance with various embodiments of the present disclosure.

[0081] FIG. 18C illustrates another side view of the exemplary insert of FIG. 18A, in accordance with various embodiments of the present disclosure.

[0082] FIG. 18D illustrates another side view of the exemplary insert of FIG. 18A, in accordance with various embodiments of the present disclosure.

[0083] FIG. 19A illustrates a plan view of a die cut of an exemplary carton of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0084] FIG. 19B illustrates a plan view of a die cut of an exemplary insert of an exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure.

[0085] FIG. 20A illustrates a perspective view of the exemplary carton of FIG. 19A in an assembled configuration with an exemplary tray in a first position, in accordance with various embodiments of the present disclosure.

[0086] FIG. 20B illustrates a perspective view of the exemplary carton of FIG. 20A with the exemplary tray in a second position, in accordance with various embodiments of the present disclosure.

[0087] FIG. **21**A illustrates a perspective view of the exemplary insert of FIG. **19**B in an assembled configuration, in accordance with various embodiments of the present disclosure.

[0088] FIG. **21**B illustrates a side view of the exemplary insert of FIG. **21**A, in accordance with various embodiments of the present disclosure.

[0089] FIG. **21**C illustrates another side view of the exemplary insert of FIG. **21**A, in accordance with various embodiments of the present disclosure.

[0090] FIG. **21**D illustrates another side view of the exemplary insert of FIG. **21**A, in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0091] The present disclosure generally relates to systems for packaging one or more containers for clinical, medical, or pharmaceutical use. The containers disclosed herein may include but are not limited to vials, such as glass vials, for holding medicament. By way of example, the containers may include glass vials that are adaptable for cc vials and that are configured to hold medicament, such as drug products in lyophilized form or liquid form. The containers may have standard capacities and dimensions as specified by ISO 8362-1. The systems disclosed herein may comprise a carton and an insert. The carton and/or the insert may comprise features configured to protect the containers packaged within the system from physical damage or breakage in all three dimensions (e.g., protect the containers from impact in the x, y, and z axes). The carton and/or the insert may be shipped in a storage configuration and may be folded into an assembled configuration from a storage configuration during assembly. Accordingly, many cartons and inserts may be stacked on top of each other in their storage configurations, thereby decreasing their profile and storage footprint while increasing efficiency and cost in shipping the systems disclosed herein and storing the systems in warehouses. During assembly, the insert may also be placed inside the carton to form the systems disclosed. The carton and the insert may be manufactured using sustainable materials that can be completely recyclable, which is beneficial to the environment compared to conventional packaging systems made of plastic trays or foam inserts. Moreover, inserts in the systems disclosed herein may be interchangeable. Accordingly, the carton may be configured to accommodate an insert for holding up to six containers, by way of example, and an insert for holding up to twelve containers, by way of example. Therefore, the systems disclosed herein may be modular in that they may be able to accommodate various numbers of containers depending on the insert placed inside the carton. These and other advantages and benefits will be apparent to one of ordinary skill in the art reviewing the present disclosure.

[0092] As used herein, the term "carton" may refer to the carton of the system for packaging one or more containers. The term "container" may refer to any component that may be housed, disposed, and/or accommodated within a packaging, such as vials, syringes, pre-filled syringes, or the like. The term "drug," as used herein, can be used interchangeably with other similar terms and can be used to refer to any type of medicament or therapeutic material including traditional and non-traditional pharmaceuticals, nutraceuticals, supplements, biologics, biologically active agents and compositions, large molecules, biosimilars, bioequivalents, therapeutic antibodies, polypeptides, proteins, small molecules and generics. Non-therapeutic injectable materials are also encompassed. The drug may be in liquid form, a lyophilized form, or in a reconstituted from lyophilized form.

[0093] As discussed above, conventional systems for packaging one or more containers do not provide the flexibility in use in that they are not suitable for accommodating multiple containers of different sizes to be packaged together into one packaging system. This limitation can have a direct consequence on the number of containers that can be packaged into each packaging system. This limitation can also have a direct consequence on the number of packaging systems having to be used to transport a given number of containers and, thus, the cost of transporting the containers from one site to another. Furthermore, conventional packaging systems that attempt to protect the containers packaged therein from damage generally include foam inserts or plastic trays. The foam inserts and plastic trays prevent containers from contacting each other and, thus, provide protection from lateral impact. However, materials such as foam or plastic are not sustainable and harmful to the environment. Moreover, conventional packaging systems do not protect the containers packaged therein from top load or bottom load. For example, when conventional packaging systems are stacked on top of each other or placed on hard surfaces, there is no way for the conventional packaging systems to protect the containers therein from top load, bottom load, and vertical impact.

[0094] Therefore, there is a need for an improved system for packaging one or more containers, such as glass vials, that protects container(s) from vertical and lateral impact. There is also a need for an improved system for packaging one or more containers that is not only manufactured using sustainable, environment-friendly material, but also modular such that one packaging system can be configured to accommodate and transport multiple container sizes and multiple container counts.

[0095] Referring now to FIGS. **1**A-**1**D, an exemplary system **200** for packaging a container **208** is shown in accordance with various embodiments of the present disclosure. As will be discussed in more detail below, system **200** may comprise a carton **202** (also interchangeably referred to as a "carton") and an insert **204** configured to be placed inside the carton **202**. Insert **204** may comprise a hole **206** configured to accommodate a container **208** therein. By way of example, the hole **206** may be die cut in the insert **204** and the hole **206** may be die cut at a center point of the insert **204** such that the hole **206** is spaced away from the sides of the insert **204**. By positioning the hole **206** away from all sides of the insert **204**, the insert **204** of the system **200** may prevent the container **208** therein from contacting the sides of the system **200** during packaging or transport. Additionally, the hole **206** may be sized to accommodate multiple sizes of containers **208**. For example, the hole **206** may be sized to accommodate one or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials.

[0096] In some embodiments, the carton **202** and the insert **204** may be manufactured and formed separately. The carton **202** and the insert **204** may be manufactured using sustainable materials that can be recycled completely. By way of example, the carton **202** and the insert **204** may be made from paperboard material or other Forest Stewardship Council (FSC)-certified material. In other embodiments, the carton **202** and the insert **204** may be manufactured and formed integrally. In some embodiments, as shown in FIGS. **1A-1C**, the carton **202** may comprise a plurality of sidewalls **213**, a top wall **214**, a bottom wall **216** (shown in FIG. **2E**), and a back flap **215** extending from the top wall **214**. The carton **202** may further comprise a first flap **207***a,* extending from a first sidewall **213***a* of a plurality of sidewalls **213**, and a second flap **207***b,* extending from a second sidewall **213***b* opposite the first sidewall **213***a* and facing the first sidewall **213***a*. The first flap **207***a* and the second flap **207***b* may define a first set of cavities **209**, as illustrated in FIGS. **1B** and **1C**. In some embodiments, the first set of cavities **209** may comprise substantially rectangular air cavities. In other embodiments, the first set of cavities **209** may comprise triangular, circular, or elliptical air cavities. The first set of cavities **209** may be configured to protect the container **208** from top load impact. By way of example, when the flaps **207***a,* **207***b* are folded down, as shown in FIG. **1C**, the first set of cavities **209** may provide air cushioning such that, when there is a top load placed on top of the carton **202**, the space created within the first set of cavities **209** may be configured to cushion the container **208** and protect the container **208** from any vertical impact or damage. In some embodiments, the sidewalls **213**, the top wall **214**, the bottom wall **216**, the back flap **215**, the first flap **207***a,* and the second flap **207***b* may be formed integrally.

[0097] FIGS. **2A-2E** illustrate the carton **202** and the insert **204** of the system **200** of FIGS. **1A-1D** in more detail. In particular, FIGS. **2A-2E** illustrate how the carton **202** is folded from a storage configuration to an assembled configuration prior to placing the insert **204** inside the carton **202**. As discussed above with reference to FIGS. **1A-1D**, the carton **202** may comprise a plurality of sidewalls **213**, a top wall **214**, a bottom wall **216** (shown in FIG. **2E**), and a back flap **215** extending from the top wall **214**. Prior to assembly, the carton **202** may be in a storage configuration. For example, when the carton **202** is first shipped or stored in a particular warehouse and before the carton **202** is assembled to hold one or more containers therein, the carton **202** may be in a storage configuration. Accordingly, because the carton **202** can be shipped or stored in a storage configuration, multiple cartons **202** can be easily stacked on top of each other for efficient shipment and warehousing.

[0098] During assembly, the carton **202** may be folded from a storage configuration to an assembled configuration. For example, the carton **202** may comprise one or more perforations, scores, or cuts along the edges of the sidewalls **213**, the top wall **214**, the bottom wall **216**, a first flap **207***a,* a second flap **207***b,* and/or back flap **215** so that the carton **202** can be easily folded from a storage configuration to an assembled configuration. FIG. **2A** illustrates the carton **202** in a storage configuration before the carton **202** is folded to an assembled configuration. FIGS. **2B-2E** illustrates the carton **202** when the carton **202** is folded from a storage configuration to an assembled configuration. As shown in FIGS. **2B-2E**, for example, when in the assembled configuration, the carton **202** may comprise a rectangular carton comprising four sidewalls **213**, top wall **214**, and bottom wall **216**. In the assembled configuration, at least the sidewalls **213** and the bottom wall **216** of the carton **202** may define a cavity **217** inside the carton **202**. The cavity **217** may be configured to accommodate the insert **204** during assembly. As will be discussed in more detail below, the insert **204** may also be folded from a storage configuration to an assembled configuration prior to placing the insert **204** inside the cavity **217** of the carton **202**.

[0099] As discussed above, the carton **202** may further comprise a first flap **207***a,* extending from a first sidewall **213***a* of a plurality of sidewalls **213**, and a second flap **207***b,* extending from a second sidewall **213***b* opposite the first sidewall **213***a* and facing the first sidewall **213***a.* During assembly, the first flap **207***a* and the second flap **207***b* may be folded down (as shown in FIG. **2B**). When folded down, the first and second flaps **207***a,* **207***b* may define a first set of cavities **209** that may be configured to protect a container therein, such as container **208**, from top load impact. By way of example, when the first and second flaps **207***a,* **207***b* are folded down, as shown in FIG. **2B**, the first set of cavities **209** may provide air cushioning such that, when there is a top load placed on the top wall **214** of the carton **202**, the space created within the first set of cavities **209** may be configured to cushion and protect any container therein from any vertical impact or damage. Additionally, the back flap **215** extending from the top wall **214** of the carton **202** may be folded (as shown in FIG. **2B**) such that the back flap **215** tucks into a front sidewall of the plurality of sidewalls **213** to close and secure the system **200** (as shown in FIG. **1D**). In some embodiments, the bottom wall **216** of the carton **202** may be pre-configured to allow for automatic assembly. For example, as shown in FIG. **2E**, the bottom wall **216** of the carton **202** may be pre-configured such that when the carton **202** is folded from a storage configuration to an assembled configuration, the bottom wall **216** may automatically straighten out into a flat surface to define the bottom surface of the carton **202**.

[0100] FIGS. **3A-3D** illustrate the insert **204** configured to be placed inside the carton **202** of FIGS. **1A-1D** and **2A-2E**. As illustrated in FIG. **3A**, prior to assembling the insert **204** and placing the insert **204** inside the carton **202**, the insert **204** may be in a storage configuration. For example, when the insert **204** is first shipped or stored in a particular warehouse, the insert **204** may be in a storage configuration. Accordingly, because the insert **204** can be shipped or stored in a storage configuration, multiple inserts **204** can be easily stacked on top of each other for efficient shipment and warehousing. During assembly, similar to the carton **202**, the insert **204** can be folded from a storage configuration to an assembled configuration. FIGS. **3B-3D** illustrate the insert **204** that is folded from a storage configuration to an assembled configuration.

[0101] As shown in FIGS. **3B-3D**, once the insert **204** is folded to an assembled configuration, the insert may comprise at least two opposite sidewalls **219**, a first layer **211**, a second layer **212**, and a third layer **218**. As shown in at least FIGS. **3B** and **3D**, the two sidewalls **219** may connect the first layer **211**, the second layer **212**, and the third layer **218**. The first layer **211** may define the top surface of the insert **204** and may comprise a die cut hole **206** configured to accommodate at least a portion of a container **208**, such as a glass vial. Accordingly, the hole **206** may be configured to laterally stabilize the container **208** therein. As discussed above, the hole **206** may be sized to accommodate one or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. Moreover, the hole **206** is spaced a distance away from the edges of the first layer **211** such that, when the container **208** is placed within the hole **206** of the insert **204**, the container **208** will not contact any sides of the insert **204** and any sidewalls **213** of the carton **202**. The second layer **212** may be positioned a first distance below the first layer **211** and may be configured to support a bottom surface of the container **208**. The third layer **218** may define the bottom surface of the insert **204**. As seen in FIGS. **3**B and **3**D, the second layer **212** is also positioned a second distance above the third layer **218** such that, when the container **208** is placed within the insert **204**, the bottom surface of the container **208** is raised the second distance from the bottom surface of the insert **204** and the bottom wall **216** of the carton **202**. Accordingly, because the bottom surface of the container **208** is raised from the bottom wall **216** of the carton by the insert **204** and supported by the second layer **212**, the system **200** may be configured to protect the container **208** from bottom load impact. In some embodiments, the first distance and the second distance may be the same. That is, in some embodiments, the first layer **211** and the second layer **212** may be distanced the same distance away from the second layer **212** and the third layer **218**, respectively. In other embodiments, the first distance and the second distance may be different. For example, the second layer **212** may be distanced farther away from the third layer **217** than from the first layer **211**.

[0102] In some embodiments, as shown in FIGS. **3**A-**3**D, the first layer **211** of the insert **204** may comprise a first set of flaps **210***a* extending from opposite sides of the first layer **211**, and the second layer **212** may comprise a second set of flaps **210***b* extending from opposite sides the second layer **212**. The first set of flaps **210***a* and the second set of flaps **210***b* may be configured to fold down (as shown in FIGS. **3**B and **3**C) to define two additional sidewalls of the insert **204**. When folded down, the first set of flaps **210***a* and the second set of flaps **210***b* may be configured to provide additional rigidity to the insert **204** such that the insert **204** can maintain its rectangular shape when folded to its assembled configuration. Additionally, when folded down, an inner surface of the first set of flaps **210***a* of the first layer **211** may be configured to abut an outer surface of the second set of flaps **210***b* of the second layer **212**. In some embodiments, the sidewalls **219**, the first layer **211**, the second layer **212**, the third layer **218**, the first set of flaps **210***a*, and the second set of flaps **210***b* may be formed integrally using sustainable materials that can be recycled completely, such as paperboard material or other FSC-certified material. Similar to the carton **202**, the insert **204** may comprise one or more perforations, scores, or cuts along the edges of the first layer **211**, the second layer **212**, the third layer **218**, the first set of flaps **210***a*, and the second set of flaps **210***b* so that the insert **204** can be easily folded from a storage configuration to an assembled configuration.

[0103] As discussed above, the insert **204** may be configured to be placed inside the carton **202** during assembly (as shown in FIGS. **1**A and **1**B). After the carton **202** is folded to its assembled configuration and the insert **204** is also folded to its assembled configuration, the insert **204** may be configured to be placed inside the cavity **217** (shown in FIG. **2**D) of the carton **202**. When placed inside the cavity **217** of the carton **202**, an inner surface of the first set of flaps **210***a* of the first layer **211** may be configured to abut an outer surface of the second set of flaps **210***b* of the second layer **212**, and an outer surface of the first set of flaps **210***a* of the first layer **211** may be configured to abut an inner surface of two opposite sidewalls **213** of the carton **204**. Referring back to FIGS. **1**A and **1**B, once the insert **204** is placed inside the carton **202**, and the container **208** is placed within the hole **206** of the insert **204**, the first flap **207***a* and the second flap **207***b* may be folded down, and the first set of cavities **209** may be configured to protect the container **208** from top load impact. Additionally, the back flap **215** extending from the top wall **214** of the carton **202** may be tucked into a front sidewall of the plurality of sidewalls **213** to close and secure the system **200** (as shown in FIG. **1**D). Accordingly, when fully assembled and packaged, the hole **206** of the insert **204** may be configured to secure the container **208** in an upright position and protect the container **208** from later impact, the first set of cavities **209** defined by the first flap **207***a* and the second flap **207***b* may protect the container **208** from top load impact, and the second layer **212** that is raised from the bottom surface of the insert **204** and the bottom wall **216** of the carton **202** may protect the container **208** from bottom load impact.

[0104] In some embodiments, the hole **206** in the insert **204** may be configured and sized to accommodate a neck of the container **208**, such as a neck of a glass vial. Additionally, or alternatively, the hole **206** may be configured and sized to accommodate a body of the container **208**, such as a body of a glass vial. In some embodiments, the size of the container **208** may be indicated by ISO 8362-1. For example, if the container **208** is a glass vial, ISO 8362-1 may indicate the standard capacity of the container **208**, the standard height of the container **208**, and/or the standard outer diameter of the body of the container **208**. Accordingly, in some embodiments, the hole **206** may be sized to accommodate the container **208** based on the standard dimensions indicated by ISO 8362-1.

[0105] FIGS. **1**A-**1**D and **2**A-**2**E illustrate a rectangular carton **202** comprising four sidewalls **213**, a top wall **214**, and a bottom wall **216**. FIGS. **3**A-**3**D also illustrate a rectangular insert **204** comprising four sidewalls and configured to fit inside the cavity **217** of the carton **202**. However, the carton **202** and/or the insert **204** are not limited to rectangular shapes. For example, the carton **202** may comprise a cylindrical shape or any other geometric shapes, and the insert **204** may be shaped to fit inside a cavity in the carton **202**.

[0106] Referring now to FIGS. **5**A-**5**F, an exemplary system **300** for packaging a plurality of containers **208** is shown in accordance with various embodiments of the present disclosure. System **300** may comprise a carton **302** and an insert **304** configured to be placed inside the carton **302**. Insert **304** may comprise a plurality of holes **306** configured to accommodate a plurality of containers **308** therein. In FIG. **5**A, the insert **304** comprises six holes **306**, each configured to accommodate a respective container **308**. However, the insert **304** may comprise any number of holes **306**, such as two, three, four, five, seven, eight, nine, ten, twelve, fifteen, twenty, twenty-five, or thirty holes. Holes **306** may be die cut in the insert **304** and the holes **306** may be die cut such that each hole is spaced equidistantly from each other. Additionally, the holes **306** may be die cut such that the holes **306** are spaced away from the sidewalls of the insert **304** and the sidewalls of the carton **302**. By spacing the holes **306** away from each other and also away from all sides of the insert **204** and the carton **302**, the system **300** may prevent the containers **308** therein from contacting the sides of the system **300** during packaging or transport and, thus, protect the containers **308** from lateral impact. The holes **306** may also be configured to laterally stabilize each individual container **308** packaged therein. Additionally, the holes **306** may be sized to accommodate multiple sizes of containers **308**. For example, the holes **306** may be sized to accommodate one or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. In some embodiments, the holes **306** may all be identical in size and shape. In other embodiments, some of the holes **306** may vary in size such that the insert **304** can accommodate containers **308** of various sizes.

[0107] In some embodiments, the carton **302** and the insert **304** may be manufactured and formed separately. The carton **302** and the insert **304** may be manufactured using sustainable materials that can be recycled completely, such as paperboard material or other FSC-certified material. In other embodiments, the carton **302** and the insert **304** may be manufactured and formed integrally. In some embodiments, as shown in FIG. **6**B, the carton **302** may comprise a plurality of sidewalls **313**, a top wall **314**, a bottom wall **316** (similar to bottom wall **216** of FIG. **2**E), and a back flap **315** extending from the top wall **314**. Referring back to FIGS. **5**A-**5**E, the carton **302** may further comprise a first flap **307***a*, extending from a first sidewall **313***a* of a plurality of sidewalls **313**, and a second flap **307***b*, extending from a second sidewall **313***b* opposite the first sidewall **313***a* and facing the first sidewall **313***a*. The first flap **307***a* and the second flap **307***b* may define a first set of cavities **309**, as illustrated in FIGS. **5**D and **5**E. In some embodiments, the first set of cavities **309** may comprise rectangular air cavities. In other embodiments, the first set of cavities **309** may comprise triangular, circular, or elliptical air cavities. The first set of cavities **309** may be configured to protect the containers **308** therein from top load impact. By way of example, when the flaps **307***a*, **307***b* are folded down, as shown in FIG. **5**E, the first set of cavities **309** may provide air cushioning such that, when there is a top load placed on top of the carton **302**, the space created within the first set of cavities **309** may be configured to cushion the containers **308** therein and protect the container **308** from any vertical impact or damage. In some embodiments, the sidewalls **313**, the top wall **314**, the bottom wall **316**, the back flap **315**, the first flap **307***a*, and the second flap **307***b* may be formed integrally.

[0108] FIGS. **6**A-**6**C, **7**A-**7**D, and **8** illustrate the carton **302** and the insert **304** of the system **300** of FIGS. **5**A-**5**F in more detail. In particular, FIGS. **6**A-**6**C illustrate how the carton **302** is folded from a storage configuration to an assembled configuration prior to placing the insert **304** inside the carton **302**. As seen in FIGS. **6**A-**6**C, the carton **302** may comprise a plurality of sidewalls **313**, a top wall **314**, a bottom wall **316**, and a back flap **316** extending from the top wall **314**. Prior to assembly, the carton **302** may be in a storage configuration, as shown in FIG. **6**A. For example, when the carton **302** is first shipped or stored in a particular warehouse and before the carton **302** is assembled to hold one or more containers **308** therein, the carton **302** may be in a storage configuration. Accordingly, because the carton **302** can be shipped or stored in a storage configuration, multiple cartons **302** can be easily stacked on top of each other for efficient shipment and warehousing.

[0109] During assembly, the carton **302** may be folded from a storage configuration to an assembled configuration. For example, the carton **302** may comprise one or more perforations, scores, or cuts along the edges of the sidewalls **313**, the top wall **314**, the bottom wall **316**, a first flap **307***a*, a second flap **307***b*, and/or back flap **315** so that the carton **302** can be easily folded from a storage configuration to an assembled configuration. FIG. **6**A illustrates the carton **302** in a storage configuration before the carton **302** is folded to an assembled configuration. FIG. **6**B illustrates the carton **302** when the carton **302** is folded from a storage configuration to an assembled configuration. As shown in FIG. **6**B, when in the assembled configuration, the carton **302** may comprise a rectangular carton comprising four sidewalls **313**, top wall **314**, and bottom wall **316**. In the assembled configuration, at least the sidewalls **313** and the bottom wall **316** of the carton **302** may define a cavity **317** inside the carton **302**. The cavity **317** may be configured to accommodate the insert **304** during assembly. As will be discussed in more detail below, the insert **304** may also be folded from a storage configuration to an assembled configuration prior to placing the insert **304** inside the cavity **317** of the carton **302**.

[0110] As shown in FIGS. **5**A-**5**F and **6**A-**6**C, the carton **302** may further comprise a first flap **307***a*, extending from a first sidewall **313***a* of a plurality of sidewalls **313**, and a second flap **307***b*, extending from a second sidewall **313***b*. During assembly, the first flap **307***a* and the second flap **307***b* may be folded down (as shown in FIGS. **5**D, **5**E, and **6**B). When folded down, the first and second flaps **307***a*, **307***b* may define a first set of cavities **309** (shown in FIGS. **5**D and **5**E) that may be configured to protect the containers **308** therein from top load impact. For example, when the first and second flaps **307***a*, **307***b* are folded down, as shown in FIG. **5**E, the first set of cavities **309** may provide air cushioning such that, when there is a top load placed on the top wall **314** of the carton **302**, the space created within the first set of cavities **309** may be configured to cushion and protect the containers **308** therein from any vertical impact or damage. Additionally, the back flap **315** (shown in FIGS. **5**C and **6**B) extending from the top wall **314** of the carton **302** may be folded down such that the back flap **315** tucks into a front sidewall of the plurality of sidewalls **313** to close and secure the system **300** (as shown in FIGS. **5**F and **6**C). In some embodiments, the bottom wall **316** of the carton **302** may be pre-configured to allow for automatic assembly. For example, similar to the bottom wall **216** shown in FIG. **3**E, the bottom wall **316** of the carton **302** may be pre-configured such that when the carton **302** is folded from a storage configuration to an assembled configuration, the bottom wall **316** may automatically straighten out into a flat surface to define the bottom surface of the carton **302**.

[0111] FIGS. **7**A-**7**D and **8** illustrate the insert **304** configured to be placed inside the carton **302** of FIGS. **5**A-**5**F and **6**A-**6**C. As illustrated in FIG. **7**A, prior to assembling the insert **304** and placing the insert **304** inside the carton **302**, the insert **304** may be in a storage configuration. For example, similar to the carton **302**, when the insert **304** is first shipped or stored in a particular warehouse, the insert **304** may be in a storage configuration. Accordingly, because the insert **304** can be shipped or stored in a storage configuration, multiple inserts **304** can be easily stacked on top of each other for efficient shipment and warehousing. During assembly, similar to the carton **302**, the insert **304** can be folded from a storage configuration to an assembled configuration. FIGS. **7**B-**7**D and **8** illustrate the insert **304** that is folded from a storage configuration to an assembled configuration.

[0112] As shown in FIGS. **7**B-**7**D and **8**, once the insert **304** is folded to an assembled configuration, the insert may comprise at least two opposite sidewalls **319**, a first layer **311**, a second layer **312**, a third layer **318**, and a fourth layer **320**. As shown in at least FIGS. 7B, 7C, and **8**, the two sidewalls **319** may connect the first layer **311**, the second layer **312**, the third layer **318**, and the fourth layer **320**. The first layer **311** may define the top surface of the insert **304** and may comprise die cut holes **306** configured to accommodate at least a portion of a plurality of container **308**, such as glass vials. The holes **306** may be sized to accommodate one or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. Moreover, the holes **306** may be spaced away from each other and away from the edges of the first layer **311** such that, when the containers **308** are placed within respective holes **306** of the insert **304**, the containers **308** will not contact each other, nor any sides of the insert **304**, nor any sidewalls **313** of the carton **302**. The second layer **312** may be positioned a first distance below the first layer **311** and may also comprise die cut holes **306** configured to accommodate at least a portion of the plurality of containers **308**, such as glass vials. The holes **306** in the second layer **312** may be vertically aligned with the holes **306** in the first layer **311** such that when the containers **308** are placed within respective holes **306**, each container **308** will fit through the respective hole **306** in the first layer **311** as well as the respective hole **306** in the second layer **312**. The third layer **318** may be positioned a second distance below the second layer **312** and may be configured to support a bottom surface of each container **308**. The fourth layer **320** may be positioned a third distance below the third layer **318** and may define the bottom surface of the insert **304**.

[0113] As seen in FIG. **8**, the third layer **318** is also positioned a third distance above the fourth layer **320** such that, when the containers **308** are placed within the insert **304**, the bottom surfaces of the containers **308** are raised the third distance from the bottom surface of the insert **304** and the bottom wall **316** of the carton **302**. Accordingly, because the bottom surfaces of the containers **308** are raised from the bottom wall **316** of the carton **302** by the insert **304** and supported by the third layer **318**, the system **300** may be configured to protect the containers **308** from bottom load impact. In some embodiments, the first distance, the second distance, and the third distance may be the same such that the first layer **311**, the second layer **312**, the third layer **318**, and the fourth layer **320** are all spaced equidistant from each other. In other embodiments, the first distance may be different from at least one of the second distance or the third distance. For example, the distance between the third layer **318** and the fourth layer **320** may be greater than the distance between the third layer **318** and the second layer **312**.

[0114] In some embodiments, the insert **304** may further comprise a vertical rib support **321** extending between the third layer **318** and the fourth layer **320**, as shown in FIG. **8**. The vertical rib support **321** may be formed integrally with the insert **304**. The vertical rib support **321** may be configured to maintain the third distance between the third layer **318** and the fourth layer **320** such that the containers **308** are consistently raised from the bottom wall **316** of the carton **302** and supported by the third layer **318** of the insert **304**. Accordingly, the vertical rib support **321** may ensure that the containers **308** are protected from bottom load impact. Additionally, the vertical rib support **321** may be configured to help the third layer **318** in supporting the weight of the containers **308**.

[0115] Referring back to FIGS. **7**B and **7**C, the first layer **311** of the insert **304** may comprise a first set of flaps **310***a* extending from opposite sides of the first layer **311**, and the second layer **312** may comprise a second set of flaps **310***b* extending from opposite sides the second layer **312**. The first set of flaps **310***a* and the second set of flaps **310***b* may be configured to fold down (as shown in FIGS. **7**C and **7**D) to define two additional sidewalls of the insert **304**. When folded down, the first set of flaps **310***a* and the second set of flaps **310***b* may be configured to provide additional rigidity to the insert **304** such that the insert **304** can maintain its rectangular shape when folded to its assembled configuration. Additionally, when folded down, an inner surface of the first set of flaps **310***a* of the first layer **311** may be configured to abut an outer surface of the second set of flaps **310***b* of the second layer **312**. In some embodiments, the sidewalls **319**, the first layer **311**, the second layer **312**, the third layer **318**, the fourth layer **320**, the first set of flaps **310***a*, and the second set of flaps **310***b* may be formed integrally using sustainable materials that can be recycled completely, such as paperboard material or other FSC-certified material.

[0116] In some embodiments, the insert **304** may be configured to be placed inside the carton **302** during assembly (as shown in FIGS. **5**C-**5**E). After the carton **302** is folded to its assembled configuration and the insert **304** is also folded to its assembled configuration, the insert **304** may be configured to be placed inside the cavity **317** (shown in FIG. **6**B) of the carton **302**. When placed inside the cavity **317** of the carton **302**, an inner surface of the first set of flaps **310***a* of the first layer **311** may be configured to abut an outer surface of the second set of flaps **310***b* of the second layer **312**, and an outer surface of the first set of flaps **310***a* of the first layer **311** may be configured to abut an inner surface of two opposite sidewalls **313** of the carton **304**. Once the insert **304** is placed inside the carton **302**, and the containers **308** are placed within the respective holes **306** of the insert **304**, the first flap **307***a* and the second flap **307***b* may be folded down (as shown in FIGS. **5**D and **5**E), and the first set of cavities **309** may be configured to protect the containers **308** therein from top load impact. Additionally, the back flap **315** extending from the top wall **314** of the carton **302** may be tucked into a front sidewall of the plurality of sidewalls **313** to close and secure the system **300** (as shown in FIG. **5**F). Accordingly, when fully assembled and packaged, the holes **306** in the first layer **311** and the second layer **312** of the insert **304** may be configured to secure the containers **308** in an upright position and protect the containers **308** from later impact, the first set of cavities **309** defined by the first flap **307***a* and the second flap **307***b* may protect the containers **308** from top load impact, and the third layer **318** that is raised from the bottom surface of the insert **304** and the bottom wall **316** of the carton **302** may protect the containers **308** from bottom load impact. Similar to the carton **302**, the insert **304** may comprise one or more perforations, scores, or cuts along the edges of the first layer **311**, the second layer **312**, the third layer **318**, the fourth layer **320**, the first set of flaps **310***a*, and the second set of flaps **310***b* so that the insert **304** can be easily folded from a storage configuration to an assembled configuration.

[0117] In some embodiments, the holes **306** in the insert **304** may be configured and sized to accommodate a neck of each respective container **308**, such as a neck of a glass vial. Additionally, or alternatively, the holes **306** may be configured and sized to accommodate a body of each respective container **308**, such as a body of a glass vial. By way of example, FIG. **8** illustrates a hole **306** in the first layer **311** of the insert **304** accommodating a first portion of the body of the container **308**, and a hole **306** in the second layer **312** of the insert **304** accommodating a second portion of the body of the container **308** below the first portion. However, in other embodiments, the holes **306** in the first layer **311** of the insert **304** may be configured to accommodate the necks of the containers **308**, while the holes **306** in the second layer **312** of the insert **304** may be configured to accommodate the bodies of the containers **308**. The holes **306** in the first layer **311** and the second layer **312** of the insert **304** may be configured to secure the containers **308** in their upright positions and protect the containers **308** from later impact or shifting. Although system **300** of FIGS. **5**A-**5**F, **6**A-**6**C, **7**A-**7**D, and **8** comprise six holes **306** in each of the first layer **311** and the second layer **312** of the insert **304** such that the insert **304** can accommodate up to six containers **308** therein, it is not limited to six holes **306**. For example, as will be discussed in more detail below, inserts may comprise any number of die cut holes, such as two, three, four, five, seven, eight, nine, ten, twelve, fifteen, twenty, twenty-five, or thirty holes, so that systems can accommodate various numbers of containers **308** therein.

[0118] In some embodiments, the size of the containers **308** may be indicated by ISO 8362-1. For example, if the containers **308** are glass vials, ISO 8362-1 may indicate the standard capacities of the containers **308**, the standard heights of the containers **308**, and/or the standard outer diameters of the bodies of the containers **308**. Accordingly, in some embodiments, the holes **306** may be sized to accommodate the containers **308** based on the standard dimensions indicated by ISO 8362-1. Additionally, or alternatively, the containers **308** packaged inside the system **300** may comprise glass vials, such as ISO 2R vials and 4 cc vials, ISO 6R vials, and/or ISO 10R and 10 cc vials. In some embodiments, the holes **306** in each insert **304** may be sized and configured to accommodate only one size of containers **308**, such as one of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. In other embodiments, the holes **306** in each insert **304** may be sized and configured to be able to accommodate multiple sizes of containers **308**, such as two or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. In yet another embodiment, the insert **304** may be made separately from the carton **302** and the insert may be interchangeably placed inside the carton **302**. Accordingly, one insert **304** with holes **306** sized to accommodate ISO 6R vials can be placed inside the carton **302** and can also be replaced with a different insert **304** with holes **306** sized to accommodate ISO 10R vials. Therefore, one carton **302** can be interchangeably used to package multiple sizes of containers **308**, depending on the insert **304** placed therein.

[0119] FIGS. **5**A-**5**F and **6**A-**6**C illustrate a rectangular carton **302** comprising four sidewalls **313**, a top wall **314**, and a bottom wall **316**. FIGS. **7**A-**7**D also illustrate a rectangular insert **304** comprising four sidewalls and configured to fit inside the cavity **317** of the carton **302**. However, the carton **302** and/or the insert **304** are not limited to rectangular shapes. For example, the carton **302** may comprise a cylindrical shape or any other geometric shapes, and the insert **304** may be shaped to fit inside a cavity in the carton **302**.

[0120] FIG. **4**A illustrates a plan view of a die cut of the carton **302**, and FIG. **4**B illustrates a plan view of a die cut of the insert **304**, in accordance with various embodiments of the present disclosure. The carton **302** and the insert **304** may be made from a unitary material. For example, the carton **302** and the insert **304** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the carton **302** and the insert **304** may be made of a combination of natural and synthetic materials. In other embodiments, the carton **302** and/or the insert **304** may be made from several pieces glued or otherwise secured together.

[0121] As seen in FIG. **4**A, the carton **302** includes a plurality of fold lines **330** diving the carton **302** into a plurality of walls and flaps. FIG. **4**A illustrates the carton **302** in its storage configuration where the carton **302** is substantially flat. The carton **302** can be folded into an assembled configuration. For example, when each of the plurality of fold lines **330** is folded, the carton **302** can be folded into the assembled configuration so as to form a box-shaped carton, as shown in FIGS. **5**B-**5**F, **6**B, and **6**C.

[0122] As shown in FIG. **4**A, the flaps **316***a-d* can be secured together to form the bottom wall **316** of the carton **302**. The carton **302** can be folded along each of the fold lines **330**, as shown in FIG. **4**A, and the flap **326** can be secured to the inner surface of the second sidewall **313***b* by glue, adhesive, or any other method of securing two surfaces together to form a box-shaped carton **302**. Moreover, the flap **325***a* can be secured to the inner surface of flap **307***a*, and the flap **325***b* can be secured to the inner surface of flap **307***b* by glue, adhesive, or any other method of securing two surfaces together to form the first and second flaps **307***a*, **307***b* that define the first set of cavities **309**, as shown in FIGS. **5**D and **5**E, for example.

[0123] Similarly, the insert **304** includes a plurality of fold lines **331** diving the insert **304** into a plurality of walls and flaps. FIG. **4**B illustrates the insert **304** in its storage configuration where the insert **304** is substantially flat. The insert **304** can be folded into an assembled configuration. For example, when each of the plurality of fold lines **331** is folded, the insert **304** can be folded into the assembled configuration so as to form the assembled insert **304**, as shown in FIG. **8**. The insert **304** can be folded along each of the fold lines **331**, the flap **327** can be secured to the inner surface of one of the sidewalls **319**, and the flap **328** can be secured to the inner surface of the other of the sidewalls **319** (as shown in FIG. **8**) to form the assembled insert **304**. The flaps **327**, **328** can be secured to the inner surfaces of the opposite sidewalls **319** by glue, adhesive, or other methods of securing two surfaces together to form the assembled insert **304**.

[0124] Referring now to FIGS. **10**A-**10**D and **11**, another exemplary system **400** for packaging a plurality of containers **408** is shown, in accordance with various embodiments of the present disclosure. Similar to the system **300** discussed above, system **400** may comprise a carton **402** and an insert **404** configured to be placed inside the carton **402**. It is understood that the components of system **400** are similar in detail to the components of system **300** discussed above. For the sake of brevity, the components of system **400** will not be discussed in substantial detail.

[0125] As shown in FIGS. **10**A and **10**B, insert **404** may comprise a plurality of holes **406** configured to accommodate a plurality of containers **408** therein. The insert **404** may comprise twelve holes **406**, each configured to accommodate a respective container **408**. In other embodiments, the insert **404** may comprise any number of holes **406**, such as fifteen, sixteen, twenty, twenty-five, or thirty holes. Similar to holes **306** of system **300**, holes **406** of system **400** are die cut and spaced away from each other and away from the sidewalls of the insert **404** and the sidewalls of the carton **402** to protect the containers **308** therein from lateral impact. The holes **406** may also be configured to laterally stabilize each individual container **408** packaged therein. Additionally, the holes **406** may be sized to accommodate multiple sizes of containers **408**. For example, the holes **406** may be sized to accommodate one or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials.

[0126] In some embodiments, the carton **402** and the insert **404** may be manufactured using sustainable materials that can be recycled completely, such as paperboard material or other FSC-certified material. As shown in FIGS. **10**A-**10**D, the carton **402** may comprise a rectangular carton having a plurality of sidewalls **413**, a top wall **414**, a bottom wall **416** (similar to bottom wall **216** of system **200** and bottom wall **316** of system **300**), and a back flap **415** extending from the top wall **414**. The carton **402** may further comprise a first flap **407***a*, extending from a first sidewall **413***a* of a plurality of sidewalls **414**, and a second flap **407***b*, extending from a second sidewall **413***b* opposite the first sidewall **413***a* and facing the first sidewall **413***a*. The first flap **407***a* and the second flap **407***b* may define a first set of cavities **409**. In some embodiments, the first set of cavities **409** may comprise rectangular air cavities, and, when folded down (as shown in FIG. **10**C), the first set of cavities **409** may be configured to protect the containers **408** therein from top load impact. The back flap **415** may be folded down as well such that the back flap **415** tucks into a front sidewall of the plurality of sidewalls **413** to close and secure the system **400** (as shown in FIG. **10**D). In some embodiments, the sidewalls **413**, the top wall **414**, the bottom wall **416**, the back flap **415**, the first flap **407***a*, and the second flap **407***b* may be formed integrally. In some embodiments, similar to the bottom wall **216** of system **200** and bottom wall **316** of system **300**, the bottom wall **416** may be pre-configured such that when the carton **402** is folded from a storage configuration to an assembled configuration, the bottom wall **416** may automatically straighten out into a flat surface to define the bottom surface of the carton **402**.

[0127] Although not shown in FIGS. **10**A-**10**D, system **400** may be folded from a storage configuration to an assembled configuration, similar to system **200** and system **300** discussed above. Accordingly, prior to assembly, both the carton **402** and the insert **404** of system **400** may be shipped and stored in a storage configuration. During assembly, the carton **402** and the insert **404** may be folded from a storage configuration to an assembled configuration. FIGS. **10**A-**10**D illustrate the system **400** when both the carton **402** and the insert **404** are folded to an assembled configuration. As shown in FIG. **10**A, in the assembled configuration, the carton **402** may comprise a cavity **417** defined by at least the sidewalls **413** and the bottom wall **416**. The cavity **417** may be configured to accommodate the insert **404**, as shown in FIG. **10**B.

[0128] FIG. **11** illustrates a side view of the insert **404** configured to be placed inside the carton **402**. As discussed above, the insert **404** may be folded from a storage configuration to an assembled configuration during assembly. As shown in FIG. **11**, once the insert **404** is folded to an assembled configuration, the insert may comprise at least two opposite sidewalls **419**, a first layer **411**, a second layer **412**, a third layer **418**, and a fourth layer **420**. The two sidewalls **419** may connect the first layer **411**, the second layer **412**, the third layer **418**, and the fourth layer **420**. The first layer **411** may define the top surface of the insert **404** and may comprise die cut holes **406** configured to accommodate at least a portion of a plurality of container **408**, such as glass vials. The second layer **412** may be positioned a first distance below the first layer **411** and may also comprise die cut holes **406** configured to accommodate at least a portion of the plurality of containers **408**, such as glass vials. The holes **406** in the second layer **412** may be vertically aligned with the holes **406** in the first layer **411** such that when the containers **408** are placed within respective holes **406**, each container **408** will fit through the respective hole **406** in the first layer **411** as well as the respective hole **406** in the second layer **412**. The third layer **418** may be positioned a second distance below the second layer **412** and may be configured to support a bottom surface of each container **408**. The fourth layer **420** may be positioned a third distance below the third layer **418** and may define the bottom surface of the insert **404**. The third layer **318** may be positioned a third distance above the fourth layer **420** to protect the containers **408** therein from bottom load impact. In some embodiments, the first distance, the second distance, and the third distance may be the same such that the first layer **411**, the second layer **412**, the third layer **418**, and the fourth layer **420** are all spaced equidistant from each other. In other embodiments, the first distance may be different from at least one of the second distance or the third distance. For example, the distance between the third layer **418** and the fourth layer **420** may be greater than the distance between the third layer **418** and the second layer **412**.

[0129] As illustrated in FIG. **11**, the insert **404** may further comprise a vertical rib support **421** extending between the third layer **418** and the fourth layer **420**. The vertical rib support **421** may be formed integrally with the insert **404** and may be configured to maintain the third distance between the third layer **418** and the fourth layer **420**. The vertical rib support **421** may ensure that the containers **408** are protected from bottom load impact and may be configured to help the third layer **418** in supporting the weight of the containers **408**. In some embodiments, the first layer **411** of the insert **404** may comprise a first set of flaps **410***a* extending from opposite sides of the first layer **411**, and the second layer **412** may comprise a second set of flaps **410***b* extending from opposite sides the second layer **412**. The first set of flaps **410***a* and the second set of flaps **410***b* may be configured to fold down to define two additional sidewalls of the insert **404**. When folded down, the first set of flaps **410***a* and the second set of flaps **410***b* may be configured to provide additional rigidity to the insert **404** such that the insert **404** can maintain its rectangular shape when folded to its assembled configuration. Additionally, when folded down, an inner surface of the first set of flaps **410***a* of the first layer **411** may be configured to abut an outer surface of the second set of flaps **410***b* of the second layer **412**. In some embodiments, the sidewalls **419**, the first layer **411**, the second layer **412**, the third layer **418**, the fourth layer **420**, the first set of flaps **410***a*, and the second set of flaps **410***b* may be formed integrally using sustainable materials that can be recycled completely, such as paperboard material or other FSC-certified material.

[0130] FIG. **11** illustrates a hole **406** in the first layer **411** of the insert **404** accommodating a first portion of the body of the container **408**, and a hole **406** in the second layer **412** of the insert **404** accommodating a second portion of the body of the container **408** below the first portion. However, in other embodiments, the holes **406** in the first layer **411** of the insert **404** may be configured to accommodate the necks of the containers **408**, while the holes **406** in the second layer **412** of the insert **404** may be configured to accommodate the bodies of the containers **408**. The holes **406** may be configured to secure the containers **408** in their upright positions and protect the containers **408** from later impact or shifting.

[0131] In some embodiments, the size of the containers **408** may be indicated by ISO 8362-1. For example, when the containers **408** are glass vials, ISO 8362-1 may indicate the standard capacities of the containers **408**, the standard heights of the containers **408**, and/or the standard outer diameters of the bodies of the containers **408**. The containers **408** packaged inside the system **400** may comprise glass vials, such as ISO 2R vials and 4 cc vials, ISO 6R vials, and/or ISO 10R and 10 cc vials. In some embodiments, the holes **406** in each insert **404** may be sized and configured to accommodate only one size of containers **408**, such as one of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. In other embodiments, the holes **406** may be sized and configured to be able to accommodate multiple sizes of containers **408**, such as two or more of ISO 2R vials and 4 cc vials, ISO 6R vials, and ISO 10R and 10 cc vials. In yet another embodiment, the insert **404** may be made separately from the carton **402** and the insert may be interchangeably placed inside the carton **402**. Accordingly, one insert **404** with holes **406** sized to accommodate ISO 6R vials can be placed inside the carton **402** and can also be replaced with a different insert **404** with holes **406** sized to accommodate ISO 10R vials. Therefore, one carton **402** can be interchangeably used to package multiple sizes of containers **408**, depending on the insert **404** placed therein.

[0132] FIG. **9**A illustrates a plan view of a die cut of the carton **402**, and FIG. **9**B illustrates a plan view of a die cut of the insert **404**, in accordance with various embodiments of the present disclosure. The carton **402** and the insert **404** may be made from a unitary material. For example, the carton **402** and the insert **404** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the carton **402** and the insert **404** may be made of a combination of natural and synthetic materials. In other embodiments, the carton **402** and/or the insert **404** may be made from several pieces glued or otherwise secured together.

[0133] As seen in FIG. **9**A, the carton **402** includes a plurality of fold lines **430** diving the carton **402** into a plurality of walls and flaps. FIG. **9**A illustrates the carton **402** in its storage configuration where the carton **402** is substantially flat. The carton **402** can be folded into an assembled configuration. For example, when each of the plurality of fold lines **430** is folded, the carton **402** can be folded into the assembled configuration so as to form a box-shaped carton, as shown in FIGS. **10**A-**10**D.

[0134] As shown in FIG. **9**A, the flaps **416***a-d* can be secured together to form the bottom wall **416** of the carton **402**. The carton **402** can be folded along each of the fold lines **430**, as shown in FIG. **9**A, and the flap **426** can be secured to the inner surface of the second sidewall **413***b* by glue, adhesive, or any other method of securing two surfaces together to form a box-shaped carton **402**. Moreover, the flap **425***a* can be secured to the inner surface of flap **407***a*, and the flap **425***b* can be secured to the inner surface of flap **407***b* by glue, adhesive, or any other method of securing two surfaces together to form the first and second flaps **407***a*, **407***b* that define the first set of cavities **409**, as shown in FIG. **10**C, for example.

[0135] Similarly, the insert **404** includes a plurality of fold lines **431** diving the insert **404** into a plurality of walls and flaps. FIG. **9**B illustrates the insert **404** in its storage configuration where the insert **404** is substantially flat. The insert **404** can be folded into an assembled configuration. For example, when each of the plurality of fold lines **431** is folded, the insert **404** can be folded into the assembled configuration so as to form the assembled insert **404**, as shown in FIG. **11**. The insert **404** can be folded along each of the fold lines **431**, the flap **427** can be secured to the inner surface of one of the sidewalls **419**, and the flap **428** can be secured to the inner surface of the other of the sidewalls **419** (as shown in FIG. **11**) to form the assembled insert **404**. The flaps **427**, **428** can be secured to the inner surfaces of the opposite sidewalls **419** by glue, adhesive, or other methods of securing two surfaces together to form the assembled insert **404**.

[0136] FIG. **12**A is a plan view of a die cut of a carton **502** for another exemplary system for packaging a container, in accordance with various embodiments of the present disclosure. The carton **502** may be made from a unitary material. For example, the carton **502** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the carton **502** may be made of a combination of natural and synthetic materials. In other embodiments, the carton **502** can be made from several pieces glued or otherwise secured together. The carton **502** includes a plurality of fold lines **530**, dividing the carton **502** into a plurality of walls and flaps. FIG. **12**A illustrates the carton **502** in a storage configuration where the carton **502** is substantially or completely flat. The carton **502** is erectable or foldable from a storage configuration where the carton **502** is substantially flat (shown in FIG. **12**A) to an assembled configuration (shown in FIG. **13**A). For example, when each of the plurality of fold lines **530** is folded, the carton **502** is erectable into the assembled configuration so as to form a box-shaped carton, as shown in FIG. **13**A.

[0137] Referring to FIGS. **12**A, **13**A, and **13**B, the carton **502** includes a bottom box **503** (shown in FIG. **13**A) having a first bottom wall **516** (shown in FIG. **13**B) made from several flaps **516***a-d* (shown in FIG. **12**A), a first sidewall **513***b*, a second sidewall **513***d*, a first back wall **513***a*, and a first front wall **513***c*. The flaps **516***a-d* can be secured to each other via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons to form the first bottom wall **516** when the carton **502** is assembled. The first back wall **513***a* includes a flap **526** that can be glued, for example, to an inner surface of the second sidewall **513***d* to form the bottom box **503**. In some embodiments, the first back wall **513***a* can be secured to the second sidewall **513***d* via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons. As a result, the first sidewall **513***b*, the second sidewall **513***d*, the first back wall

513*a,* the first front wall **513***c,* and the first bottom wall **516** are all secured together to form the bottom box **503**.

[0138] Furthermore, the carton **502** includes a lid **514**, hingedly connected or coupled to the first back wall **513***a*. In some embodiments, the lid **514** is an overlid that is movable between an open position and a closed position and includes a flap **515**. When the carton **502** is assembled, the lid **514** can be moved from an open position where the internal cavity of the bottom box **503** is accessible to a closed position where the internal cavity of the bottom box **503** is not accessible. Additionally, the flap **515** of the lid **514** can be folded and tucked behind the first front wall **513***c* to secure the carton **502** in a closed position and prevent accidental opening of the carton **502**.

[0139] Moreover, the carton **502** may include a first flap **507***a* and a second flap **507***b* extending from the first sidewall **513***b* and the second sidewall **513***d,* respectively, of the bottom box **503**. As shown in FIGS. **12**A and **13**A, the second sidewall **513***d* is positioned opposite the first sidewall **513***b* and faces the first sidewall **513***b*. In other words, the first sidewall **513***b* and the second sidewall **513***d* are opposing walls of the bottom box **503**. The first flap **507***a* may include a tab **525***a* at an end thereof, and the second flap **507***b* may also include a tab **525***b* at an end thereof. When the carton **502** is folded along the plurality of fold lines **530** into the assembled configuration, the tabs **525***a* and **525***b* can be secured to the inner surfaces of the first sidewall **513***b* and the second sidewall **513***d,* respectively, such that the first flap **507***a* and the second flap **507***b* define a set of cavities **509** (shown in FIG. **13**A). In some embodiments, the tabs **525***a* and **525***b* can be glued to the inner surfaces of the first sidewall **513***b* and the second sidewall **513***d,* respectively. In other embodiments, the tabs **525***a* and **525***b* can be secured to the inner surfaces of the first sidewall **513***b* and the second sidewall **513***d,* respectively, via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons. When the carton **502** is assembled, the set of cavities **509** defined by the first flap **507***a* and the second flap **507***b* may comprise rectangular air cavities. In some embodiments, the set of cavities **509** may be configured to protect one or more container disposed within the bottom box **503** from top load impact.

[0140] In some embodiments, the bottom box **503** of the carton **502** includes a release tab **562**. As shown in FIG. **12**A, for example, the first front wall **513***c* includes a perforated line **560** about a release tab **562** adjacent a bottom edge **564** of the first front wall **513***c*. The release tab **562** is enclosed by a perforated line **560** and the bottom edge **564**. The release tab **562** includes a first kiss cut **566**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on at least a portion of the release tab **562**. The first kiss cut **566** may improve securement between the release tab **562** and the adhesive. The perforated line **560** allows the release tab **562** to be detached from the first front wall **513***c* when sufficient force is applied to the release tab **562**. As shown in FIG. **12**A, the release tab **562** is partially rectangular with a rounded side, however, the release tab **562** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0141] Additionally, or alternatively, as shown in FIG. **12**A, the lid **514** of the bottom box **503** may also include a perforated line **580** about a release tab **582** adjacent an edge **584** between the lid **514** and the flap **515** of the lid. The release tab **582** is enclosed by the perforated line **580** and the edge **584**. The perforated line **580** allows the release tab **582** to be detached from the lid **514** when sufficient force is applied to the release tab **582**. As shown in FIG. **12**A, the release tab **582** is partially rectangular with a rounded side, however, the release tab **582** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0142] In some embodiments, when the carton **502** is folded into the assembled configuration and the lid **540** is placed over the bottom box **503**, the position of the release tab **582** on the lid **514** may align with the position of the release tab **562** on the first front wall **513***c* of the bottom box **503** such that the release tabs **582** and **562** are positioned perpendicular to each other. The edge **584** and the edge **564** that partially define the release tabs **582** and **562**, respectively, may align with each other and be positioned adjacent to each other. In some embodiments, the release tab **582** and/or the release tab **562** may be a push tab that detaches the release tab from the lid **514** and/or the first front wall **513***c* of the bottom box **503**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on the release tab **582** and/or the release tab **562**. Accordingly, the release tab **582** and/or the release tab **562** may include a kiss cut **568**, **566** to improve securement between the release tab and the adhesive.

[0143] The release tab **582** and the release tab **562** may be partially circumscribed by perforated lines **580** and **560**, respectively. The perforated lines **580** and **560** may each consist of a series of cuts. The kiss cuts **568** and **566** may each be a partial cut through the release tabs **582** and **562**, respectively, that improve the bonding between the release tabs **582**, **562** and a tamper evident seal by increasing the surface area the tamper evident seal can adhere to. The tamper evident seal can, for example, be an adhesive seal such as an adhesive or closure label to seal the carton **502**. For example, an adhesive can be placed over the kiss cut **568** on the lid **514** and the kiss cut **566** on the first front wall **513***c* of the bottom box **503** to seal the carton **502** in a closed position. The adhesive may act as a tamper evident seal. That is, any attempt to pull the adhesive off the carton **502** will result in a visible damage to the carton **502** and/or the adhesive. For example, when the lid **514** is in the closed position and the carton **502** is sealed, the release tab **582** and the release tab **562** may be secured to the lid **514** and the first front wall **513***c,* respectively. When a person attempts to pull the adhesive off the carton **502** and the lid **514** is in the open position, one or both of the release tabs **582** and **562** may be separated along the respective perforated line(s) **580**, **560** from the lid **514** and/or the first front wall **513***c*. In some embodiments, when a person attempts to pull the adhesive off the carton **502**, one of the release tabs **582** and **562** may be separated along a respective one of the perforated lines **580**, **560** from one of the lid **514** and the first front wall **513***c* and may be secured to the other one of the lid **514** and the first front wall **513***c* via the adhesive.

[0144] FIG. **12**B is a plan view of a die cut of an insert **504** configured to be disposed within the bottom box **503** of the carton **502**, in accordance with various embodiments of the present disclosure. The insert **504** may be made from a unitary material. For example, the insert **504** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the insert **504** may be made of a combination of natural and synthetic materials. In other embodiments, the insert **504** can be made from several pieces glued or otherwise secured together. As shown in FIG. **12**B, the insert **504** includes a plurality of fold lines **531**, dividing the insert **504** into a plurality of walls **510***a-g*. FIG. **12**B illustrates the insert **504** in a flat configuration. Similar to the carton **502**, the insert **504** is erectable from a storage configuration where the insert **504** is substantially flat (shown in FIG. **12**B) to an assembled configuration (shown in FIG. **13**C). For example, when each of the plurality of fold lines **531** is folded, the insert **504** is erectable into the assembled configuration shown in FIG. **13**C.

[0145] As shown in FIGS. **12**B, **13**B, and **13**C, the insert **504** includes a top wall **510***b*, a second bottom wall **510***d*, a second front wall **510***c*, a second back wall **510***a*, and a middle wall **510***f* positioned between the top wall **510***b* and the second bottom wall **510***d*. Once each of the walls **510***a-g* are folded, the wall **510***e* can be secured to the inner surface of the second back wall **510***a*, and the wall **510***g* can be secured to the inner surface of the second front wall **510***c* to secure the insert **504** in the assembled configuration. In some embodiments, the walls **510***e*, **510***g* can be glued to the inner surfaces of the second back wall **510***a* and the second front wall **510***c*, respectively. In other embodiments, the walls **510***e*, **510***g* can be secured to the inner surfaces of the second back wall **510***a* and the second front wall **510***c*, respectively, via interlocking slits, other adhesives, or other known methods of assembling boxes and inserts.

[0146] When the insert **504** is folded from its storage configuration where the insert **504** is substantially flat into its assembled configuration, the insert **504** may be configured to be disposed within the cavity **505** of the bottom box **503** of the carton **502**, as shown in FIG. **13**B. When the insert **504** is disposed within the cavity **505** of the bottom box **503** of the carton **502**, the second bottom wall **510***d* is configured to abut the first bottom wall **516**, the second back wall **510***a* is configured to abut the first back wall **513***a*, and the second front wall **510***c* is configured to abut the first front wall **513***c*. In some embodiments, the insert **504** may be removably disposed within the cavity **505** of the bottom box **503**. In other embodiments, the insert **504** may be secured to the cavity **505** of the bottom box **503** via glue, other adhesives, or other known methods of assembling boxes, cartons, and trays.

[0147] The top wall **510***b* of the insert **504** may define a cavity **506** dimensioned, sized, and configured to house a container, such as container **508**. When the container **508** is disposed within the cavity **506** of the insert **504**, the bottom surface **508***a* of the container **508** may abut the middle wall **510***f* of the insert **504**. Accordingly, the middle wall **510***f* of the insert **504** may be configured to support the container **508** within the insert **504**. In some embodiments, the cavity **506** may be dimensioned, sized, and configured to accommodate a neck of the container **508**, such as a neck of a glass vial. Additionally, or alternatively, as shown in FIG. **13**B, the cavity **506** may be dimensioned, sized, and configured to accommodate a body of the container **508**, such as a body of a glass vial. In some embodiments, the size of the container **508** may be indicated by ISO 8362-1. For example, the container **508** may include a drug vial, and the drug vial may be one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In some embodiments, the drug vial may be filled or pre-filled with a drug. The drug may comprise, for example, tarlatamab (AMG 757) or another product containing a half-life extended (HLE) anti-delta-like ligand 3 (DLL3) x anti-CD3 bispecific T cell engager construct.

[0148] In some embodiments, when the insert **504** is disposed within the bottom box **503** of the carton **502**, one or more leaflets **540** (shown in FIGS. **13**A and **13**B) may be disposed within the cavity **505** of the bottom box **503**. One or more leaflets **540** may include, for example, a user manual, an instruction manual, a patient information leaflet, consumer medicine information, or any other document that provides information about a drug or medication contained inside the container **508** or delivery of such drug or medication.

[0149] FIG. **14**A is a plan view of a die cut of a carton **602** for another exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure. The carton **602** may be made from a unitary material. For example, the carton **602** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the carton **602** may be made of a combination of natural and synthetic materials. In other embodiments, the carton **602** can be made from several pieces glued or otherwise secured together. The carton **602** includes a plurality of fold lines **630**, dividing the carton **602** into a plurality of walls and flaps. FIG. **14**A illustrates the carton **602** in a storage configuration where the carton **602** is substantially or completely flat. The carton **602** is erectable or foldable from a storage configuration where the carton **602** is substantially flat (shown in FIG. **14**A) to an assembled configuration (shown in FIG. **15**A). For example, when each of the plurality of fold lines **630** is folded, the carton **602** is erectable into the assembled configuration so as to form a box-shaped carton, as shown in FIG. **15**A.

[0150] Referring to FIGS. **14**A, **15**A, **15**B, the carton **602** includes a bottom box **603** (shown in FIG. **15**A) having a first bottom wall **616** made from several flaps **616***a-d*, a first sidewall **613***b*, a second sidewall **613***d*, a first back wall **613***a*, and a first front wall **613***c*. The flaps **616***a-d* can be secured to each other via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons to form the first bottom wall **616** when the carton **602** is assembled. The first back wall **613***a* includes a flap **626** that can be glued, for example, to an inner surface of the second sidewall **613***d* to form the bottom box **603**. In some embodiments, the first back wall **613***a* can be secured to the second sidewall **613***d* via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons. As a result, the first sidewall **613***b*, the second sidewall **613***d*, the first back wall **613***a*, the first front wall **613***c*, and the first bottom wall **616** are all secured together to form the bottom box **603**.

[0151] Furthermore, the carton **602** includes a lid **614**, hingedly connected or coupled to the first back wall **613***a*. In some embodiments, the lid **614** is an overlid that is movable between an open position and a closed position and includes a flap **615**. When the carton **602** is assembled, the lid **614** can be moved from an open position where the internal cavity of the bottom box **603** is accessible to a closed position where the internal cavity of the bottom box **603** is not accessible. Additionally, the flap **615** of the lid **614** can be folded and tucked behind the first front wall **613***c* to secure the carton **602** in a closed position and prevent accidental opening of the carton **602**.

[0152] Moreover, the carton **602** may include a first flap **607***a* and a second flap **607***b* extending from the first sidewall **613***b* and the second sidewall **613***d*, respectively, of the bottom box **603**. The second sidewall **613***d* is positioned opposite the first sidewall **613***b* and faces the first sidewall **613***b*. In other words, the first sidewall **613***b* and the second sidewall **613***d* are opposing walls of the bottom box **603**. The first flap **607***a* may include a tab **625***a* at an end thereof, and the second flap **607***b* may also include a tab **625***b* at an end thereof. When the carton **602** is folded along the plurality of fold lines **630** into the assembled configuration, the tabs **625***a* and **625***b* can be secured to the inner surfaces of the first sidewall **613***b* and the second sidewall **613***d*, respectively, such that the first flap **607***a* and the second flap **607***b* define a set of cavities **609** (shown in FIG. **15**A). In some embodiments, the tabs **625***a* and **625***b* can be glued to the inner surfaces of the first sidewall **613***b* and the second sidewall **613***d*, respectively. In other embodiments, the tabs **625***a* and **625***b* can be secured to the inner surfaces of the first sidewall **613***b* and the second sidewall **613***d*, respectively, via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons. When the carton **602** is assembled, the set of cavities **609** defined by the first flap **607***a* and the second flap **607***b* may comprise rectangular air cavities. In some embodiments, the set of cavities **609** may be configured to protect one or more container disposed within the bottom box **603** from top load impact.

[0153] In some embodiments, the bottom box **603** of the carton **602** includes a release tab **662**. As shown in FIG. **14**A, for example, the first front wall **613***c* includes a perforated line **660** about a release tab **662** adjacent a bottom edge **664** of the first front wall **613***c*. The release tab **662** is enclosed by a perforated line **660** and the bottom edge **664**. The release tab **662** includes a first kiss cut **666**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on at least a portion of the release tab **662**. The first kiss cut **666** may improve securement between the release tab **662** and the adhesive. The perforated line **660** allows the release tab **662** to be detached from the first front wall **613***c* when sufficient force is applied to the release tab **662**. As shown in FIG. **14**A, the release tab **662** is partially rectangular with a rounded side, however, the release tab **662** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0154] Additionally, or alternatively, as shown in FIG. **14**A, the lid **614** of the bottom box **603** may also include a perforated line **680** about a release tab **682** adjacent an edge **684** between the lid **614** and the flap **615** of the lid. The release tab **682** is enclosed by the perforated line **680** and the edge **684**. The perforated line **680** allows the release tab **682** to be detached from the lid **614** when sufficient force is applied to the release tab **682**. As shown in FIG. **14**A, the release tab **682** is partially rectangular with a rounded side, however, the release tab **682** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0155] In some embodiments, when the carton **602** is folded into the assembled configuration and the lid **640** is placed over the bottom box **603**, the position of the release tab **682** on the lid **614** may align with the position of the release tab **662** on the first front wall **613***c* of the bottom box **603** such that the release tabs **682** and **662** are positioned perpendicular to each other. The edge **684** and the edge **664** that partially define the release tabs **682** and **662**, respectively, may align with each other and be positioned adjacent to each other. In some embodiments, the release tab **682** and/or the release tab **662** may be a push tab that detaches the release tab from the lid **614** and/or the first front wall **613***c* of the bottom box **603**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on the release tab **682** and/or the release tab **662**. Accordingly, the release tab **682** and/or the release tab **662** may include a kiss cut **668**, **666** to improve securement between the release tab and the adhesive.

[0156] The release tab **682** and the release tab **662** may be partially circumscribed by perforated lines **680** and **660**, respectively. The perforated lines **680** and **660** may each consist of a series of cuts. The kiss cuts **668** and **666** may each be a partial cut through the release tabs **682** and **662**, respectively, that improve the bonding between the release tabs **682**, **662** and a tamper evident seal by increasing the surface area the tamper evident seal can adhere to. The tamper evident seal can, for example, be an adhesive seal such as an adhesive or closure label to seal the carton **602**. For example, an adhesive can be placed over the kiss cut **668** on the lid **614** and the kiss cut **666** on the first front wall **613***c* of the bottom box **603** to seal the carton **602** in a closed position. The adhesive may act as a tamper evident seal. That is, any attempt to pull the adhesive off the carton **602** will result in a visible damage to the carton **602** and/or the adhesive. For example, when the lid **614** is in the closed position and the carton **602** is sealed, the release tab **682** and the release tab **662** may be secured to the lid **614** and the first front wall **613***c*, respectively. When a person attempts to pull the adhesive off the carton **602** and the lid **614** is in the open position, one or both of the release tab **682** and **662** may be separated along the respective perforated line(s) **680**, **660** from the lid **614** and/or the first front wall **613***c*. In some embodiments, when a person attempts to pull the adhesive off the carton **602**, one of the release tabs **682** and **662** may be separated along a respective one of the perforated lines **680**, **660** from one of the lid **614** and the first front wall **613***c* and may be secured to the other one of the lid **614** and the first front wall **613***c* via the adhesive.

[0157] FIG. **14**B is a plan view of a die cut of an insert **604** configured to be disposed within the bottom box **603** of the carton **602**, in accordance with various embodiments of the present disclosure. The insert **604** may be made from a unitary material. For example, the insert **604** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the insert **604** may be made of a combination of natural and synthetic materials. In other embodiments, the insert **604** can be made from several pieces glued or otherwise secured together. As shown in FIG. **14**B, the insert **604** includes a plurality of fold lines **631**, dividing the insert **604** into a plurality of walls **610***a-h*. FIG. **14**B illustrates the insert **604** in a flat configuration. Similar to the carton **602**, the insert **604** is erectable from a storage configuration where the insert **604** is substantially flat (shown in FIG. **14**B) to an assembled configuration (shown in FIG. **15**B). For example, when each of the plurality of fold lines **631** is folded, the insert **604** is erectable into the assembled configuration shown in FIG. **15**B.

[0158] The insert **604** includes a top wall **610***b*, a second bottom wall **610***d*, a second front wall **610***c*, a second back wall **610***a*, a first middle wall **610***f* positioned between the top wall **610***b* and the second bottom wall **610***d*, and a second middle wall **610***h* positioned between the top wall **610***b* and the second bottom wall **610***d*. The height of the first middle wall **610***f* and the height of the second middle wall **610***h* may be offset. For example, when assembled, the first middle wall **610***f* may be positioned higher than the second middle wall **610***h*. Once each of the walls **610***a-h* are folded, the wall **610***e* can be secured to the inner surface of the second back wall **610***a*, and the wall **610***g* can be secured to the inner surface of the second front wall **610***c* to secure the insert **604** in the assembled configuration. In some embodiments, the walls **610***e*, **610***g* can be glued to the inner surfaces of the second back wall **610***a* and the second front wall **610***c*, respectively. In other embodiments, the walls **610***e*, **610***g* can be secured to the inner surfaces of the second back wall **610***a* and the second front wall **610***c*, respectively, via interlocking slits, other adhesives, or other known methods of assembling boxes and inserts.

[0159] When the insert **604** is folded from its storage configuration where the insert **604** is substantially flat into its assembled configuration, the insert **604** may be configured to be disposed within the cavity **605** of the bottom box **603** of the carton **602**, as shown in FIG. **15**B. When the insert **604** is disposed within the cavity **605** of the bottom box **603** of the carton **602**, the second bottom wall **610***d* is configured to abut the first bottom wall **616**, the second back wall **610***a* is configured to abut the first back wall **613***a*, and the second front wall **610***c* is configured to abut the first front wall **613***c*. In some embodiments, the insert **604** may be removably disposed within the cavity **605** of the bottom box **603**. In other embodiments, the insert **604** may be secured to the cavity **605** of the bottom box **603** via glue, other adhesives, or other known methods of assembling boxes, cartons, and trays.

[0160] The top wall **610***b* of the insert **604** may define a first cavity **606***a* dimensioned, sized, and configured to house a first container, such as container **608***a*, and a second cavity **606***b* dimensioned, sized, and configured to house a second container, such as container **608***b*. When the first and second containers **608***a*, **608***b* are disposed within the first and second cavities **606***a*, **606***b*, respectively, the bottom surface of the first container **608***a* may abut the first middle wall **610***f* and the bottom surface of the second container **608***b* may abut the second middle wall **610***h*. Accordingly, the first and second middle walls **610***f*, **610***h* of the insert **604** may be configured to support the first and second containers **608***a*, **608***b*, respectively, within the insert **604**.

[0161] As shown in FIG. **14**B, the diameter of the first cavity **606***a* may be smaller than the diameter of the second cavity **606***b* to accommodate containers of different sizes within a single packaging. For example, the diameter of the first container **608***a* may be smaller than the diameter of the second container **608***b*. The first container **608***a* may be one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial, and the second container **608***b* may be another one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In one embodiment, for instance, the first container **608***a* may be an ISO 20R vial, and the second container **608***b* may be an ISO 50R vial. Because the dimensions (e.g., diameters, heights, etc.) of the first and second containers **608***a*, **608***b* may be different, the heights of the first middle wall **610***f* and the second middle wall **610***h* may be offset, as discussed above. For instance, the first middle wall **610***f* may be positioned higher in the insert **604** than the second middle wall **610***h*. Accordingly, even though the height of the first container **608***a* may be shorter than the height of the second container **608***b*, the first and second containers **608***a*, **608***b* may still be positioned at the same or similar height within the insert **604** because the first middle wall **610***f* and the second middle wall **610***h* supporting the first and second containers **608***a*, **608***b*, respectively, are offset.

[0162] In some embodiments, the cavities **606***a*, **606***b* may each be dimensioned, sized, and configured to accommodate a neck of the containers **608***a*, **608***b*, respectively. Additionally, or alternatively, as shown in FIGS. **15**A and **15**B, the cavities **606***a*, **606***b* may each be dimensioned, sized, and configured to accommodate a body of the containers **608***a*, **608***b*, respectively. In some embodiments, the first container **608***a* and the second container **608***b* may include drug vials that are filled or pre-filled with a drug. The drug may comprise, for example, tarlatamab (AMG 757) or another product containing a half-life extended (HLE) anti-delta-like ligand 3 (DLL3) x anti-CD3 bispecific T cell engager construct. In some embodiments, the first container **608***a* and the second container **608***b* may include drug vials that are filled or pre-filled with different drugs.

[0163] In some embodiments, when the insert **604** is disposed within the bottom box **603** of the carton **602**, one or more leaflets **640** (shown in FIG. **15**A) may be disposed within the cavity **605** (shown in FIG. **15**B) of the bottom box **603**. One or more leaflets **640** may include, for example, a user manual, an instruction manual, a patient information leaflet, consumer medicine information, or any other document that provides information about drugs or medications contained inside the containers **608***a*, **608***b* or delivery of such drugs or medications.

[0164] FIG. **16**A is a plan view of a die cut of a carton **702** for another exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure. The carton **702** may be made from a unitary material. For example, the carton **702** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the carton **702** may be made of a combination of natural and synthetic materials. In other embodiments, the carton **702** can be made from several pieces glued or otherwise secured together. The carton **702** includes a plurality of fold lines **730**, dividing the carton **702** into a plurality of walls and flaps. FIG. **16**A illustrates the carton **702** in a storage configuration where the carton **702** is substantially or completely flat. The carton **702** is erectable or foldable from a storage configuration where the carton **702** is substantially flat (shown in FIG. **16**A) to an assembled configuration (shown in FIGS. **17**A and **17**B). For example, when each of the plurality of fold lines **730** is folded, the carton **702** is erectable into the assembled configuration so as to form a box-shaped carton, as shown in FIGS. **17**A and **17**B.

[0165] Referring to FIGS. **16**A, **17**A, and **17**B, the carton **702** includes a bottom box **703** (shown in FIGS. **17**A and **17**B) having a first bottom wall **716** made from several flaps **716***a-d*, a first sidewall **713***b*, a second sidewall **713***d*, a first back wall **713***a*, and a first front wall **713***c*. The flaps **716***a-d* can be secured to each other via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons to form the first bottom wall **716** when the carton **702** is assembled. The first back wall **713***a* includes a flap **726** that can be glued, for example, to an inner surface of the second sidewall **713***d* to form the bottom box **703**. In some embodiments, the first back wall **713***a* can be secured to the second sidewall **713***d* via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons. As a result, the first sidewall **713***b*, the second sidewall **713***d*, the first back wall **713***a*, the first front wall **713***c*, and the first bottom wall **716** are all secured together to form the bottom box **703**.

[0166] Furthermore, the carton **702** includes a lid **714**, hingedly connected or coupled to the first back wall **713***a*. In some embodiments, the lid **714** is an overlid that is movable between an open position and a closed position and includes a flap **715**. When the carton **702** is assembled, the lid **714** can be moved from an open position where the internal cavity of the bottom box **703** is accessible to a closed position where the internal cavity of the bottom box **703** is not accessible. Additionally, the flap **715** of the lid **714** can be folded and tucked behind the first front wall **713***c* to secure the carton **702** in a closed position and prevent accidental opening of the carton **702**.

[0167] Moreover, the carton **702** may include a first flap **707***a* and a second flap **707***b* extending from the first sidewall **713***b* and the second sidewall **713***d*, respectively, of the bottom box **703**. The second sidewall **713***d* is positioned opposite the first sidewall **713***b* and faces the first sidewall **713***b*. In other words, the first sidewall **713***b* and the second sidewall **713***d* are opposing walls of the bottom box **703**.

[0168] In some embodiments, the bottom box **703** of the carton **702** includes a release tab **762**. As shown in FIG. **16**A, for example, the first front wall **713***c* includes a perforated line **760** about a release tab **762** adjacent a bottom edge **764** of the first front wall **713***c*. The release tab **762** is enclosed by a perforated line **760** and the bottom edge **764**. The release tab **762** includes a first kiss cut **766**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on at least a portion of the release tab **762**. The first kiss cut **766** may improve securement between the release tab **762** and the adhesive. The perforated line **760** allows the release tab **762** to be detached from the first front wall **713***c* when sufficient force is applied to the release tab **762**. As shown in FIG. **16**A, the release tab **762** is partially rectangular with a rounded side, however, the release tab **762** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0169] Additionally, or alternatively, as shown in FIG. **16**A, the lid **714** of the bottom box **703** may also include a perforated line **780** about a release tab **782** adjacent an edge **784** between the lid **714** and the flap **715** of the lid. The release tab **782** is enclosed by the perforated line **780** and the edge **784**. The perforated line **780** allows the release tab **782** to be detached from the lid **714** when sufficient force is applied to the release tab **782**. As shown in FIG. **16**A, the release tab **782** is partially rectangular with a rounded side, however, the release tab **782** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0170] In some embodiments, when the carton **702** is folded into the assembled configuration and the lid **740** is placed over the bottom box **703**, the position of the release tab **782** on the lid **714** may align with the position of the release tab **762** on the first front wall **713***c* of the bottom box **703** such that the release tabs **782** and **762** are positioned perpendicular to each other. The edge **784** and the edge **764** that partially define the release tabs **782** and **762**, respectively, may align with each other and be positioned adjacent to each other. In some embodiments, the release tab **782** and/or the release tab **762** may be a push tab that detaches the release tab from the lid **714** and/or the first front wall **713***c* of the bottom box **703**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on the release tab **782** and/or the release tab **762**. Accordingly, the release tab **782** and/or the release tab **762** may include a kiss cut **768**, **766** to improve securement between the release tab and the adhesive.

[0171] The release tab **782** and the release tab **762** may be partially circumscribed by perforated lines **780** and **760**, respectively. The perforated lines **780** and **760** may each consist of a series of cuts. The kiss cuts **768** and **766** may each be a partial cut through the release tabs **782** and **762**, respectively, that improve the bonding between the release tabs **782**, **762** and a tamper evident seal by increasing the surface area the tamper evident seal can adhere to. The tamper evident seal can, for example, be an adhesive seal such as an adhesive or closure label to seal the carton **702**. For example, an adhesive can be placed over the kiss cut **768** on the lid **714** and the kiss cut **766** on the first front wall **713***c* of the bottom box **703** to seal the carton **702** in a closed position. The adhesive may act as a tamper evident seal. That is, any attempt to pull the adhesive off the carton **702** will result in a visible damage to the carton **702** and/or the adhesive. For example, when the lid **714** is in the closed position and the carton **602** is sealed, the release tab **782** and the release tab **762** may be secured to the lid **714** and the first front wall **713***c*, respectively. When a person attempts to pull the adhesive off the carton **702** and the lid **714** is in the open position, one or both of the release tabs **782** and **762** may be separated along the respective perforated line(s) **780**, **760** from the lid **714** and/or the first front wall **713***c*. In some embodiments, when a person attempts to pull the adhesive off the carton **702**, one of the release tabs **782** and **762** may be separated along a respective one of the perforated lines **780**, **760** from one of the lid **714** and the first front wall **713***c* and may be secured to the other one of the lid **714** and the first front wall **713***c* via the adhesive.

[0172] In some embodiments, the carton **702** may also comprise a tray **791** hingedly coupled to the bottom box **703** and movable between a first position and a second position. As shown in FIGS. **16**A, **17**A, and **17**B, the tray **791** may comprise a plurality of sidewalls **791***a-d* and may be hingedly coupled to the first front wall **713***c* via sidewall **791***c*. FIG. **17**A illustrates the tray **791** is the first (open) position where a cavity **705** of the bottom box **703** is accessible, and FIG. **17**B illustrates the tray **791** is the second (closed) position where the cavity **705** of the bottom box **703** is no longer accessible. When the tray **791** is in the second (closed) position, the plurality of sidewalls **791***a-d* may be folded up along the plurality of fold lines **730** to define a second bottom box disposed within the bottom box **703**. The tray **791** may define a second bottom box configured to house or accommodate one or more objects, such as one or more leaflets. By way of example, the tray **791** may be configured to house or accommodate a user manual, an instruction manual, a patient information leaflet, consumer medicine information, or any other document that provides information about drugs or medications contained inside one or more containers packaged within the carton **702** or delivery of such drugs or medications.

[0173] In some embodiments, the tray **791** may include a perforated line **790** about a release tab **792** adjacent an edge

794 of the tray **791**. The release tab **792** is enclosed by a perforated line **790** and the bottom edge **794**. The perforated line **790** may consist of a series of cuts. The perforated line **790** allows the release tab **792** to be detached from the tray **791** when sufficient force is applied to the release tab **792**. Accordingly, the release tab **792** may be detached from the tray **791** so that the tray **791** can be lifted to access the cavity **705** of the bottom box **703**, as shown in FIG. **17**A. As shown in FIG. **16**A, the release tab **792** is partially rectangular with a rounded side, however, the release tab **792** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0174] FIG. **16**B is a plan view of a die cut of an insert **704** configured to be disposed within the cavity **705** of the bottom box **703** of the carton **702**, in accordance with various embodiments of the present disclosure. The insert **704** may be made from a unitary material. For example, the insert **704** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the insert **704** may be made of a combination of natural and synthetic materials. In other embodiments, the insert **704** can be made from several pieces glued or otherwise secured together. As shown in FIG. **16**B, the insert **704** includes a plurality of fold lines **731**, dividing the insert **704** into a plurality of walls **710***a-i*. FIG. **16**B illustrates the insert **704** in a flat configuration. Similar to the carton **702**, the insert **704** is erectable from a storage configuration where the insert **704** is substantially flat (shown in FIG. **16**B) to an assembled configuration (shown in FIG. **18**A-**18**D). For example, when each of the plurality of fold lines **731** is folded, the insert **704** is erectable into the assembled configuration shown in FIGS. **18**A-**18**D.

[0175] The insert **704** includes a top wall **710***b*, a second bottom wall **710***d*, a second front wall **710***c*, a second back wall **710***a*, a first middle wall **710***f* positioned between the top wall **710***b* and the second bottom wall **710***d*, and a second middle wall **710***h* positioned between the first middle wall **710***f* and the second bottom wall **710***d*. When assembled, the first middle wall **710***f* may be positioned higher than the second middle wall **710***h*, as shown in FIG. **18**B. Once each of the walls are folded along the fold lines **731**, the wall **710***e* can be secured to the inner surface of the second back wall **710***a*, the wall **710**g can be secured to the inner surface of the second front wall **710***c*, and the wall **710***i* can be secured to the inner surface of the wall **710***e* to secure the insert **704** in the assembled configuration. In some embodiments, the walls **710***e*, **710***g*, and **710***i* can be glued to the inner surfaces of the second back wall **710***a*, the second front wall **710***c*, and the wall **710***e*, respectively. In other embodiments, the walls **710***e*, **710***g*, and **710***i* can be secured to the inner surfaces of the second back wall **710***a*, the second front wall **710***c*, and the wall **710***e*, respectively, via interlocking slits, other adhesives, or other known methods of assembling boxes and inserts.

[0176] As shown in FIG. **16**B, the insert **704** may comprise a first pair of opposing flaps **710***n* coupled to the second front wall **710***c* and a second pair of opposing flaps **710***m* coupled to the second back wall **710***a*. Moreover, the top wall **710***b* may comprise a third pair of opposing flaps **710***j* hingedly coupled to the top wall **710***b*. The third pair of opposing flaps **710***j* may be movable between an open position and a closed position. For example, the third pair of opposing flaps **710***j* can be folded along the fold lines **731** from the open position to the closed position. The third pair of opposing flaps **710***j* may each include a flap **710***k*.

[0177] As a part of the assembly process, the first pair of opposing flaps **710***n* and the second pair of opposing flaps **710***m* can be folded along the fold lines **731**, and the third pair of opposing flaps **710***j* can be folded along the fold lines **731** to cover the first pair of opposing flaps **710***n* and the second pair of opposing flaps **710***m*. In the closed position, the third pair of opposing flaps **710***j* may define the third and fourth sidewalls of the insert **704**. Afterwards, the flaps **710***k* can be folded and tucked within the second bottom wall **710***d* (as shown in FIG. **18**D) to secure the insert **704** in the closed position, as shown in at least FIGS. **18**A and **18**D.

[0178] When the insert **704** is folded from its storage configuration where the insert **704** is substantially flat into its assembled configuration, the insert **704** may be disposed within the cavity **705** of the bottom box **703** of the carton **702**, as shown in FIG. **17**A. When the insert **704** is disposed within the cavity **705** of the bottom box **703** of the carton **702**, the second bottom wall **710***d* is configured to abut the first bottom wall **716**, the second back wall **710***a* is configured to abut the first back wall **713***a*, and the second front wall **710***c* is configured to abut the first front wall **713***c*. In some embodiments, the insert **704** may be removably disposed within the cavity **705** of the bottom box **703**. In other embodiments, the insert **704** may be secured to the cavity **705** of the bottom box **703** via glue, other adhesives, or other known methods of assembling boxes, cartons, and trays. Once the insert **704** is disposed within the cavity **705**, the tray **791** may be moved from the first position to the second position (shown in FIG. **17**B) such that the tray **791** sits on top of the insert **704** and covers the insert **704**.

[0179] The top wall **710***b* of the insert **704** may define a plurality of cavities **706***a-c*, each dimensioned, sized, and configured to house a respective container. In addition, the first middle wall **710***f* may define a plurality of cavities **706***d-e*. In some embodiments, the number of cavities defined by the top wall **710***b* may be different from the number of cavities defined by the first middle wall **710***f*. For example, as shown in FIG. **16**B, the top wall **710***b* may define three cavities **706***a-c*, while the first middle wall **710***f* may define two cavities **706***d-e*. In other embodiments, the number of cavities defined by the top wall **710***b* may be equal to the number of cavities defined by the first middle wall **710***f*. In some embodiments, each of the plurality of cavities **706***a-e* may have the same dimensions (e.g., diameters) and, thus, may be dimensioned, sized, and configured to house containers with the same dimensions (e.g., diameters, heights, etc.). In other embodiments, at least one of the plurality of cavities **706***a-e* may have different dimensions, such as a different diameter, than the remainer of the plurality of cavities **706***a-e*. By way of example, cavities **706***a*, **706***b*, **706***d*, and **706***e* may have the same diameters and, thus, may each be sized, dimensioned, and configured to house an ISO 2R vial, while the cavity **706***c* may have a greater diameter than the cavities **706***a*, **706***b*, **706***d*, and **706***e* and, thus, be sized, dimensioned, and configured to house an ISO 6R vial.

[0180] In some embodiments, the positions of the cavities **706***d*, **706***e* in the first middle wall **710***f* may vertically align with the positions of the cavities **706***a*, **706***b* in the top wall **710***b*, respectively, when the insert **704** is in the assembled configuration. Accordingly, when containers are disposed within the cavities **706***a*, **706***b*, each of the containers will fit through the cavities **706***a*, **706***b* in the top wall **710***b* as well as the respective cavities **706***d*, **706***e* in the first middle wall **710***f*. When containers are disposed within the cavities **706***a*, **706***b* in the top wall **710***b* and the respective cavities **706***d*, **706***e* in the first middle wall **710***f*, the bottom surfaces of the containers may abut the second middle wall **710***h*. When a container is disposed within the cavity **706***c* in the top wall, the bottom surface of the container may abut the first middle wall **710***f*. Accordingly, at least one of the first middle wall **710***f* or the second middle wall **710***h* may be configured to abut the bottom surfaces of the containers and support the containers when the containers are housed within the insert **704**.

[0181] In some embodiments, the insert **704** may further comprise a vertical rib support **721** extending between the second middle wall **710***h* and the second bottom wall **710***d*, as shown in FIG. **18**C. The vertical rib support **721** may be formed integrally with the insert **704**. The vertical rib support **721** may be configured to maintain a predefined distance between the second middle wall **710***h* and the second bottom wall **710***d* such that the containers housed within the insert **704** are consistently raised from the second bottom wall **710***d* of the insert **704** and supported by at least one of the first middle wall **710***f* or the second middle wall **710***h* of the insert **704**. Accordingly, the vertical rib support **721** may ensure that the containers housed within the insert **704** are protected from bottom load impact. Additionally, the vertical rib support **721** may be configured to help at least the second middle wall **710***h* in supporting the weight of the containers.

[0182] In some embodiments, the second middle wall **710***h* may comprise a set of flaps **710***p* extending from opposite sides of the second middle wall **710***h*, as shown in FIG. **16**B. Once the insert **704** is folded into the assembled configuration, the flaps **710***p* can be folded up or down (as shown in FIGS. **18**B and **18**C). When folded up or down, the flaps **710***p* may be configured to provide additional rigidity to the insert **704** such that the insert **704** can maintain its box shape when folded to its assembled configuration.

[0183] As discussed above, the diameters of the cavities **706***a-e* may be different, in some embodiments, to accommodate containers of different sizes within a single insert and a single packaging. For example, the diameter of the cavities **706***a*, **706***b*, **706***d*, and **706***e* may be smaller than the diameter of the cavity **706***c*, and, thus, the diameter of the first set of containers housed within the cavities **706***a*, **706***b*, **706***d*, and **706***e* may be smaller than the diameter of the second container housed within the cavity **706***c*. The first set of containers may, for example, be one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial, and the second container may be another one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In one embodiment, for instance, the first set of containers may be an ISO 2R vial, and the second container may be an ISO 6R vial. Because the dimensions (e.g., diameters, heights, etc.) of the containers may be different, the heights of the first middle wall **710***f* and the second middle wall **710***h* may be offset, as discussed above. For instance, the first middle wall **710***f* may be positioned higher in the insert **704** than the second middle wall **710***h*. Accordingly, even though the height of the first set of containers housed within the cavities **706***a*, **706***b*, **706***d*, and **706***e* may be greater than the height of the second container housed within the cavity **706***c*, all of the containers may still be positioned at the same or similar height within the insert **704** because the first middle wall **710***f* and the second middle wall **710***h* supporting the containers are offset.

[0184] In some embodiments, the cavities **706***a-e* may each be dimensioned, sized, and configured to accommodate a neck of the containers housed within the insert **704**. In other embodiments, the cavities **706***a-e* may each be dimensioned, sized, and configured to accommodate a body of the containers housed within the insert **704**. In yet another embodiment, some of the cavities **706***a-e* may be dimensioned, sized, and configured to accommodate the neck of the containers, while others of the cavities **706***a-e* may be dimensioned, sized, and configured to accommodate the body of the containers. In some embodiments, the containers housed within the insert **704** may include drug vials that are filled or pre-filled with a drug. The drug may comprise, for example, tarlatamab (AMG 757) or another product containing a half-life extended (HLE) anti-delta-like ligand 3 (DLL3) x anti-CD3 bispecific T cell engager construct. In some embodiments, at least one of the containers housed within the insert **704** may be a drug vial that is filled or pre-filled with a drug that is different from the drug contained in the remainder of the containers.

[0185] FIG. **19**A is a plan view of a die cut of a carton **802** for another exemplary system for packaging a plurality of containers, in accordance with various embodiments of the present disclosure. The carton **802** may be made from a unitary material. For example, the carton **802** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the carton **802** may be made of a combination of natural and synthetic materials. In other embodiments, the carton **802** can be made from several pieces glued or otherwise secured together. The carton **802** includes a plurality of fold lines **830**, dividing the carton **802** into a plurality of walls and flaps. FIG. **19**A illustrates the carton **802** in a storage configuration where the carton **802** is substantially or completely flat. The carton **802** is erectable or foldable from a storage configuration where the carton **802** is substantially flat (shown in FIG. **19**A) to an assembled configuration (shown in FIGS. **20**A and **20**B). For example, when each of the plurality of fold lines **830** is folded, the carton **802** is erectable into the assembled configuration so as to form a box-shaped carton, as shown in FIGS. **20**A and **20**B.

[0186] Referring to FIGS. **19**A, **20**A, and **20**B, the carton **802** includes a bottom box **803** (shown in FIGS. **20**A and **20**B) having a first bottom wall **816** made from several flaps **816***a-d*, a first sidewall **813***b*, a second sidewall **813***d*, a first back wall **813***a*, and a first front wall **813***c*. The flaps **816***a-d* can be secured to each other via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons to form the first bottom wall **816** when the carton **802** is assembled. The first back wall **813***a* includes a flap **826** that can be glued, for example, to an inner surface of the second sidewall **813***d* to form the bottom box **803**. In some embodiments, the first back wall **813***a* can be secured to the second sidewall **813***d* via interlocking slits, other adhesives, or other known methods of assembling boxes and cartons. As a result, the first sidewall **813***b*, the second sidewall **813***d*, the first back wall **813***a*, the first front wall **813***c*, and the first bottom wall **816** are all secured together to form the bottom box **803**.

[0187] Furthermore, the carton **802** includes a lid **814**, hingedly connected or coupled to the first back wall **813***a*. In some embodiments, the lid **814** is an overlid that is movable between an open position and a closed position and includes a flap **815**. When the carton **802** is assembled, the lid **814** can be moved from an open position where the internal cavity **805** of the bottom box **803** is accessible to a closed position where the internal cavity **805** of the bottom box **803** is not accessible. Additionally, the flap **815** of the lid **814** can be folded and tucked behind the first front wall **813***c* to secure the carton **802** in a closed position and prevent accidental opening of the carton **802**.

[0188] Moreover, the carton **802** may include a first flap **807***a* and a second flap **807***b* extending from the first sidewall **813***b* and the second sidewall **813***d*, respectively, of the bottom box **803**. The second sidewall **813***d* is positioned opposite the first sidewall **813***b* and faces the first sidewall **813***b*. In other words, the first sidewall **813***b* and the second sidewall **813***d* are opposing walls of the bottom box **803**.

[0189] In some embodiments, the bottom box **803** of the carton **802** includes a release tab **862**. As shown in FIG. **19**A, for example, the first front wall **813***c* includes a perforated line **860** about a release tab **862** adjacent a bottom edge **864** of the first front wall **813***c*. The release tab **862** is enclosed by a perforated line **860** and the bottom edge **864**. The release tab **862** includes a first kiss cut **866**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on at least a portion of the release tab **862**. The first kiss cut **866** may improve securement between the release tab **862** and the adhesive. The perforated line **860** allows the release tab **862** to be detached from the first front wall **813***c* when sufficient force is applied to the release tab **862**. As shown in FIG. **19**A, the release tab **862** is partially rectangular with a rounded side, however, the release tab **862** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0190] Additionally, or alternatively, as shown in FIG. **19**A, the lid **814** of the bottom box **803** may also include a perforated line **880** about a release tab **882** adjacent an edge **884** between the lid **814** and the flap **815** of the lid. The release tab **882** is enclosed by the perforated line **880** and the edge **884**. The perforated line **880** allows the release tab **882** to be detached from the lid **814** when sufficient force is applied to the release tab **882**. As shown in FIG. **19**A, the release tab **882** is partially rectangular with a rounded side, however, the release tab **882** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0191] In some embodiments, when the carton **802** is folded into the assembled configuration and the lid **840** is placed over the bottom box **803**, the position of the release tab **882** on the lid **814** may align with the position of the release tab **862** on the first front wall **813***c* of the bottom box **803** such that the release tabs **882** and **862** are positioned perpendicular to each other. The edge **884** and the edge **864** that partially define the release tabs **882** and **862**, respectively, may align with each other and be positioned adjacent to each other. In some embodiments, the release tab **882** and/or the release tab **862** may be a push tab that detaches the release tab from the lid **814** and/or the first front wall **813***c* of the bottom box **803**. In some embodiments, an adhesive, such as an adhesive label, may be positioned on the release tab **882** and/or the release tab **862**. Accordingly, the release tab **882** and/or the release tab **862** may include a kiss cut **868**, **866** to improve securement between the release tab and the adhesive.

[0192] The release tab **882** and the release tab **862** may be partially circumscribed by perforated lines **880** and **860**, respectively. The perforated lines **880** and **860** may each consist of a series of cuts. The kiss cuts **868** and **866** may each be a partial cut through the release tabs **882** and **862**, respectively, that improve the bonding between the release tabs **882**, **862** and a tamper evident seal by increasing the surface area the tamper evident seal can adhere to. The tamper evident seal can, for example, be an adhesive seal such as an adhesive or closure label to seal the carton **802**. For example, an adhesive can be placed over the kiss cut **868** on the lid **814** and the kiss cut **866** on the first front wall **813***c* of the bottom box **803** to seal the carton **802** in a closed position. The adhesive may act as a tamper evident seal. That is, any attempt to pull the adhesive off the carton **802** will result in a visible damage to the carton **802** and/or the adhesive. For example, when the lid **814** is in the closed position and the carton **802** is sealed, the release tab **882** and the release tab **862** may be secured to the lid **814** and the first front wall **813***c*, respectively. When a person attempts to pull the adhesive off the carton **802** and the lid **814** is in the open position, one or both of the release tabs **882** and **862** may be separated along the respective perforated line(s) **880**, **860** from the lid **814** and/or the first front wall **813***c*. In some embodiments, when a person attempts to pull the adhesive off the carton **802**, one of the release tabs **882** and **862** may be separated along a respective one of the perforated lines **880**, **860** from one of the lid **814** and the first front wall **813***c* and may be secured to the other one of the lid **814** and the first front wall **813***c* via the adhesive.

[0193] In some embodiments, the carton **802** may also comprise a tray **891** hingedly coupled to the bottom box **803** and movable between a first position and a second position. As shown in FIGS. **19**A, **20**A, and **20**B, the tray **891** may comprise a plurality of sidewalls **891***a-d* and may be hingedly coupled to the first front wall **813***c* via sidewall **891***c*. FIG. **20**A illustrates the tray **891** is the first (open) position where a cavity **805** of the bottom box **803** is accessible, and FIG. **20**B illustrates the tray **891** is the second (closed) position where the cavity **805** of the bottom box **803** is no longer accessible. When the tray **891** is in the second (closed) position, the plurality of sidewalls **891***a-d* may be folded up along the plurality of fold lines **830** to define a second bottom box disposed within the bottom box **803**. The tray **891** may define a second bottom box configured to house or accommodate one or more objects, such as one or more leaflets. By way of example, the tray **891** may be configured to house or accommodate a user manual, an instruction manual, a patient information leaflet, consumer medicine information, or any other document that provides information about drugs or medications contained inside one or more containers packaged within the carton **802** or delivery of such drugs or medications.

[0194] In some embodiments, the tray **891** may include a perforated line **890** about a release tab **892** adjacent an edge **894** of the tray **891**. The release tab **892** is enclosed by a perforated line **890** and the bottom edge **894**. The perforated line **890** may consist of a series of cuts. The perforated line **890** allows the release tab **892** to be detached from the tray **891** when sufficient force is applied to the release tab **892**. Accordingly, the release tab **892** may be detached from the tray **891** so that the tray **891** can be lifted to access the cavity **805** of the bottom box **803**, as shown in FIG. **20**A. As shown in FIG. **19**A, the release tab **892** is partially rectangular with a rounded side, however, the release tab **892** could be any of a variety of shapes including semi-circular, triangular, quadrilateral, pentagonal, etc.

[0195] FIG. **19**B is a plan view of a die cut of an insert **804** configured to be disposed within the cavity **805** of the bottom box **803** of the carton **802**, in accordance with various embodiments of the present disclosure. The insert **804** may be made from a unitary material. For example, the insert **804** may be made of a paperboard, cardboard, matboard, or similar materials. In some embodiments, the insert **804** may be made of a combination of natural and synthetic materials. In other embodiments, the insert **804** can be made from several pieces glued or otherwise secured together. As shown in FIG. **19**B, the insert **804** includes a plurality of fold lines **831**, dividing the insert **804** into a plurality of walls **810***a-i*. FIG. **19**B illustrates the insert **804** in a flat configuration. Similar to the carton **802**, the insert **804** is erectable from a storage configuration where the insert **804** is substantially flat (shown in FIG. **19**B) to an assembled configuration (shown in FIG. **21**A-**21**D). For example, when each of the plurality of fold lines **831** is folded, the insert **804** is erectable into the assembled configuration shown in FIGS. **21**A-**21**D.

[0196] The insert **804** includes a top wall **810***b*, a second bottom wall **810***d*, a second front wall **810***c*, a second back wall **810***a*, a first middle wall **810***f* positioned between the top wall **810***b* and the second bottom wall **810***d*, and a second middle wall **810***h* positioned between the first middle wall **810***f* and the second bottom wall **810***d*. When assembled, the first middle wall **810***f* may be positioned higher than the second middle wall **810***h*, as shown in FIG. **21**B. Once each of the walls are folded along the fold lines **831**, the wall **810***e* can be secured to the inner surface of the second back wall **810***a*, the wall **810***g* can be secured to the inner surface of the second front wall **810***c*, and the wall **810***i* can be secured to the inner surface of the wall **810***e* to secure the insert **804** in the assembled configuration. In some embodiments, the walls **810***e*, **810***g*, and **810***i* can be glued to the inner surfaces of the second back wall **810***a*, the second front wall **810***c*, and the wall **810***e*, respectively. In other embodiments, the walls **810***e*, **810***g*, and **810***i* can be secured to the inner surfaces of the second back wall **810***a*, the second front wall **810***c*, and the wall **810***e*, respectively, via interlocking slits, other adhesives, or other known methods of assembling boxes and inserts.

[0197] As shown in FIG. **19**B, the insert **804** may comprise a first pair of opposing flaps **810***n* coupled to the second front wall **810***c* and a second pair of opposing flaps **810***m* coupled to the second back wall **810***a*. Moreover, the top wall **810***b* may comprise a third pair of opposing flaps **810***j* hingedly coupled to the top wall **810***b*. The third pair of opposing flaps **810***j* may be movable between an open position and a closed position. For example, the third pair of opposing flaps **810***j* can be folded along the fold lines **831** from the open position to the closed position. The third pair of opposing flaps **810***j* may each include a flap **810***k*.

[0198] As a part of the assembly process, the first pair of opposing flaps **810***n* and the second pair of opposing flaps **810***m* can be folded along the fold lines **831**, and the third pair of opposing flaps **810***j* can be folded along the fold lines **831** to cover the first pair of opposing flaps **810***n* and the second pair of opposing flaps **810***m*. In the closed position, the third pair of opposing flaps **810***j* may define the third and fourth sidewalls of the insert **804**. Afterwards, the flaps **810***k* can be folded and tucked within the second bottom wall **810***d* (as shown in FIG. **21**D) to secure the insert **804** in the closed position, as shown in at least FIGS. **21**A and **21**D.

[0199] When the insert **804** is folded from its storage configuration where the insert **804** is substantially flat into its assembled configuration, the insert **804** may be disposed within the cavity **805** of the bottom box **803** of the carton **802**, as shown in FIG. **20**A. When the insert **804** is disposed within the cavity **805** of the bottom box **803** of the carton **802**, the second bottom wall **810***d* is configured to abut the first bottom wall **816**, the second back wall **810***a* is configured to abut the first back wall **813***a*, and the second front wall **810***c* is configured to abut the first front wall **813***c*. In some embodiments, the insert **804** may be removably disposed within the cavity **805** of the bottom box **803**. In other embodiments, the insert **804** may be secured to the cavity **805** of the bottom box **803** via glue, other adhesives, or other known methods of assembling boxes, cartons, and trays. Once the insert **804** is disposed within the cavity **805**, the tray **891** may be moved from the first position to the second position (shown in FIG. **20**B) such that the tray **891** sits on top of the insert **804** and covers the insert **804**.

[0200] The top wall **810***b* of the insert **804** may define a plurality of cavities **806***a-c*, each dimensioned, sized, and configured to house a respective container. In addition, the first middle wall **810***f* may define a plurality of cavities **806***d-f*. In some embodiments, the number of cavities defined by the top wall **810***b* may be different from the number of cavities defined by the first middle wall **810***f*. In some embodiments, the number of cavities defined by the top wall **810***b* may be equal to the number of cavities defined by the first middle wall **810***f*. For example, as shown in FIG. **19**B, the top wall **810***b* may define three cavities **806***a-c*, and the first middle wall **810***f* may also define three cavities **806***d-f*. In some embodiments, each of the plurality of cavities **806***a-f* may have the same dimensions (e.g., diameters) and, thus, may be dimensioned, sized, and configured to house containers with the same dimensions (e.g., diameters, heights, etc.). In other embodiments, at least one of the plurality of cavities **806***a-f* may have different dimensions, such as a different diameter, than the remainer of the plurality of cavities **806***a-f*. As shown in FIG. **19**B, for example, the cavities **806***a*, **806***b*, **806***d*, and **806***e* may be dimensioned, sized, and configured to house containers with the same dimensions, while the cavities **806***c* and **806***f* may be dimensioned, sized, and configured to house containers with dimensions that are different than the containers accommodated within the cavities **806***a*, **806***b*, **806***d*, and **806***e*. By way of example, cavities **806***a*, **806***b*, **806***d*, and **806***e* may have the same diameters and, thus, may each be sized, dimensioned, and configured to house an ISO 2R vial, while the cavities **806***c* and **806***f* may have a greater diameter than the cavities **806***a*, **806***b*, **806***d*, and **806***e* and, thus, be sized, dimensioned, and configured to house an ISO 6R vial. In some embodiments, some of the cavities may have additional cut-outs to allow the cavities to accommodate containers of various dimensions. For example, as shown in FIGS. **19**B and **21**A, the cavities **806***c* and **806***f* may each have additional cut-outs **806***c*' and **806***f*', respectively. The cut-outs **806***c*' and **806***f* allows flexibility in the cavities **806***c* and **806***f* such that the cavities **806***c* and **806***f* are shaped, sized, dimensioned, and configured to accommodate containers of various dimensions.

[0201] In some embodiments, the positions of the cavities **806***d*, **806***e*, **806***f* in the first middle wall **810***f* may vertically align with the positions of the cavities **806***a*, **806***b*, **806***c* in the top wall **810***b*, respectively, when the insert **804** is in the assembled configuration. Accordingly, when containers are disposed within the cavities **806***a*, **806***b*, **806***c*, each of the containers will fit through the cavities **806***a*, **806***b*, **806***c* in the top wall **810***b* as well as the respective cavities **806***d*, **806***e*, **806***f* in the first middle wall **810***f*. When containers are disposed within the cavities **806***a*, **806***b*, **806***c* in the top wall **810***b* and the respective cavities **806***d*, **806***e*, **806***f* in the first middle wall **810***f*, the bottom surfaces of at least one of the containers may abut the second middle wall **810***h*. Accordingly, the second middle wall **810***h* may be configured to abut the bottom surfaces of at least one of the containers and support the container(s) when the container(s) are housed within the insert **804**. Although FIG. **19**B illustrates the cavities **806***a*, **806***b*, **806***d*, and **806***e* as circular cavities, the cavities **806***a-f* could be any of a variety of shapes including semi-circular, rectangular, triangular, quadrilateral, pentagonal, etc. In some embodiments, the cavities **806***a-f* may all have the same shape. In other embodiments, the cavities **806***a-f* may have different shapes.

[0202] In some embodiments, the insert **804** may further comprise a vertical rib support **821** extending between the second middle wall **810***h* and the second bottom wall **810***d*, as shown in FIG. **21**C. The vertical rib support **821** may be formed integrally with the insert **804**. The vertical rib support **821** may be configured to maintain a predefined distance between the second middle wall **810***h* and the second bottom wall **810***d* such that the containers housed within the insert **804** are consistently raised from the second bottom wall **810***d* of the insert **804** and supported by at least one of the first middle wall **810***f* or the second middle wall **810***h* of the insert **804**. Accordingly, the vertical rib support **821** may ensure that the containers housed within the insert **804** are protected from bottom load impact. Additionally, the vertical rib support **821** may be configured to help at least the second middle wall **810***h* in supporting the weight of the containers.

[0203] In some embodiments, the second middle wall **810***h* may comprise a set of flaps **810***p* extending from opposite sides of the second middle wall **810***h*, as shown in FIG. **19**B. Once the insert **804** is folded into the assembled configuration, the flaps **810***p* can be folded up or down (as shown in FIGS. **21**B and **21**C). When folded up or down, the flaps **810***p* may be configured to provide additional rigidity to the insert **804** such that the insert **804** can maintain its box shape when folded to its assembled configuration.

[0204] As discussed above, the diameters of the cavities **806***a-e* may be different, in some embodiments, to accommodate containers of different sizes within a single insert and a single packaging. For example, the diameter of the cavities **806***a*, **806***b*, **806***d*, and **806***e* may be smaller than the diameter of the cavity **806***c*, and, thus, the diameter of the first set of containers housed within the cavities **806***a*, **806***b*, **806***d*, and **806***e* may be smaller than the diameter of the second container housed within the cavity **806***c*. The first set of containers may, for example, be one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial, and the second container may be another one of an ISO 2R vial, an ISO 6R vial, an ISO 10R vial, an ISO 20R vial, or an ISO 50R vial. In one embodiment, for instance, the first set of containers may be an ISO 2R vial, and the second container may be an ISO 6R vial. Because the dimensions (e.g., diameters, heights, etc.) of the containers may be different, the heights of the first middle wall **810***f* and the second middle wall **810***h* may be offset, as discussed above. For instance, the first middle wall **810***f* may be positioned higher in the insert **804** than the second middle wall **810***h*. Accordingly, even though the height of the first set of containers housed within the cavities **806***a*, **806***b*, **806***d*, and **806***e* may be greater than the height of the second container housed within the cavity **806***c*, all of the containers may still be positioned at the same or similar height within the insert **804** because the first middle wall **810***f* and the second middle wall **810***h* supporting the containers are offset.

[0205] In some embodiments, the cavities **806***a-e* may each be dimensioned, sized, and configured to accommodate a neck of the containers housed within the insert **804**. In other embodiments, the cavities **806***a-e* may each be dimensioned, sized, and configured to accommodate a body of the containers housed within the insert **804**. In yet another embodiment, some of the cavities **806***a-e* may be dimensioned, sized, and configured to accommodate the neck of the containers, while others of the cavities **806***a-e* may be dimensioned, sized, and configured to accommodate the body of the containers. In some embodiments, the containers housed within the insert **804** may include drug vials that are filled or pre-filled with a drug. The drug may comprise, for example, tarlatamab (AMG 757) or another product containing a half-life extended (HLE) anti-delta-like ligand 3 (DLL3) x anti-CD3 bispecific T cell engager construct. In some embodiments, at least one of the containers housed within the insert **804** may be a drug vial that is filled or pre-filled with a drug that is different from the drug contained in the remainder of the containers.

[0206] The presently disclosed systems for packaging one or more containers are not limited to the specific types or numbers of containers explicitly described herein. For example, the packaging systems according to the present disclosure may be designed and/or manufactured to accommodate any desired number of containers and any desired combination of containers, including standard glass vials specified in ISO 8362-1, non-standard glass vials, and/or non-standard, non-glass containers.

[0207] All features described herein, including in the specification, claims, abstract, and drawings, and all the steps in any method or process described herein, may be combined in any combination, except combinations where one or more of the features and/or steps are mutually exclusive.

[0208] The above description describes various systems for packaging containers for clinical, pharmaceutical, or medical use. The systems described herein can comprise or be used with a drug including but not limited to those drugs identified below as well as their generic and biosimilar counterparts. The term drug, as used herein, can be used interchangeably with other similar terms and can be used to refer to any type of medicament or therapeutic material including traditional and non-traditional pharmaceuticals, nutraceuticals, supplements, biologics, biologically active agents and compositions, large molecules, biosimilars, bioequivalents, therapeutic antibodies, polypeptides, proteins, small molecules and generics. Non-therapeutic injectable materials are also encompassed. The drug may be in liquid form, a lyophilized form, or in a reconstituted from lyophilized form. The following example list of drugs should not be considered as all-inclusive or limiting.

[0209] In some embodiments, the containers may be filled with colony stimulating factors, such as granulocyte colony-stimulating factor (G-CSF). Such G-CSF agents include but are not limited to Neulasta® (pegfilgrastim, pegylated filgastrim, pegylated G-CSF, pegylated hu-Met-G-CSF) and Neupogen® (filgrastim, G-CSF, hu-MetG-CSF), UDENYCA® (pegfilgrastim-cbqv), Ziextenzo® (LA-EP2006; pegfilgrastim-bmez), or FULPHILA (pegfilgrastim-bmez).

[0210] In some embodiments, the containers described herein may contain an erythropoiesis stimulating agent (ESA), which may be in liquid or lyophilized form. An ESA is any molecule that stimulates erythropoiesis. In some embodiments, an ESA is an erythropoiesis stimulating protein. As used herein, "erythropoiesis stimulating protein" means any protein that directly or indirectly causes activation of the erythropoietin receptor, for example, by binding to and causing dimerization of the receptor. Erythropoiesis stimulating proteins include erythropoietin and variants, analogs, or derivatives thereof that bind to and activate erythropoietin receptor; antibodies that bind to erythropoietin receptor and activate the receptor; or peptides that bind to and activate erythropoietin receptor. Erythropoiesis stimulating proteins include, but are not limited to, Epogen® (epoetin alfa), Aranesp® (darbepoetin alfa), Dynepo® (epoetin delta), Mircera® (methyoxy polyethylene glycol-epoetin beta), Hematide®, MRK-2578, INS-22, Retacrit® (epoetin zeta), Neorecormon® (epoetin beta), Silapo® (epoetin zeta), Binocrit® (epoetin alfa), epoetin alfa Hexal, Abseamed® (epoetin alfa), Ratioepo® (epoetin theta), Eporatio® (epoetin theta), Biopoin® (epoetin theta), epoetin alfa, epoetin beta, epoetin iota, epoetin omega, epoetin delta, epoetin zeta, epoetin theta, and epoetin delta, pegylated erythropoietin, carbamylated erythropoietin, as well as the molecules or variants or analogs thereof.

[0211] Among particular illustrative proteins are the specific proteins set forth below, including fusions, fragments, analogs, variants or derivatives thereof: OPGL specific antibodies, peptibodies, related proteins, and the like (also referred to as RANKL specific antibodies, peptibodies and the like), including fully humanized and human OPGL specific antibodies, particularly fully humanized monoclonal antibodies; Myostatin binding proteins, peptibodies, related proteins, and the like, including myostatin specific peptibodies; IL-4 receptor specific antibodies, peptibodies, related proteins, and the like, particularly those that inhibit activities mediated by binding of IL-4 and/or IL-13 to the receptor; Interleukin 1-receptor 1 ("IL1-R1") specific antibodies, peptibodies, related proteins, and the like; Ang2 specific antibodies, peptibodies, related proteins, and the like; NGF specific antibodies, peptibodies, related proteins, and the like; CD22 specific antibodies, peptibodies, related proteins, and the like, particularly human CD22 specific antibodies, such as but not limited to humanized and fully human antibodies, including but not limited to humanized and fully human monoclonal antibodies, particularly including but not limited to human CD22 specific IgG antibodies, such as, a dimer of a human-mouse monoclonal hLL2 gamma-chain disulfide linked to a human-mouse monoclonal hLL2 kappa-chain, for example, the human CD22 specific fully humanized antibody in Epratuzumab, CAS registry number 501423-23-0; IGF-1 receptor specific antibodies, peptibodies, and related proteins, and the like including but not limited to anti-IGF-1R antibodies; B-7 related protein 1 specific antibodies, peptibodies, related proteins and the like ("B7RP-1" and also referring to B7H2, ICOSL, B7h, and CD275), including but not limited to B7RP-specific fully human monoclonal IgG2 antibodies, including but not limited to fully human IgG2 monoclonal antibody that binds an epitope in the first immunoglobulin-like domain of B7RP-1, including but not limited to those that inhibit the interaction of B7RP-1 with its natural receptor, ICOS, on activated T cells; IL-15 specific antibodies, peptibodies, related proteins, and the like, such as, in particular, humanized monoclonal antibodies, including but not limited to HuMax IL-15 antibodies and related proteins, such as, for instance, 145c7; IFN gamma specific antibodies, peptibodies, related proteins and the like, including but not limited to human IFN gamma specific antibodies, and including but not limited to fully human anti-IFN gamma antibodies; TALL-1 specific antibodies, peptibodies, related proteins, and the like, and other TALL specific binding proteins; Parathyroid hormone ("PTH") specific antibodies, peptibodies, related proteins, and the like; Thrombopoietin receptor ("TPO-R") specific antibodies, peptibodies, related proteins, and the like; Hepatocyte growth factor ("HGF") specific antibodies, peptibodies, related proteins, and the like, including those that target the HGF/SF: cMet axis (HGF/SF: c-Met), such as fully human monoclonal antibodies that neutralize hepatocyte growth factor/scatter (HGF/SF); TRAIL-R2 specific antibodies, peptibodies, related proteins and the like; Activin A specific antibodies, peptibodies, proteins, and the like; TGF-beta specific antibodies, peptibodies, related proteins, and the like; Amyloid-beta protein specific antibodies, peptibodies, related proteins, and the like; c-Kit specific antibodies, peptibodies, related proteins, and the like, including but not limited to proteins that bind c-Kit and/or other stem cell factor receptors; OX40L specific antibodies, peptibodies, related proteins, and the like, including but not limited to proteins that bind OX40L and/or other ligands of the OX40 receptor; Activase® (alteplase, tPA); Aranesp® (darbepoetin alfa) Erythropoietin [30-asparagine, 32-threonine, 87-valine, 88-asparagine, 90-threonine], Darbepoetin alfa, novel erythropoiesis stimulating protein (NESP); Epogen® (epoetin alfa, or erythropoietin); GLP-1, Avonex® (interferon beta-1a); Bexxar® (tositumomab, anti-CD22 monoclonal antibody); Betaseron® (interferon-beta); Campath® (alemtuzumab, anti-CD52 monoclonal antibody); Dynepo® (epoetin delta); Velcade® (bortezomib); MLN0002 (anti-$\alpha 4\beta 7$ mAb); MLN1202 (anti-CCR2 chemokine receptor mAb); Enbrel® (etanercept, TNF-receptor/Fc fusion protein, TNF blocker); Eprex® (epoetin alfa); Erbitux® (cetuximab, anti-EGFR/HER1/c-ErbB-1); Genotropin® (somatropin, Human Growth Hormone); Herceptin® (trastuzumab, anti-HER2/neu (erbB2) receptor mAb); Kanjinti™ (trastuzumab-anns) anti-HER2 monoclonal antibody, biosimilar to Herceptin®, or another product containing trastuzumab for the treatment of breast or gastric cancers; Humatrope® (somatropin, Human Growth Hormone); Humira® (adalimumab); Vectibix® (panitumumab), Xgeva® (denosumab), Prolia® (denosumab), Immunoglobulin G2 Human Monoclonal Antibody to RANK Ligand, Enbrel® (etanercept, TNF-receptor/Fc fusion protein, TNF blocker), Nplate® (romiplostim), rilotumumab, ganitumab, conatumumab, brodalumab, insulin in solution; Infergen® (interferon alfacon-1); Natrecor® (nesiritide; recombinant human B-type natriuretic peptide (hBNP); Kineret® (anakinra); Leukine® (sargamostim, rhuGM-CSF); LymphoCide® (epratuzumab, anti-CD22 mAb); Benlysta™ (lymphostat B, belimumab, anti-BlyS mAb); Metalyse® (tenecteplase, t-PA analog); Mircera® (methoxy polyethylene glycol-epoetin beta); Mylotarg® (gemtuzumab ozogamicin); Raptiva® (efalizumab); Cimzia® (certolizumab pegol, CDP 870); Soliris™ (eculizumab); pexelizumab (anti-C5 complement); Numax® (MEDI-524); Lucentis® (ranibizumab); Panorex® (17-1A, edrecolomab); Trabio® (lerdelimumab); TheraCim hR3 (nimotuzumab); Omnitarg (pertuzumab, 2C4); Osidem® (IDM-1); OvaRex® (B43.13); Nuvion® (visilizumab); cantuzumab mertansine (huC242-DM1); NeoRecormon® (epoetin beta); Neumega® (oprelvekin, human interleukin-11); Orthoclone OKT3® (muromonab-CD3, anti-CD3 monoclonal antibody); Procrit® (epoetin alfa); Remicade® (infliximab, anti-TNFα monoclonal antibody); Reopro® (abciximab, anti-GP IIb/IIia receptor monoclonal antibody); Actemra® (anti-IL6 Receptor mAb); Avastin® (bevacizumab), HuMax-CD4 (zanolimumab); Mvasi™ (bevacizumab-awwb); Rituxan® (rituximab, anti-CD20 mAb); Tarceva® (erlotinib); Roferon-A®-(interferon alfa-2a); Simulect® (basiliximab); Prexige@ (lumiracoxib); Synagis® (palivizumab); 145c7-CHO (anti-IL15 antibody, see U.S. Pat. No. 7, 153,507); Tysabri® (natalizumab, anti-α4integrin mAb); Valortim® (MDX-1303, anti-B. anthracis protective antigen mAb); ABthrax™; Xolair® (omalizumab); ETI211 (anti-MRSA mAb); IL-1 trap (the Fc portion of human IgG1 and the extracellular domains of both IL-1 receptor components (the Type I receptor and receptor accessory protein)); VEGF trap (Ig domains of VEGFR1 fused to IgG1 Fc); Zenapax® (daclizumab); Zenapax® (daclizumab, anti-IL-2Rα mAb); Zevalin® (ibritumomab tiuxetan); Zetia® (ezetimibe); Orencia® (atacicept, TACI-Ig); anti-CD80 monoclonal antibody (galiximab); anti-CD23 mAb (lumiliximab); BR2-Fc (huBR3/huFc fusion protein, soluble BAFF antagonist); CNTO 148 (golimumab, anti-TNFα mAb); HGS-ETR1 (mapatumumab; human anti-TRAIL Receptor-1 mAb); HuMax-CD20 (ocrelizumab, anti-CD20 human mAb); HuMax-EGFR (zalutumumab); M200 (volociximab, anti-α5β1 integrin mAb); MDX-010 (ipilimumab, anti-CTLA-4 mAb and VEGFR-1 (IMC-18F1); anti-BR3 mAb; anti-C. difficile Toxin A and Toxin B C mAbs MDX-066 (CDA-1) and MDX-1388); anti-CD22 dsFv-PE38 conjugates (CAT-3888 and CAT-8015); anti-CD25 mAb (HuMax-TAC); anti-CD3 mAb (NI-0401); adecatumumab; anti-CD30 mAb (MDX-060); MDX-1333 (anti-IFNAR); anti-CD38 mAb (HuMax CD38); anti-CD40L mAb; anti-Cripto mAb; anti-CTGF Idiopathic Pulmonary Fibrosis Phase I Fibrogen (FG-3019); anti-CTLA4 mAb; anti-eotaxin1 mAb (CAT-213); anti-FGF8 mAb; anti-ganglioside GD2 mAb; anti-ganglioside GM2 mAb; anti-GDF-8 human mAb (MYO-029); anti-GM-CSF Receptor mAb (CAM-3001); anti-HepC mAb (HuMax HepC); anti-IFNa mAb (MEDI-545, MDX-198); anti-IGF1R mAb; anti-IGF-1R mAb (HuMax-Inflam); anti-IL12 mAb (ABT-874); anti-IL12/IL23 mAb (CNTO 1275); anti-IL13 mAb (CAT-354); anti-IL2Ra mAb (HuMax-TAC); anti-IL5 Receptor mAb; anti-integrin receptors mAb (MDX-018, CNTO 95); anti-IP10 Ulcerative Colitis mAb (MDX-1100); BMS-66513; anti-Mannose Receptor/hCGß mAb (MDX-1307); anti-mesothelin dsFv-PE38 conjugate (CAT-5001); anti-PD1mAb (MDX-1106 (ONO-4538)); anti-PDGFRα antibody (IMC-3G3); anti-TGFB mAb (GC-1008); anti-TRAIL Receptor-2 human mAb (HGS-ETR2); anti-TWEAK mAb; anti-VEGFR/FIt-1 mAb; and anti-ZP3 mAb (HuMax-ZP3).

[0212] In some embodiments, the containers described herein may contain a sclerostin antibody, such as but not limited to romosozumab, blosozumab, BPS 804 (Novartis), Evenity™ (romosozumab-aqqg), another product containing romosozumab for treatment of postmenopausal osteoporosis and/or fracture healing and in other embodiments, a monoclonal antibody (IgG) that binds human Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9). Such PCSK9 specific antibodies include, but are not limited to, Repatha® (evolocumab) and Praluent® (alirocumab). In other embodiments, the containers may contain rilotumumab, bixalomer, trebananib, ganitumab, conatumumab, motesanib diphosphate, brodalumab, vidupiprant or panitumumab. In some embodiments, the containers may be filled with IMLYGIC® (talimogene laherparepvec) or another oncolytic HSV for the treatment of melanoma or other cancers including but are not limited to OncoVEXGALV/CD; OrienX010; G207, 1716; NV1020; NV12023; NV1034; and NV1042. In some embodiments, the containers may contain endogenous tissue inhibitors of metalloproteinases (TIMPs) such as but not limited to TIMP-3. In some embodiments, the containers may contain Aimovig® (erenumab-aooe), anti-human CGRP-R (calcitonin gene-related peptide type 1 receptor) or another product containing erenumab for the treatment of migraine headaches. Antagonistic antibodies for human calcitonin gene-related peptide (CGRP) receptor such as but not limited to erenumab and bispecific antibody molecules that target the CGRP receptor and other headache targets may also be contained in the containers of the present disclosure. Additionally, bispecific T cell engager (BiTE®) molecules such as but not limited to BLINCYTO® (blinatumomab) can be contained in the containers of the present disclosure. In some embodiments, the containers may contain an APJ large molecule agonist such as but not limited to apelin or analogues thereof. In some embodiments, a therapeutically effective amount of an anti-thymic stromal lymphopoietin (TSLP) or TSLP receptor antibody is used in the containers of the present disclosure. In some embodiments, the containers may contain Avsola™ (infliximab-axxq), anti-TNF a monoclonal antibody, biosimilar to Remicade® (infliximab) (Janssen Biotech, Inc.) or another product containing infliximab for the treatment of autoimmune diseases. In some embodiments, the containers may contain Kyprolis® (carfilzomib), (2S)-N-((S)-1-((S)-4-methyl-1-((R)-2-methyloxiran-2-yl)-1-oxopentan-2-ylcarbamoyl)-2-phenylethyl)-2-((S)-2-(2-morpholinoacetamido)-4-phenylbutanamido)-4-methylpentanamide, or another product containing carfilzomib for the treatment of multiple myeloma. In some embodiments, the containers may contain Otezla® (apremilast), N-[2-[(1S)-1-(3-ethoxy-4-methoxyphenyl)-2-(methylsulfonyl)ethyl]-2,3-dihydro-1,3-dioxo-1H-isoindol-4-yl]acetamide, or another product containing apremilast for the treatment of various inflammatory diseases. In some embodiments, the containers may contain Parsabiv™ (etelcalcetide HCl, KAI-4169) or another product containing etelcalcetide HCl for the treatment of secondary hyperparathyroidism (sHPT) such as in patients with chronic kidney disease (KD) on hemodialysis. In some embodiments, the containers may contain ABP 798 (rituximab), a biosimilar candidate to Rituxan®/MabThera™, or another product containing an anti-CD20 monoclonal antibody. In some embodiments, the containers may contain a VEGF antagonist such as a non-antibody VEGF antagonist and/or a VEGF-Trap such as aflibercept (Ig domain 2 from VEGFR1 and Ig domain 3 from VEGFR2, fused to Fc domain of IgG1). In some embodiments, the containers may contain ABP 959 (eculizumab), a biosimilar candidate to Soliris®, or another product containing a monoclonal antibody that specifically binds to the complement protein C5. In some embodiments, the containers may contain Rozibafusp alfa (formerly AMG 570) is a novel bispecific antibody-peptide conjugate that simultaneously blocks ICOSL and BAFF activity. In some embodiments, the containers may contain Omecamtiv mecarbil, a small molecule selective cardiac myosin activator, or myotrope, which directly targets the contractile mechanisms of the heart, or another product containing a small molecule selective cardiac myosin activator. In some embodiments, the containers may contain Sotorasib (formerly known as AMG 510), a $KRAS^{G12C}$ small molecule inhibitor, or another product containing a $KRAS^{G12C}$ small molecule inhibitor. In some embodiments, the containers may contain Tezepelumab, a human monoclonal antibody that inhibits the action of thymic stromal lymphopoietin (TSLP), or another product containing a human monoclonal antibody that inhibits the action of TSLP. In some embodiments, the drug delivery device may contain or be used with rocatinlimab (AMG 451), a human anti-OX40 monoclonal antibody that is expressed on activated T cells and blocks OX40 to inhibit and/or reduce the number of OX40 pathogenic T cells that are responsible for driving system and local atopic dermatitis inflammatory responses. In some embodiments, the containers may contain AMG 714, a human monoclonal antibody that binds to Interleukin-15 (IL-15) or another product containing a human monoclonal antibody that binds to Interleukin-15 (IL-15). In some embodiments, the containers may contain AMG 890, a small interfering RNA (siRNA) that lowers lipoprotein(a), also known as Lp(a), or another product containing a small interfering RNA (siRNA) that lowers lipoprotein(a). In some embodiments, the containers may contain ABP 654 (human IgG1 kappa antibody), a biosimilar candidate to Stelara®, or another product that contains human IgG1 kappa antibody and/or binds to the p40 subunit of human cytokines interleukin (IL)-12 and IL-23. In some embodiments, the containers may contain Amjevita™ or Amgevita™ (formerly ABP 501) (mab anti-TNF human IgG1), a biosimilar candidate to Humira®, or another product that contains human mab anti-TNF human IgG1. In some embodiments, the containers may contain AMG 160, or another product that contains a half-life extended (HLE) anti-prostate-specific membrane antigen (PSMA) x anti-CD3 BITER (bispecific T cell engager) construct. In some embodiments, the containers may contain AMG 119, or another product containing a delta-like ligand 3 (DLL3) CAR T (chimeric antigen receptor T cell) cellular therapy. In some embodiments, the containers may contain AMG 119, or another product containing a delta-like ligand 3 (DLL3) CAR T (chimeric antigen receptor T cell) cellular therapy. In some embodiments, the containers may contain AMG 133, or another product containing a gastric inhibitory polypeptide receptor (GIPR) antagonist and GLP-1R agonist. In some embodiments, the containers may contain AMG 171 or another product containing a Growth Differential Factor 15 (GDF15) analog. In some embodiments, the dr containers may contain AMG 176 or another product containing a small molecule inhibitor of myeloid cell leukemia 1 (MCL-1). In some embodiments, the containers may contain AMG 199 or another product containing a half-life extended (HLE) bispecific T cell engager construct (BiTE®). In some embodiments, the containers may contain AMG 256 or another product containing an anti-PD-1 x IL21 mutein and/or an IL-21 receptor agonist designed to selectively turn on the Interleukin 21 (IL-21) pathway in programmed cell death-1 (PD-1) positive cells. In some embodiments, the containers may contain AMG 330 or another product containing an anti-CD33 x anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the containers may contain AMG 404 or another product containing a human anti-programmed cell death-1 (PD-1) monoclonal antibody being investigated as a treatment for patients with solid tumors. In some embodiments, the containers may contain AMG 427 or another product containing a half-life extended (HLE) anti-fms-like tyrosine kinase 3 (FLT3) x anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the containers may contain AMG 430 or another product containing an anti-Jagged-1 monoclonal antibody. In some embodiments, the containers may contain AMG 506 or another product containing a multi-specific FAP x 4-1BB-targeting DARPin® biologic under investigation as a treatment for solid tumors. In some embodiments, the containers may contain AMG 509 or another product containing a bivalent T-cell engager and is designed using XmAb® 2+1 technology. In some embodiments, the containers may contain AMG 562 or another product containing a half-life extended (HLE) CD19 x CD3 BITER (bispecific T cell engager) construct. In some embodiments, the containers may contain Efavaleukin alfa (formerly AMG 592) or another product containing an IL-2 mutein Fc fusion protein. In some embodiments, the containers may contain AMG 596 or another product containing a CD3 x epidermal growth factor receptor vIII (EGFRvIII) BiTE® (bispecific T cell engager) molecule. In some embodiments, the containers may contain AMG 673 or another product containing a half-life extended (HLE) anti-CD33 x anti-CD3 BITE® (bispecific T cell engager) construct. In some embodiments, the containers may contain AMG 701 or another product containing a half-life extended (HLE) anti-B-cell maturation antigen (BCMA) x anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the containers may contain tarlatamab (AMG 757) or another product containing a half-life extended (HLE) anti-delta-like ligand 3 (DLL3) x anti-CD3 BiTE® (bispecific T cell engager) construct. In some embodiments, the containers may contain AMG 910 or another product containing a half-life extended (HLE) epithelial cell tight junction protein claudin 18.2 x CD3 BITE® (bispecific T cell engager) construct.

[0213] Although the systems for packaging one or more containers, such as glass vials, have been described in terms of exemplary embodiments, they are not limited thereto. The detailed description is to be construed as exemplary only and does not describe every possible embodiment of the present disclosure. Numerous alternative embodiments could be implemented, using either current technology or technology developed after the filing date of this patent that would still fall within the scope of the claims defining the invention(s) disclosed herein.

[0214] Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above described embodiments without departing from the spirit and scope of the invention (s) disclosed herein, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept(s).

1. A system for packaging one or more containers, comprising:

a carton having a storage configuration where the carton is substantially flat, and an assembled configuration, the carton comprising:

a bottom box having a first bottom wall, a first sidewall, a second sidewall, a first back wall, and a first front wall when the carton is in the assembled configuration;

a lid coupled to the bottom box and movable between an open position and a closed position;

a first flap extending from the first sidewall of the bottom box; and

a second flap extending from the second sidewall of the bottom box, the second sidewall being opposite the first sidewall and facing the first sidewall, wherein the first flap and the second flap define a set of cavities; and

an insert configured to be disposed within the bottom box of the carton, the insert having a storage configuration where the insert is substantially flat and an assembled configuration, and where the insert defines at least one cavity dimensioned to house at least one container,

wherein, in the assembled configuration, the insert comprises a top wall defining the at least one cavity, a second bottom wall, a second back wall, a second front wall, and a middle wall positioned between the top wall and the second bottom wall.

2. The system of claim **1**, wherein the insert defines a first cavity dimensioned to house a first container and a second cavity dimensioned to house a second container.

3. The system of claim **2**, wherein a diameter of the first cavity is smaller than a diameter of the second cavity, and wherein a diameter of the first container is smaller than a diameter of the second container.

4. (canceled)

5. The system of claim **1**, wherein the carton further comprises a release tab comprising a portion of at least one of the lid and the bottom box and being defined by a perforated line in at least one of the lid and the bottom box, and wherein the release tab is a push tab that detaches the release tab from at least one of the lid and the bottom box.

6. The system of claim **5**, wherein:

when the lid is in the closed position, the release tab is secured to at least one of the lid and the bottom box via the perforation, and

when the lid is in the open position, the release tab is separated along the perforated line from at least one of the lid and the bottom box.

7. (canceled)

8. (canceled)

9. The system of claim **1**, wherein, when the insert is disposed within the bottom box of the carton, the second bottom wall is configured to abut the first bottom wall, the second back wall is configured to abut the first back wall, and the second front wall is configured to abut the first front wall.

10. The system of claim **1**, wherein the at least one container includes a drug vial, wherein the drug vial is filled or pre-filled with a drug, and wherein the drug comprises tarlatamab.

11. (canceled)

12. (canceled)

13. (canceled)

14. The system of claim **1**, wherein the carton and the insert are made of paperboard or a combination of synthetic and natural materials.

15. A system for packaging a plurality of containers, comprising:

a carton having a storage configuration where the carton is substantially flat, and an assembled configuration, the carton comprising:

a bottom box having a first bottom wall, a first sidewall, a second sidewall, a first back wall, and a first front wall when the carton is in the assembled configuration;

a lid coupled to the bottom box and movable between an open position and a closed position;

a tray coupled to the bottom box and movable between a first position and a second position;

a first flap extending from the first sidewall of the bottom box; and

a second flap extending from the second sidewall of the bottom box, the second sidewall being opposite the first sidewall and facing the first sidewall; and

an insert configured to be disposed within the bottom box of the carton, the insert having a storage configuration where the insert is substantially flat and an assembled configuration, and where the insert defines a plurality of cavities, each cavity dimensioned to house a respective one of a plurality of containers,

wherein, in the assembled configuration, the insert comprises a top wall, a second bottom wall, a third sidewall, a fourth sidewall, a second back wall, a second front wall, and a first middle wall positioned between the top wall and the second bottom wall, and

wherein the top wall and the first middle wall define the plurality of cavities.

16. The system of claim **15**, wherein the tray is configured to define a second bottom box disposed within the bottom box when the tray is in the second position.

17. The system of claim **15**, wherein the insert further comprises a first pair of opposing flaps coupled to the second front wall and a second pair of opposing flaps coupled to the second back wall.

18. The system of claim **15**, wherein the insert further comprises a third pair of opposing flaps coupled to the top wall, wherein the third pair of opposing flaps are movable between an open position and a closed position, and wherein, in the closed position, the third pair of opposing flaps define the third sidewall and the fourth sidewall of the insert.

19. (canceled)

20. The system of claim **15**, wherein, when the insert is disposed within the bottom box of the carton, the second bottom wall is configured to abut the first bottom wall, the third sidewall is configured to abut the first sidewall, the fourth sidewall is configured to abut the second sidewall, the second back wall is configured to abut the first back wall, and the second front wall is configured to abut the first front wall.

21. (canceled)

22. (canceled)

23. The system of claim **15**, wherein the insert further comprises a vertical rib support extending between the second middle wall and the second bottom wall, wherein the vertical rib support is configured to maintain a predefined distance between the second middle wall and the second bottom wall.

24. The system of claim **15**, wherein:

a number of cavities defined by the top wall is different from a number of cavities defined by the first middle wall; or

a number of cavities defined by the top wall is equal to a number of cavities defined by the first middle wall.

25. (canceled)

26. The system of claim **15**, wherein:

each of the plurality of cavities have equal dimensions; or

at least one of the plurality of cavities has different dimensions than a remainder of the plurality of cavities.

27. (canceled)

28. The system of claim **15**, wherein the carton further comprises a release tab comprising a portion of at least one of the lid and the bottom box and being defined by a perforated line in at least one of the lid and the bottom box, wherein the release tab is a push tab that detaches the release tab from at least one of the lid and the bottom box.

29. The system of claim **28**, wherein:

when the lid is in the closed position, the release tab is secured to at least one of the lid and the bottom box via the perforation, and

when the lid is in the open position, the release tab is separated along the perforated line from at least one of the lid and the bottom box.

30. (canceled)

31. (canceled)

32. The system of claim **15**, wherein at least one of the plurality of containers includes a drug vial, wherein the drug vial is filled or pre-filled with a drug, and wherein the drug comprises tarlatamab.

33. (canceled)

34. (canceled)

35. (canceled)

36. The system of claim **15**, wherein the carton and the insert are made of paperboard or a combination of synthetic and natural materials.

37-62. (canceled)

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